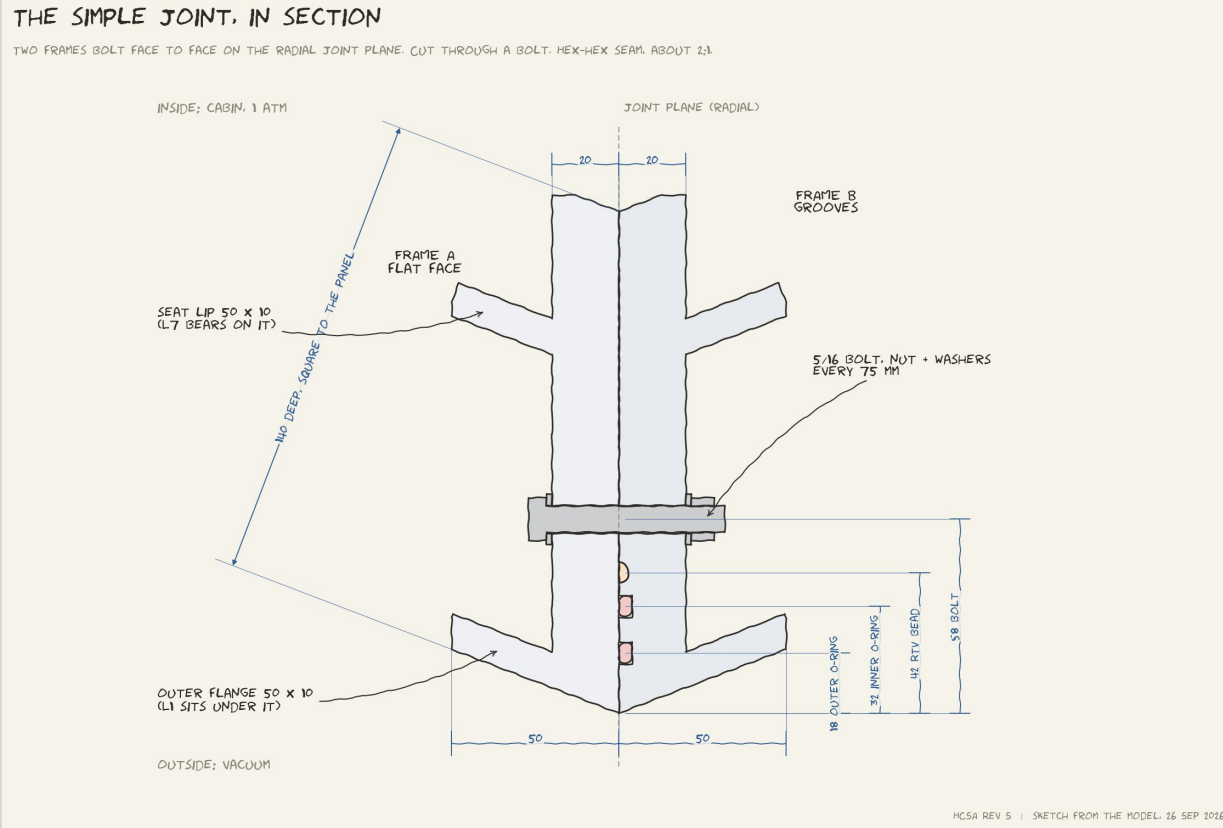
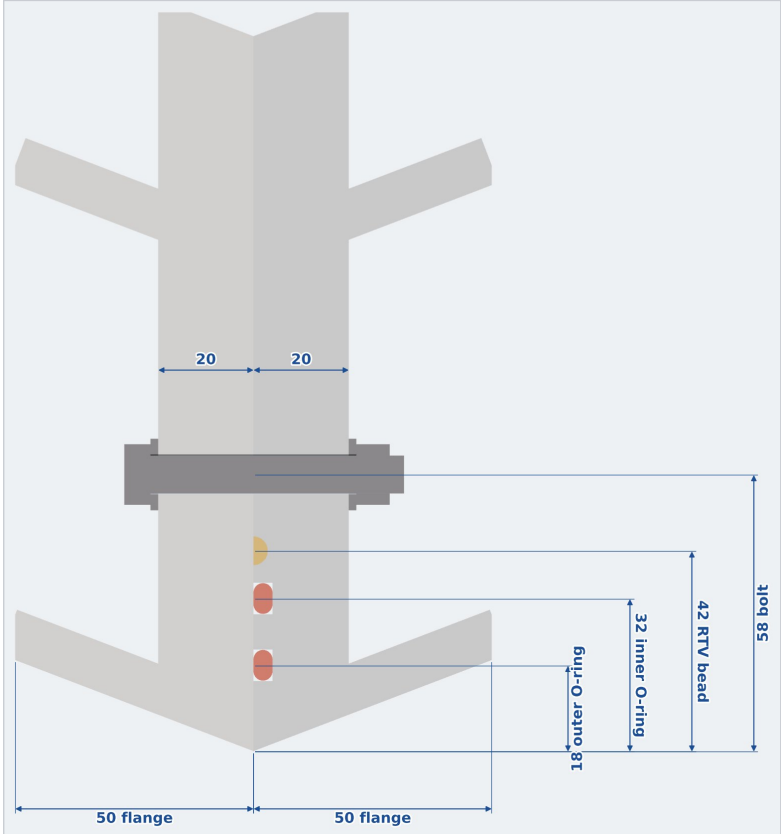


HCSA Rev S: what each drawing shows, and why

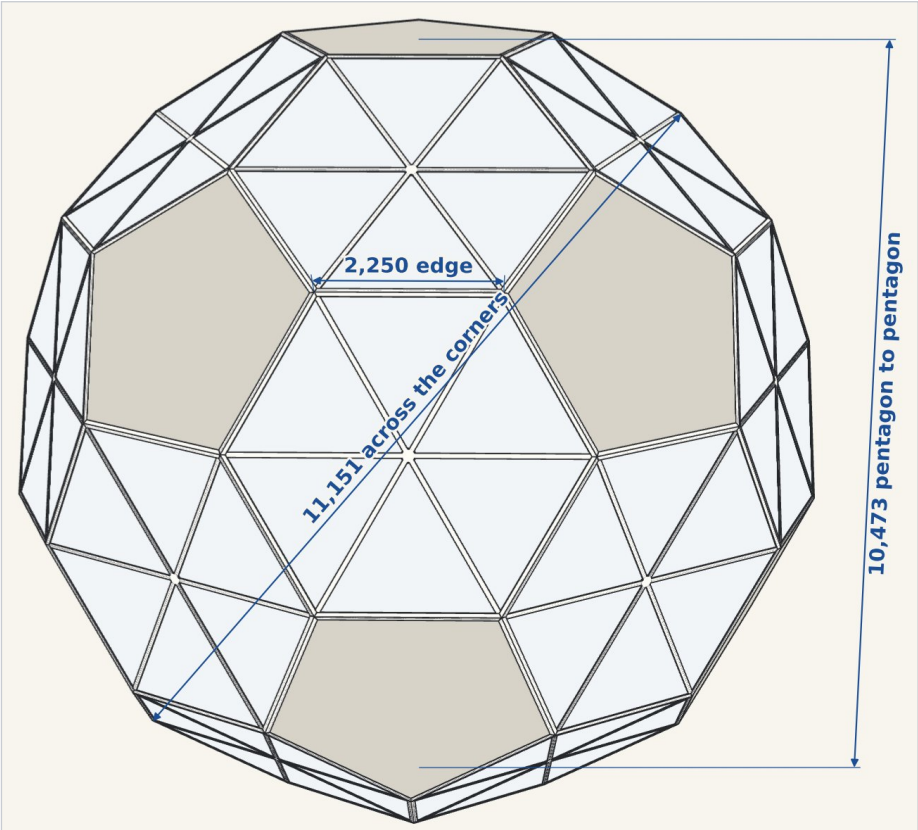
Every sketch and render of the simple joint, the whole shell and the test articles, with dimensions. Manos Theofanous, Brown ENGN 1972, 26 Sep 2026.



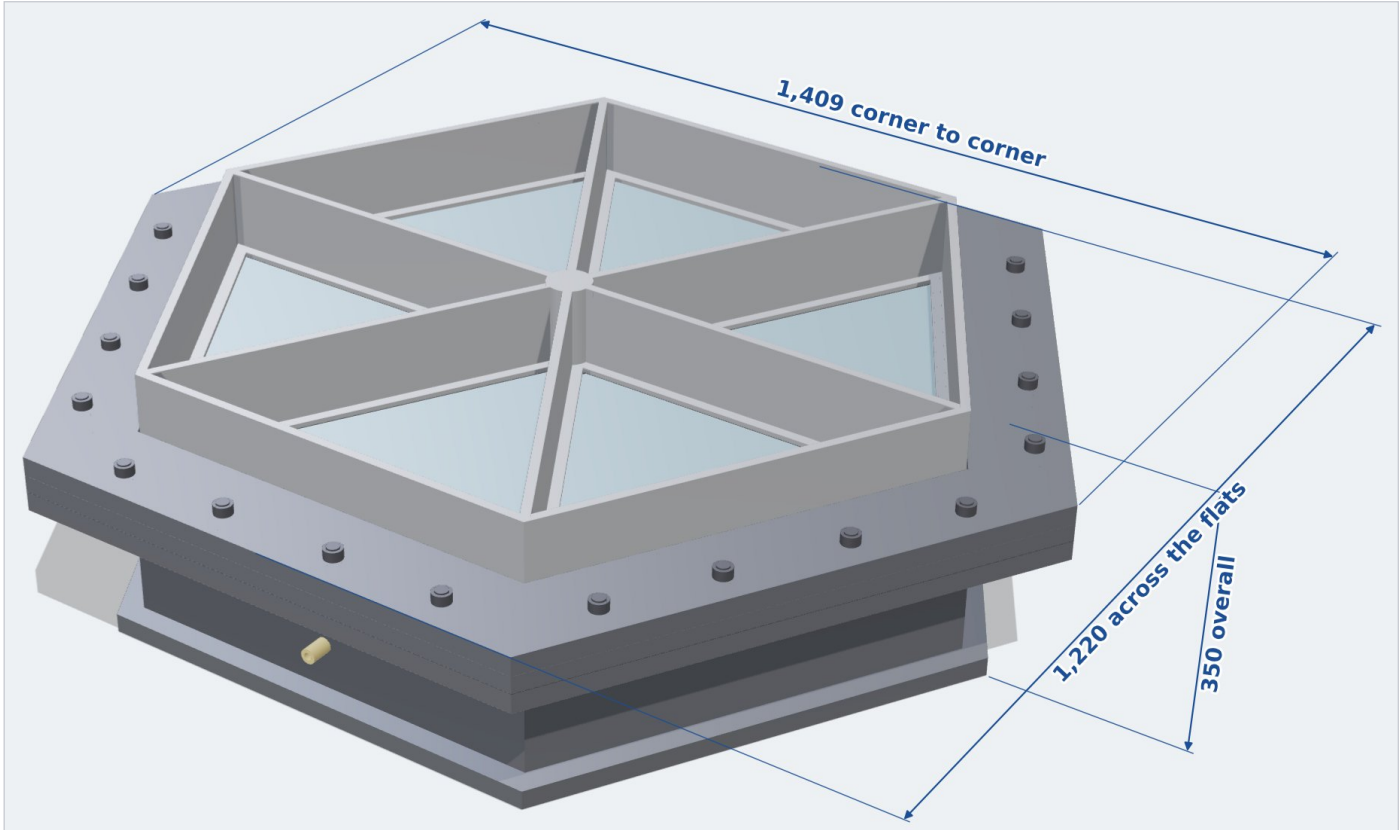
2D sketch



2D render



3D sketch



3D render

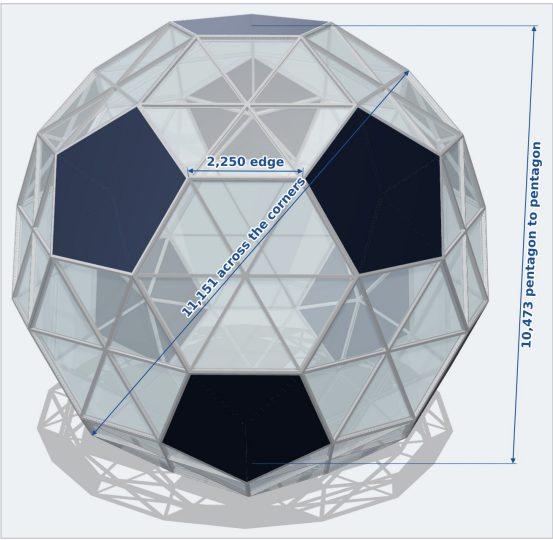
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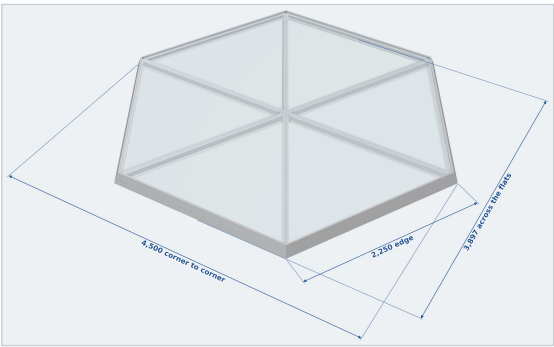
How it all fits together

The facility is a closed shell 11.15 m across, the shape of a football: 20 glass hexagons, which are the windows, and 12 solid pentagons, which house the solar panels and the deployable shields that fold out over the windows. Inside is air at 1 atm; outside is vacuum. So the air pushes out on every panel, about 1.33 MN on each hexagon, and the shell is in tension. It fails by breaking glass, leaking at a seam or pulling a seam apart, not by buckling.

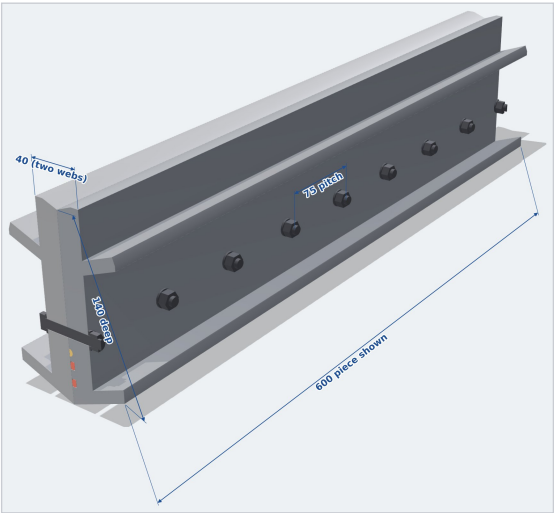
The parts, from big to small



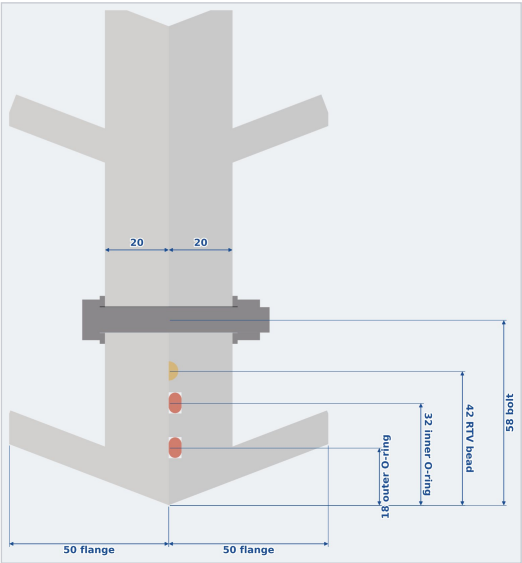
The shell
32 panels, 90 seams, 60 corners



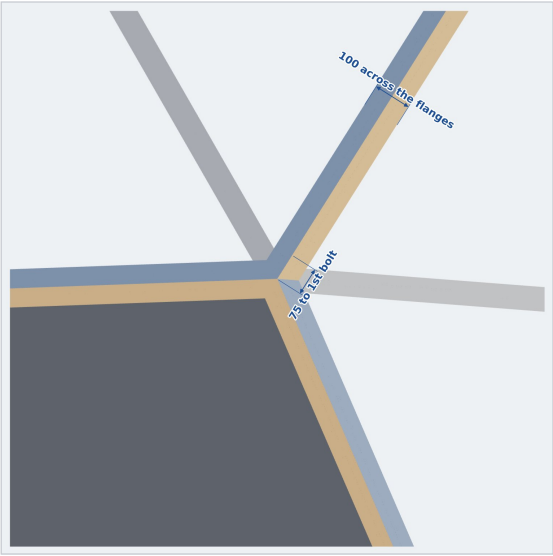
A panel
frames, ribs and panes



A seam
two frames bolted face to face

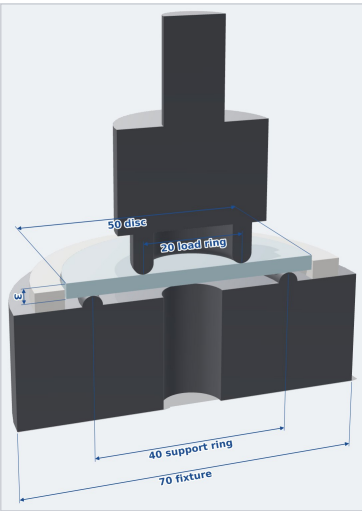


The joint
bolts, two O-rings, RTV

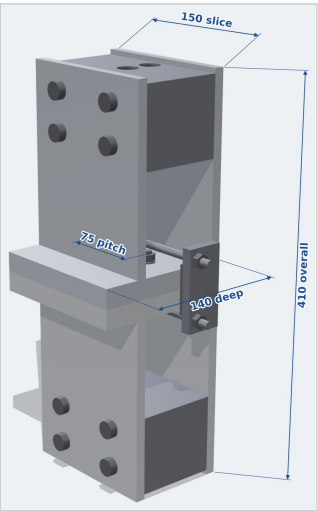


A corner
three seams meet

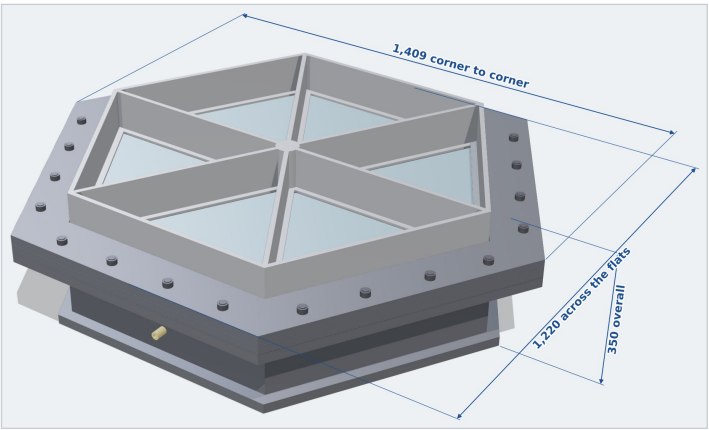
The tests, smallest first



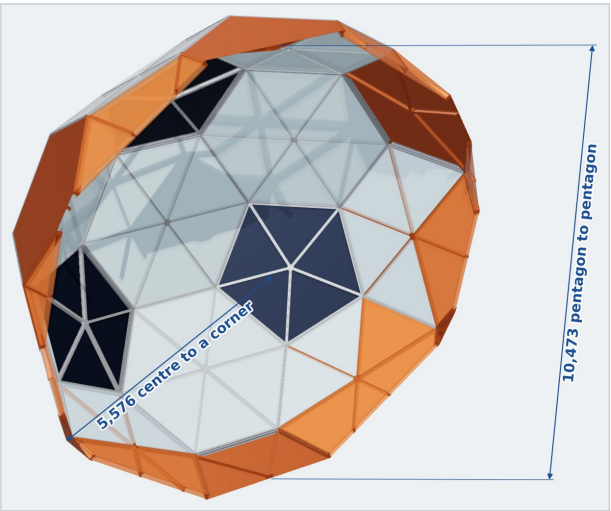
T0 glass coupons
How strong the glass is



T1 seam slices
Bolts and seals under the seam pull



T2 one panel at 1:4
How a panel bends onto its frame and ribs



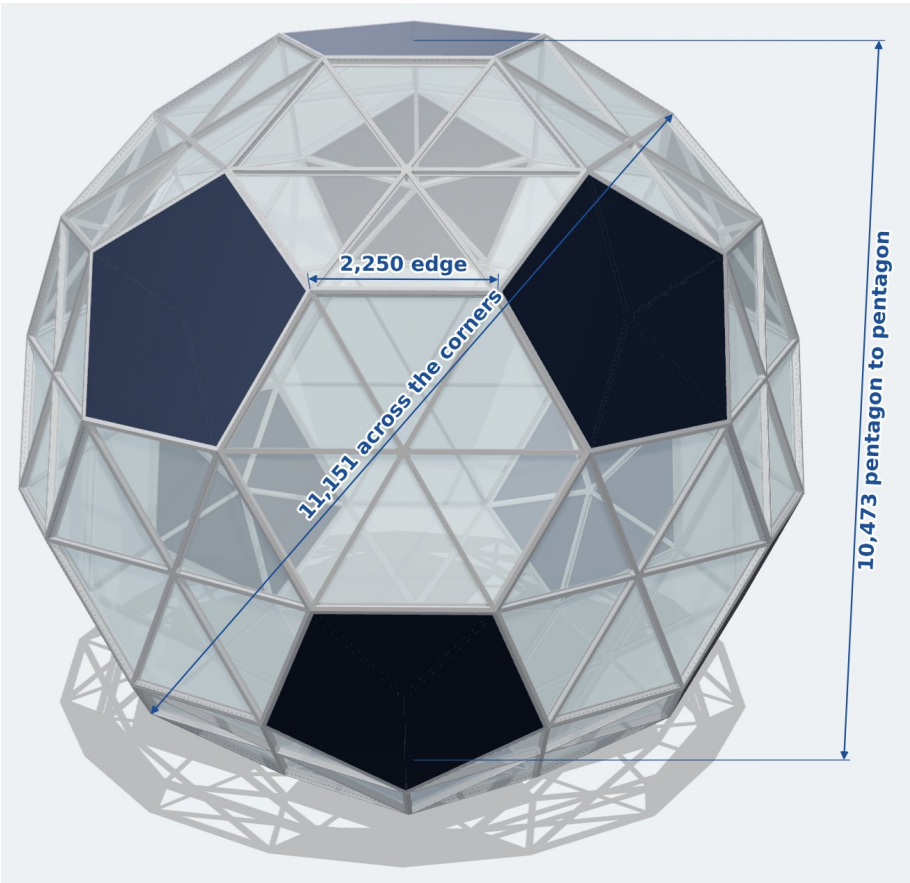
FEA size study
How the whole shell shares the load

How to read the pages

- 2D sketch: a hand-style drawing cut from the model.
- 2D render: a flat, shaded view or section.

- 3D sketch and 3D render: the same tilted view as a line drawing and as a render, shown one above the other.
- Blue lines are dimensions in millimetres, taken from the model. A dimension drawn on a tilted view still gives the true length.

The whole shell



What it is

The complete pressure shell: a truncated icosahedron, 11,151 mm across the corners, with every edge 2,250 mm long. 32 flat panels joined along 90 seams that meet at 60 corners: 20 glass hexagons, which are the windows, and 12 solid pentagons, which house the solar panels and the deployable shields. Every seam has 29 bolts, so the shell has 2,610 of them.

Why it is designed like this

- A sphere is the best shape for holding air in: the pressure pulls on the wall evenly everywhere. This shape is close to a sphere but is built only from flat panels, and flat glass is what glass makers can make at this size.
- It uses just two panel shapes, so every hexagon is the same part and every pentagon is the same part. That keeps the design modular.
- The 2,250 mm edge sets the panel size: a hexagon 4.5 m corner to corner.
- The pentagons are not windows. Each one houses solar panels and a deployable shield that folds out to cover the hexagons around it. Every hexagon borders three pentagons, so the twelve pentagons can reach every window. In the model each pentagon is still only its structure (frames, ribs and a 4 mm plate), because the housing is not designed yet.

Where it sits in the whole

This is the whole structure. Every other page is a piece of it.

How we test it

- No physical test of the whole shell this year.
- The open question is how the shell passes the load between panels. Depending on the path, the pull across a seam is anywhere from 35 to 277 kN per metre. An FEA size study (one panel, a seven-panel cap, the full shell) settles it first.
- A 1:10 closed shell under water (1.12 m across) only if the study says a cap is not enough. Spring 2027 at the earliest.

Numbers to remember

- Inside 1 atm (101.3 kPa), outside vacuum
- 1.33 MN pushing out on each hexagon
- Up to 277 kN/m across each seam (upper bound)
- Model mass as drawn about 26,700 kg (not an estimate: the ribs are still the old, too-weak ones)

2D SKETCH

shell elevation

What you are looking at

The whole shell seen from the side, drawn by hand from the model. The shaded pentagons are the solid ones that house the solar panels and shields; dashed lines are edges at the back; the lines inside each hexagon are its ribs. The pentagon at the top (PENT-00) is seen edge-on, which is why the top is flat.

Dimensions shown (mm)

- 10,753 across in this view (11,151 across the corners)
- 10,473 from the top pentagon to the bottom one

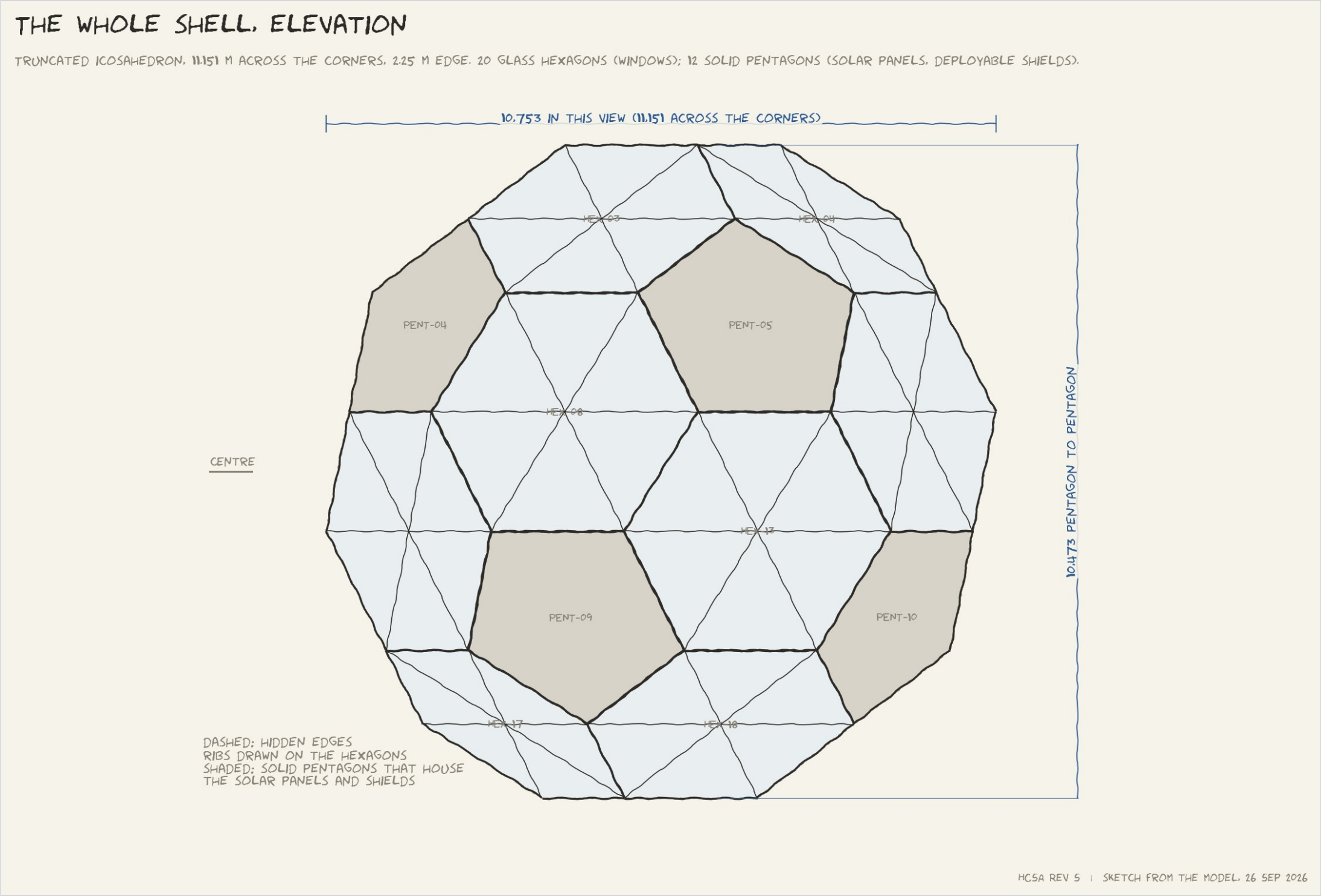
Part of

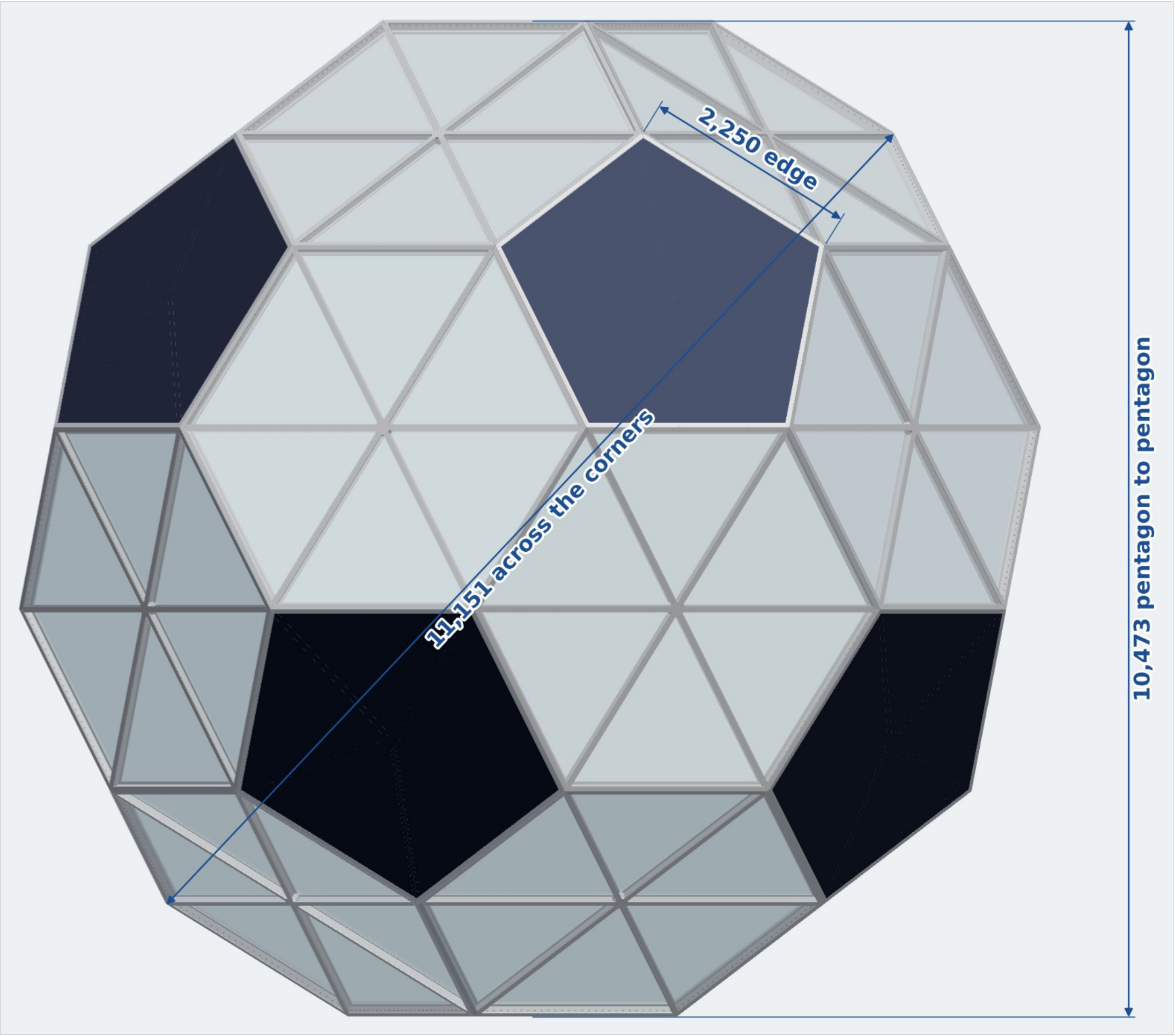
This is the whole structure. Every other page is a piece of it.

Testing

No physical test this year; an FEA size study comes first.

Why it is designed like this: see the page “The whole shell” at the start of this part.





2D RENDER

shell elevation

What you are looking at

The same side view, shaded. The glass hexagons are see-through, so you can see the far side of the shell through them. The dark pentagons are the solid solar panel and shield housings. The blue line through the middle is the size across two opposite corners.

Dimensions shown (mm)

- 11,151 across the corners
- 10,473 pentagon to pentagon
- 2,250 edge

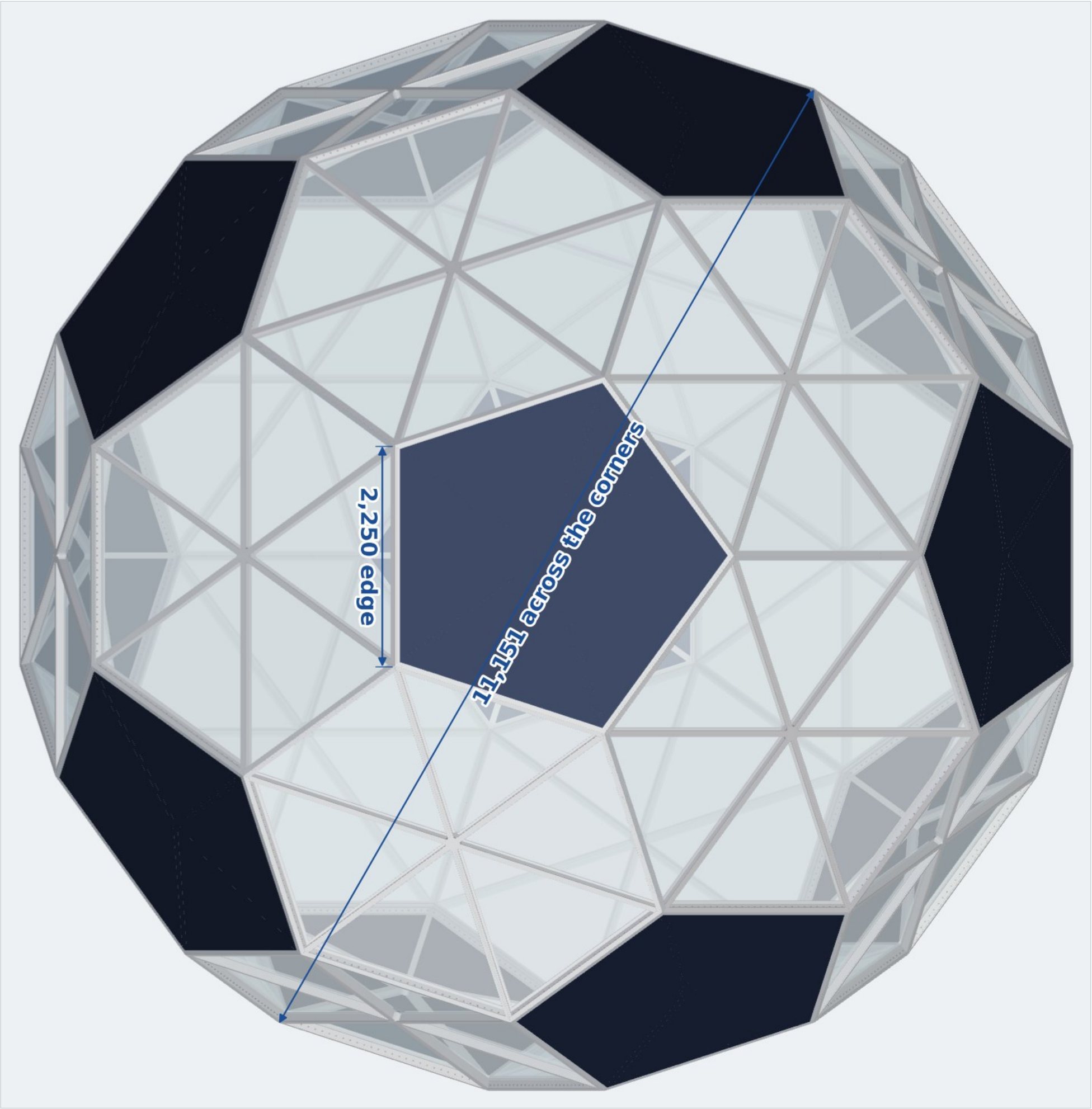
Part of

This is the whole structure. Every other page is a piece of it.

Testing

No physical test this year; an FEA size study comes first.

Why it is designed like this: see the page “The whole shell” at the start of this part.



2D RENDER

shell plan top

What you are looking at

Looking straight down on the top. PENT-00 is in the middle, with five hexagons around it. This is the view that shows the pattern: every pentagon is ringed by hexagons, and no two pentagons touch.

Dimensions shown (mm)

- 11,151 across the corners
- 2,250 edge

Part of

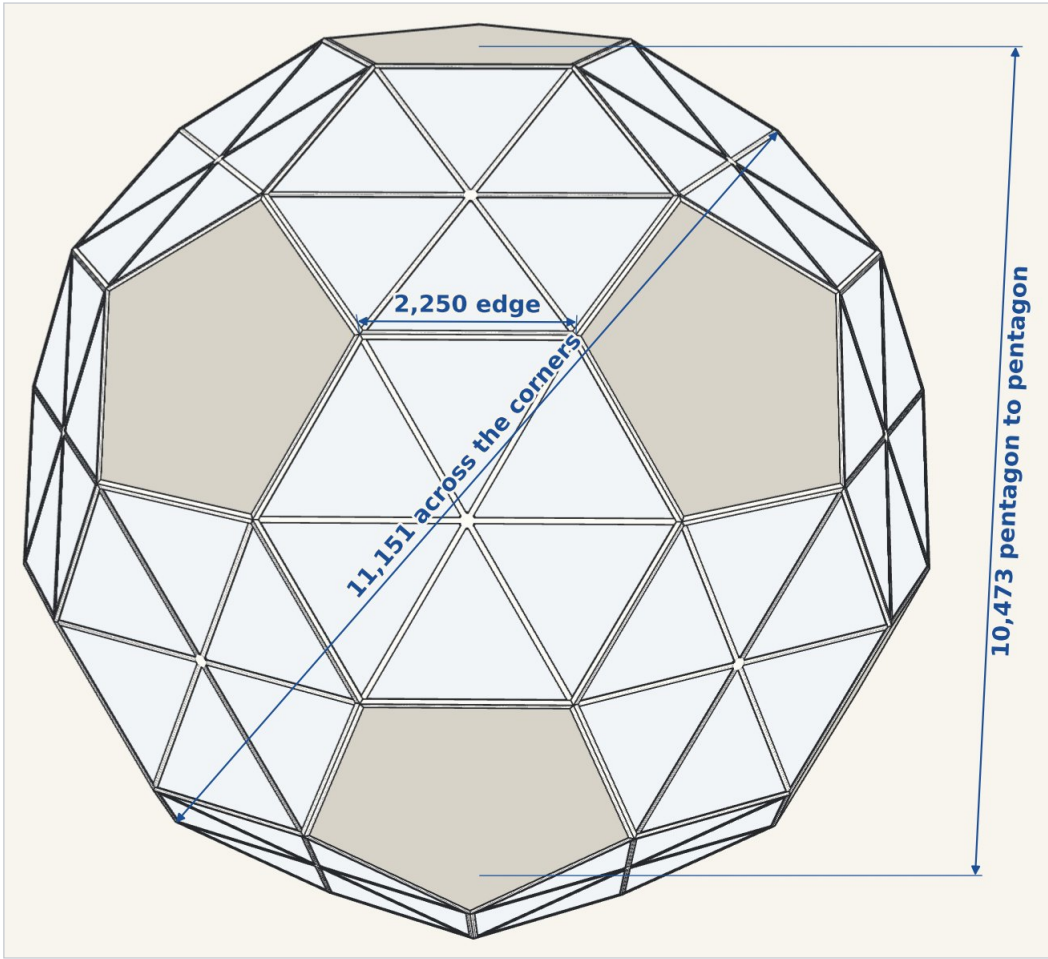
This is the whole structure. Every other page is a piece of it.

Testing

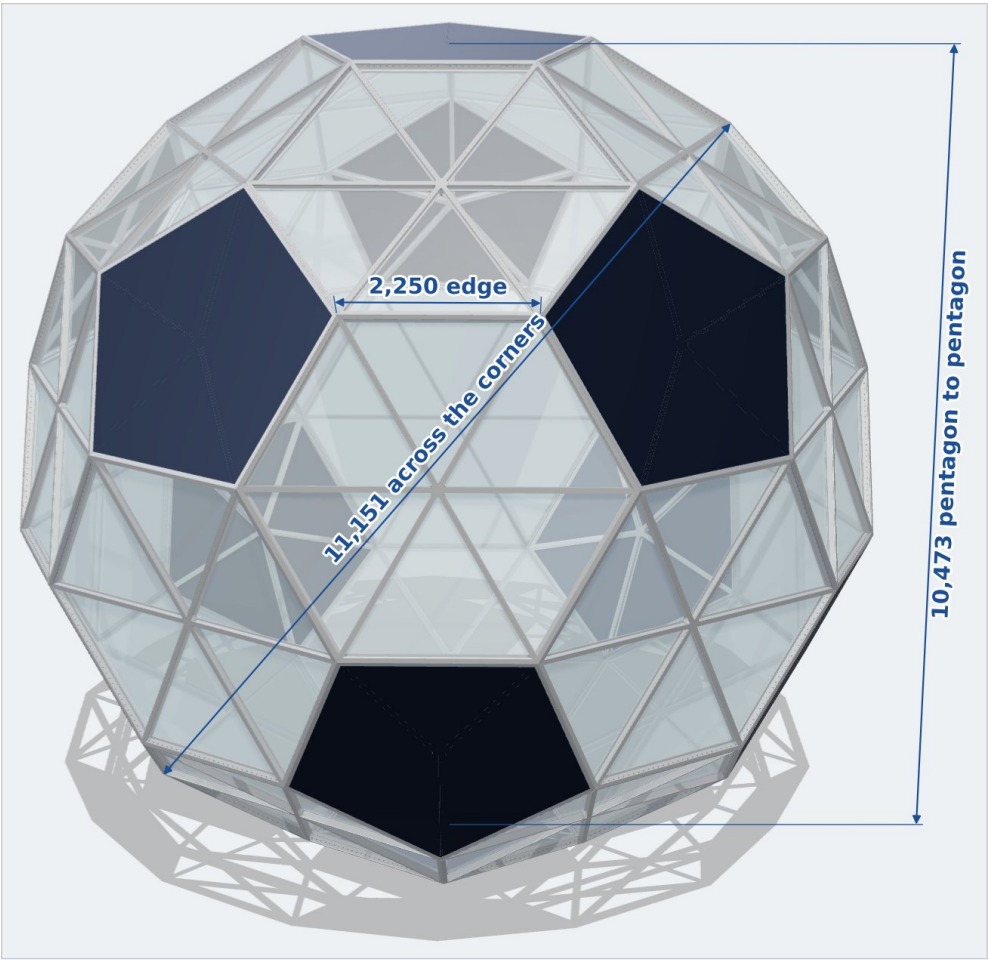
No physical test this year; an FEA size study comes first.

Why it is designed like this: see the page “The whole shell” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

shell exterior

What you are looking at

The whole shell from outside, as a line drawing and as a render. The glass hexagons are the windows; the solid pentagons (shaded in the drawing, dark in the render) house the solar panels and the shields. The dimensions are the size across the corners, the height from the top pentagon to the bottom one, and one edge.

Dimensions shown (mm)

- 11,151 across the corners
- 10,473 pentagon to pentagon
- 2,250 edge

Part of

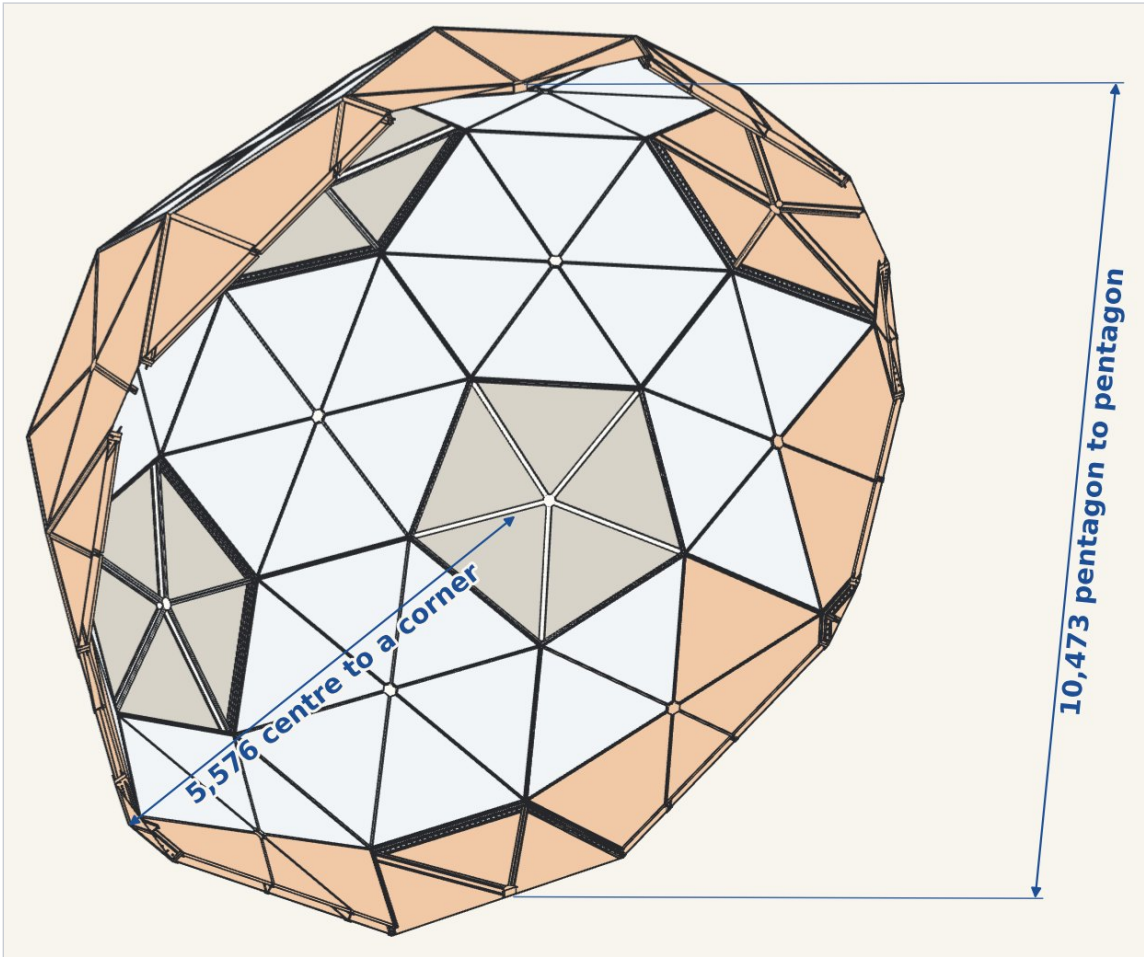
This is the whole structure. Every other page is a piece of it.

Testing

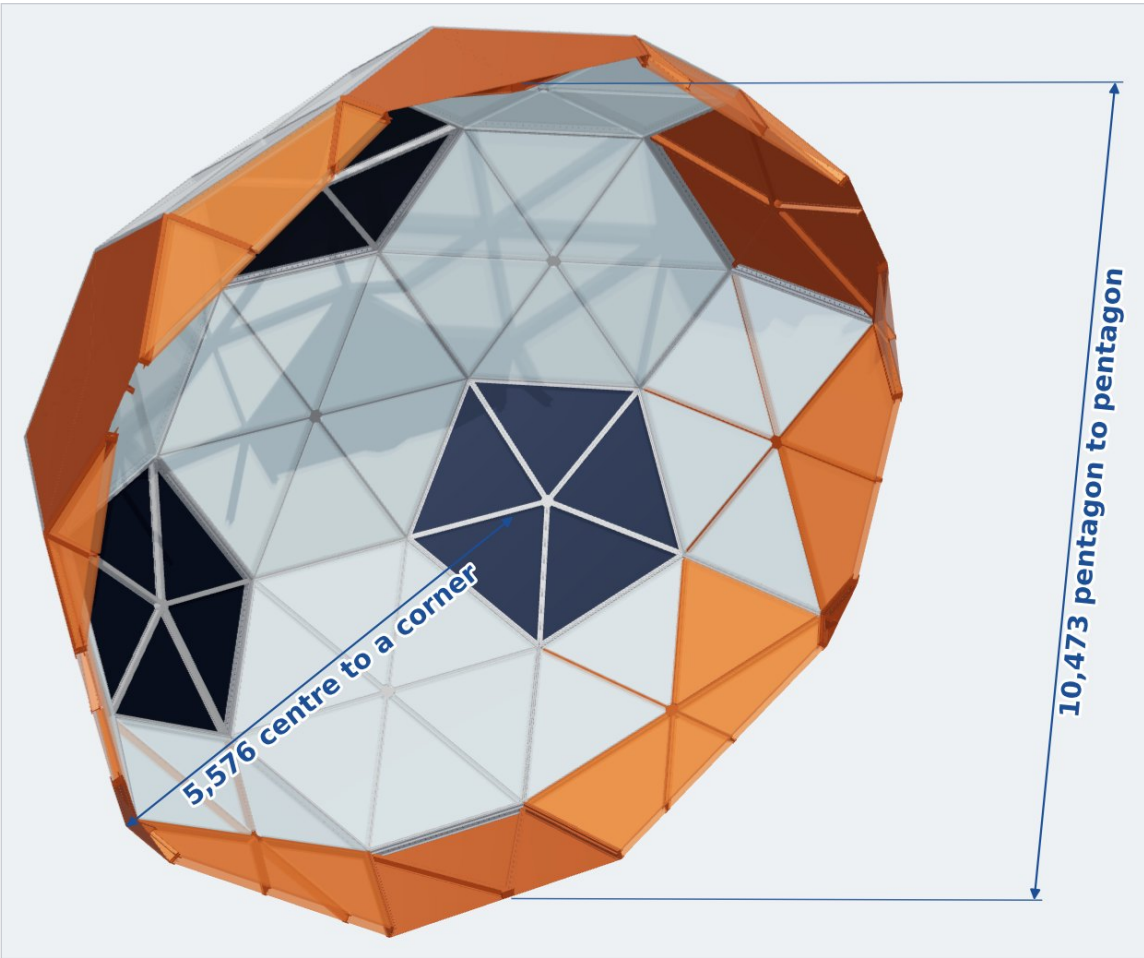
No physical test this year; an FEA size study comes first.

Why it is designed like this: see the page “The whole shell” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

shell cutaway

What you are looking at

The shell cut in half, seen from the side that was removed, so you look into the far half like a bowl. Orange marks the panels and frames the cut goes through, so the rim of the cut is easy to follow. Inside you can see the edge frames and ribs of each panel, the seams between them and the backs of the pentagons. The blue line runs from the centre of the shell to one corner.

Dimensions shown (mm)

- 5,576 centre to a corner
- 10,473 pentagon to pentagon

Part of

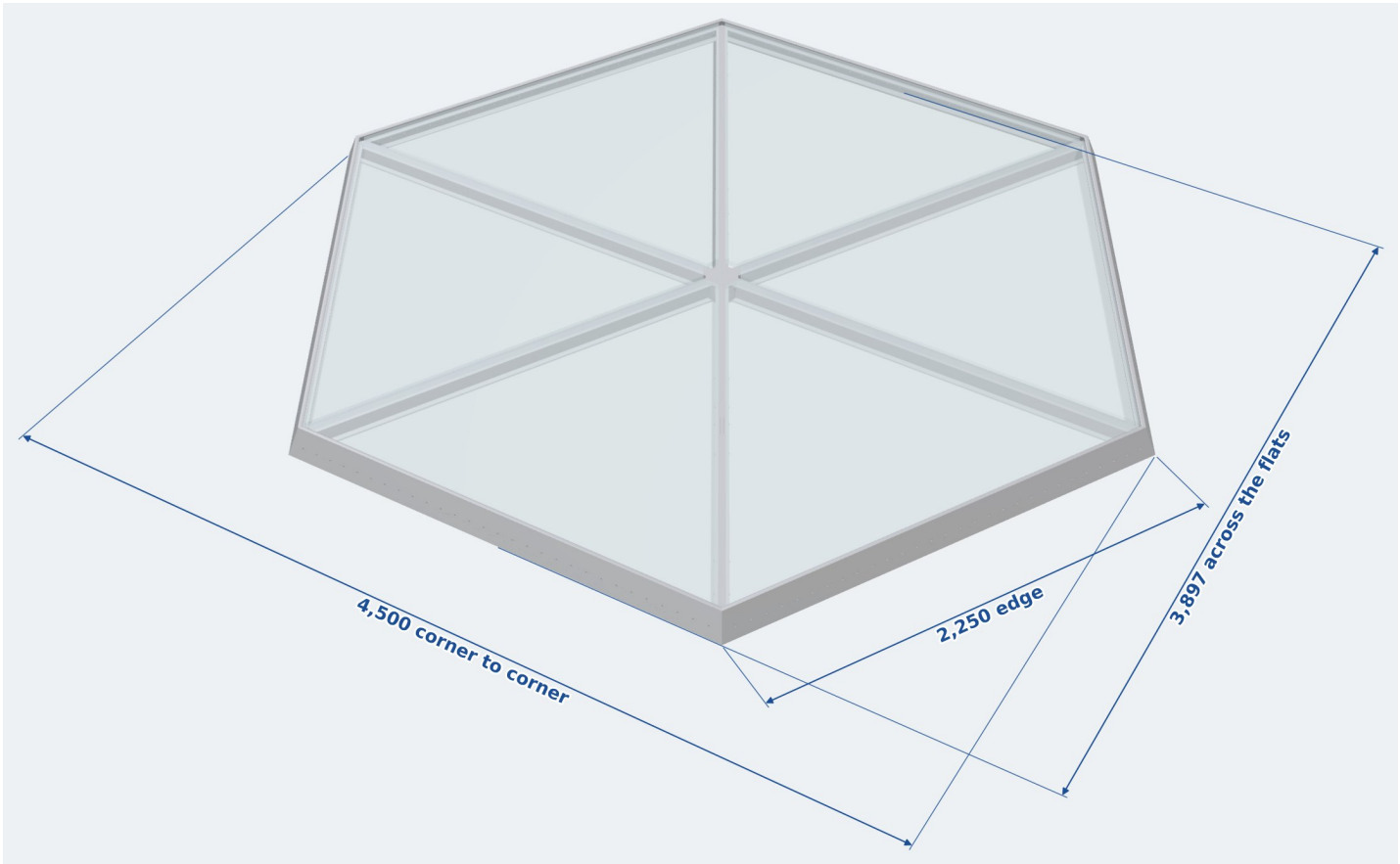
This is the whole structure. Every other page is a piece of it.

Testing

No physical test this year; an FEA size study comes first.

Why it is designed like this: see the page "The whole shell" at the start of this part.

The hexagon panel



What it is

One glass panel: 4,500 mm corner to corner, 3,897 mm across the flats, 140 mm deep. Six edge frames around the outside, six ribs running from a 160 mm hub to the corners, and six triangular windows. In this model each window has two panes: L1, a 6 mm fused silica outer pane under the outer flange, and L7, the 31 mm borosilicate pressure pane that sits on the seat lip. The layers between them (gas gap, solar and tint glass, shutters) are left out of this model.

Why it is designed like this

- One pane 4.5 m across would be far too thick and heavy. The ribs split the window into six triangles, each about 2.2 square metres.
- The pressure pane sits on the cabin side of its seat lip, so the cabin air presses it onto the seat instead of pushing it off.
- The frame is 140 mm deep because it has to hold the whole glazing stack.
- The ribs are spokes: nothing holds the hub, so each rib carries its share of the pane load all the way to a corner. That is why the hand numbers say the current rib is far too weak (about 2,900 MPa at 1 atm). The week 6 rib study makes them deeper or adds a ring.

Where it sits in the whole

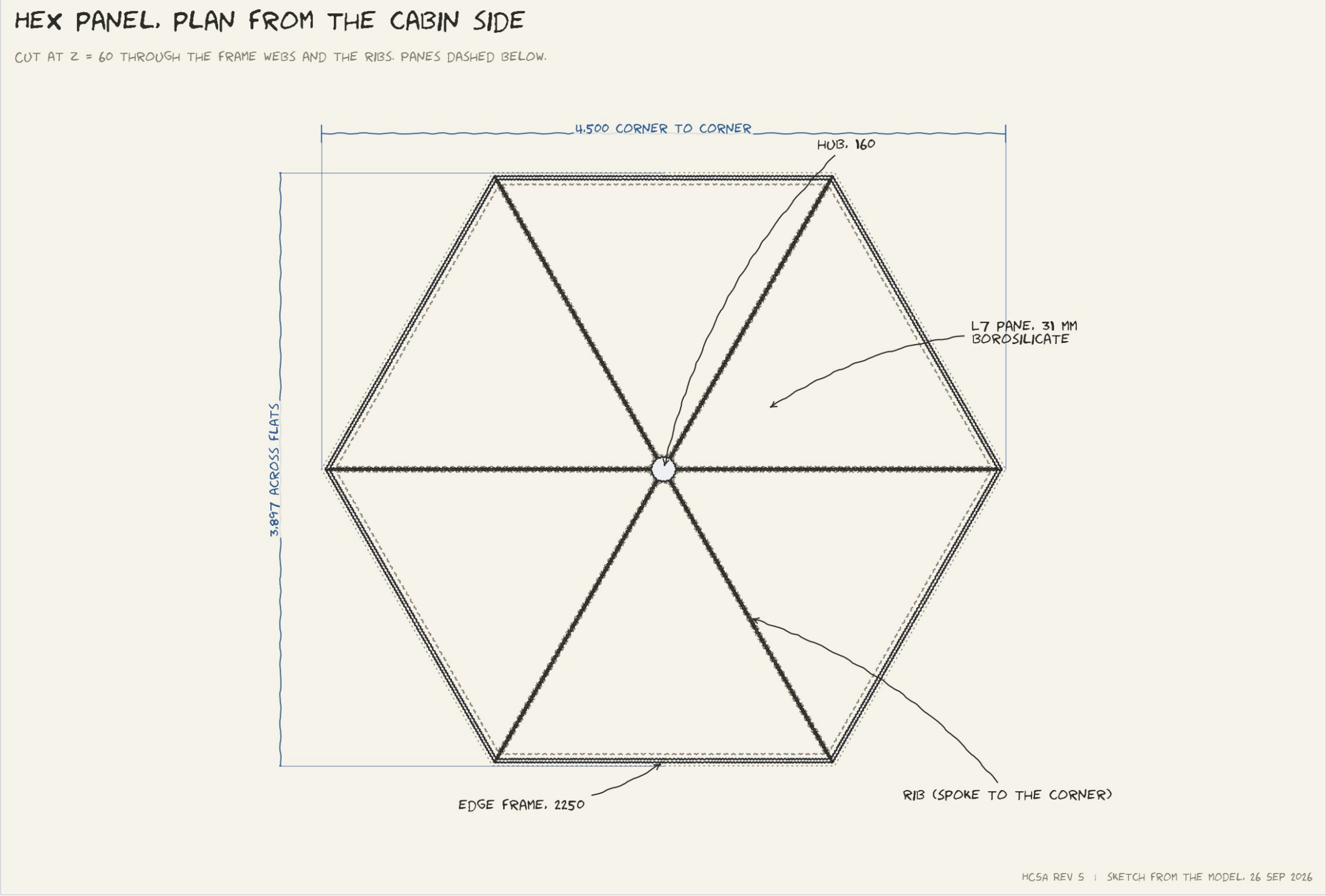
20 of these make up most of the shell.

How we test it

- T2 tests one at 1:4 scale, with aluminium panes in place of glass, to see how the panes bend onto the frame and ribs.
- T0 measures how strong the glass really is.
- A full-size panel test only with an industrial partner.

Numbers to remember

- Pane stress about 53 MPa at 1 atm
- Pane bending 7.2 mm against a 3.1 mm limit if the seat acts like a hinge
- 1.33 MN on the panel at 1 atm



2D SKETCH

hex panel plan

What you are looking at

The hexagon panel from the cabin side, cut 60 mm into the frame so the edge frames and ribs show as outlines. The pressure panes are dashed below. The ribs run from the hub to the six corners.

Dimensions shown (mm)

- 4,500 corner to corner
- 3,897 across the flats
- hub 160 across
- edge frame 2,250

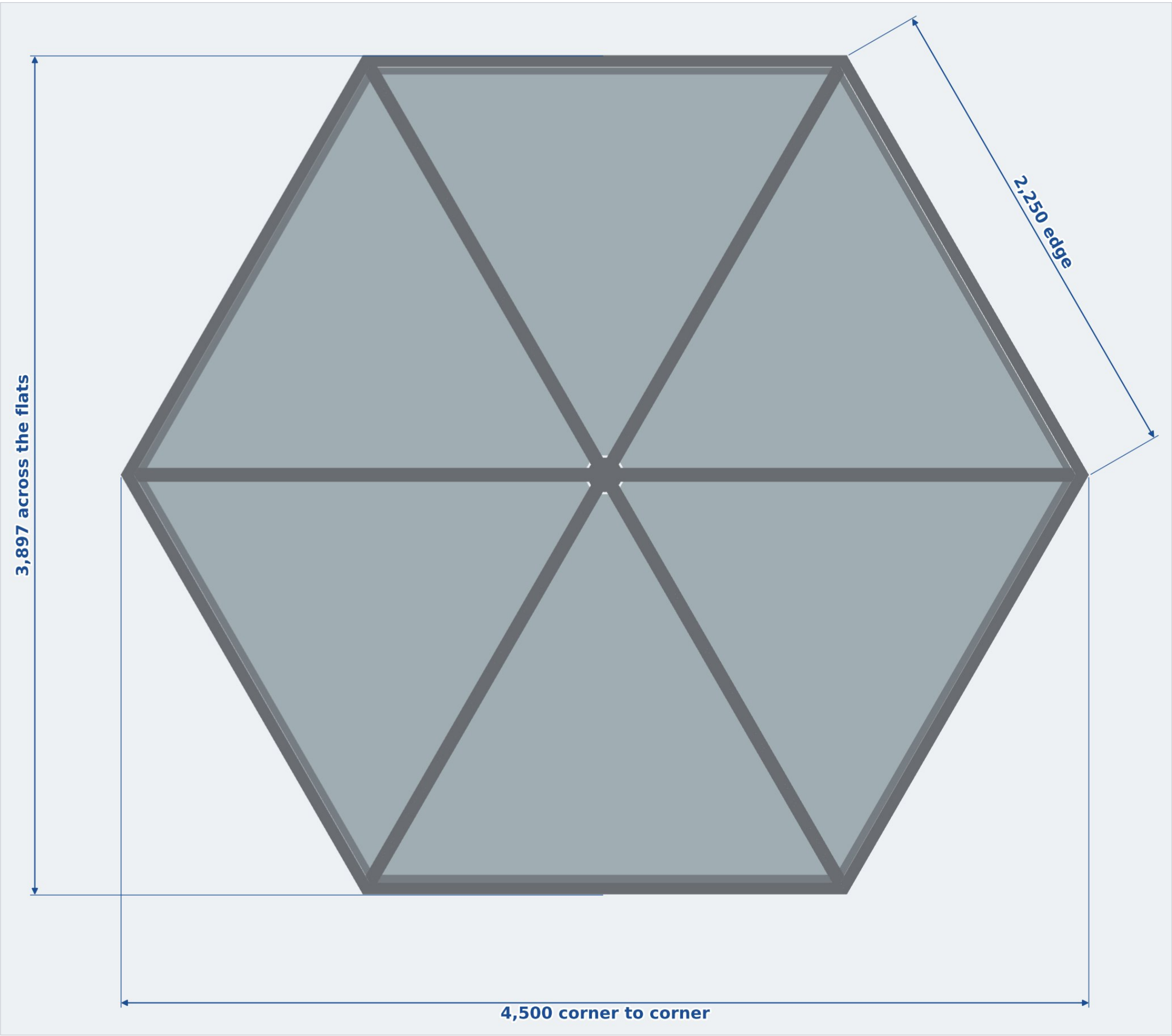
Part of

20 of these make up most of the shell.

Testing

T2 tests one at 1:4 scale; T0 tests the glass.

Why it is designed like this: see the page “The hexagon panel” at the start of this part.



2D RENDER

hex panel plan exterior

What you are looking at

The panel from outside. The six triangles are the L1 outer panes; the grey bands are the outer flanges of the edge frames and the ribs.

Dimensions shown (mm)

- 4,500 corner to corner
- 3,897 across the flats
- 2,250 edge

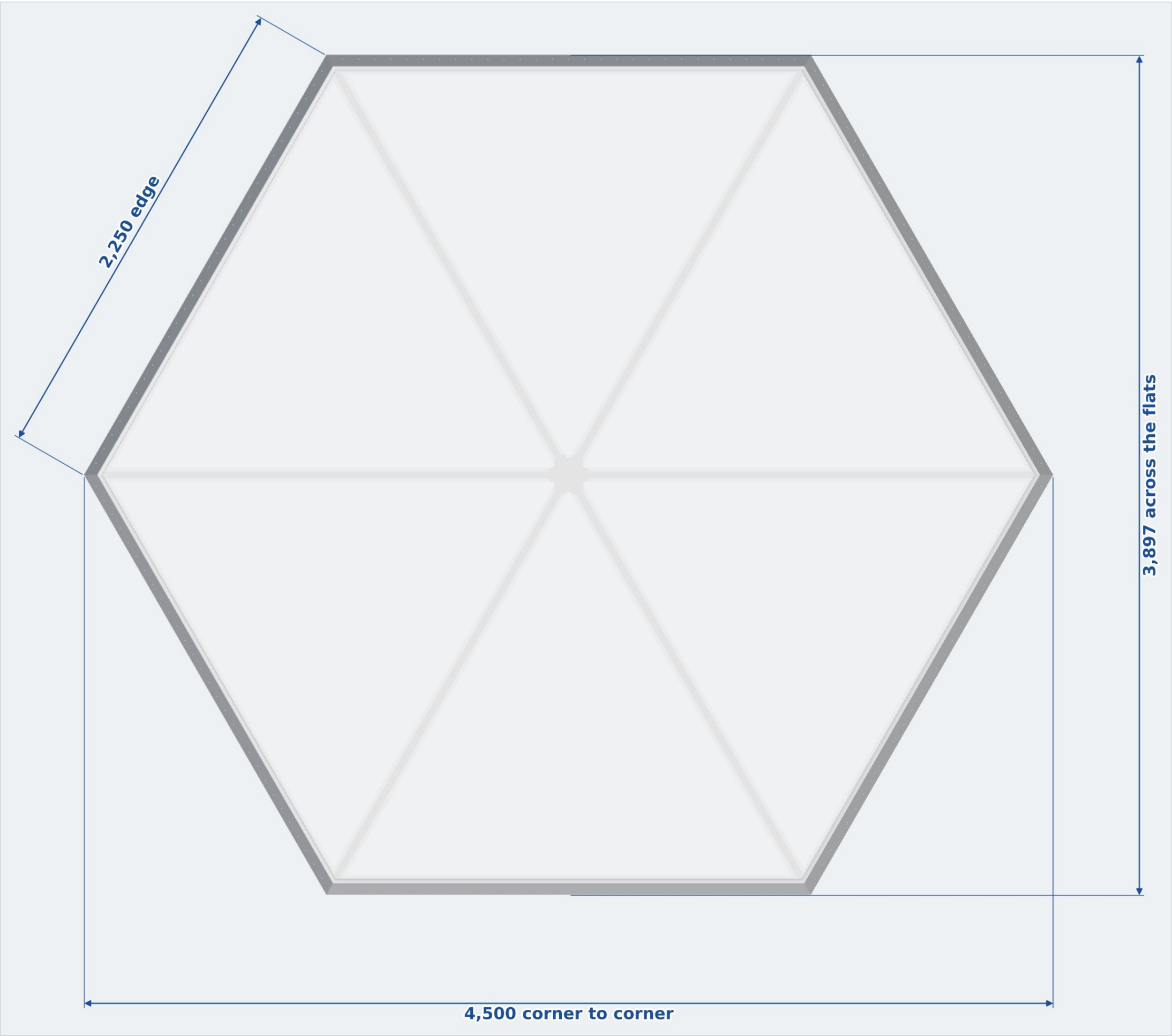
Part of

20 of these make up most of the shell.

Testing

T2 tests one at 1:4 scale; T0 tests the glass.

Why it is designed like this: see the page “The hexagon panel” at the start of this part.



2D RENDER

hex panel plan cabin

What you are looking at
The panel from the cabin side. What you see here are the L7 pressure panes, the ones that hold the air in.

Dimensions shown (mm)

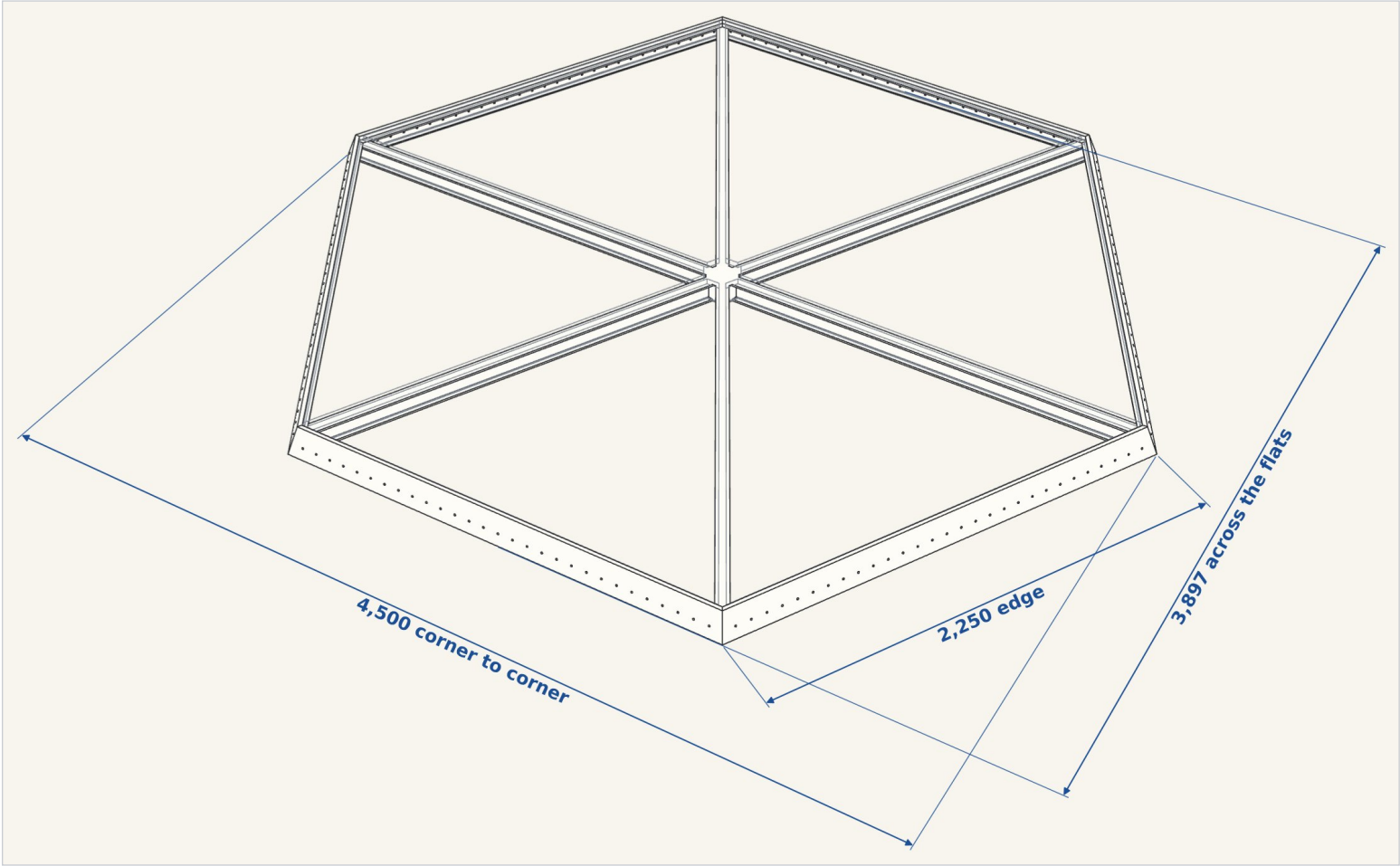
- 4,500 corner to corner
- 3,897 across the flats
- 2,250 edge

Part of
20 of these make up most of the shell.

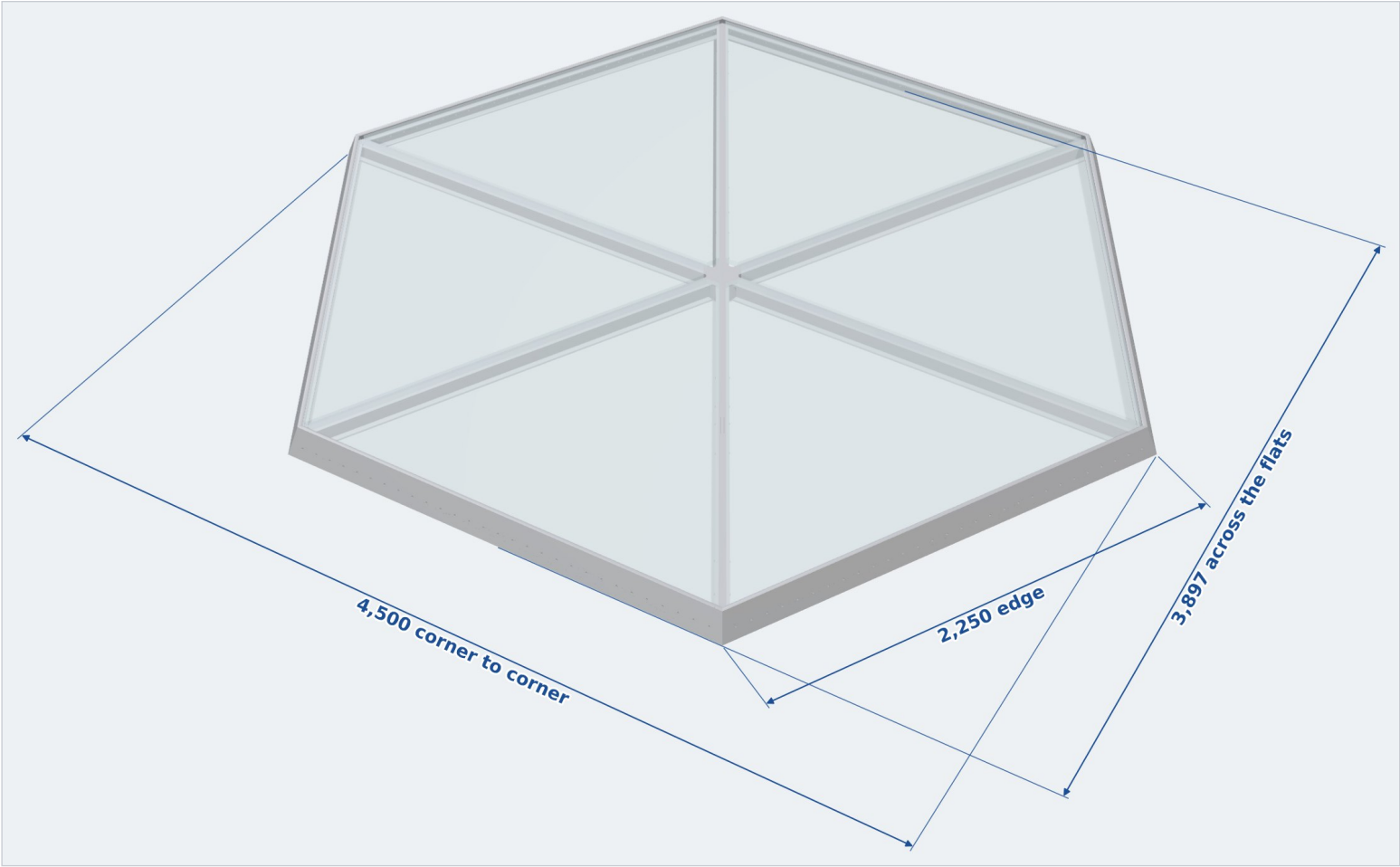
Testing
T2 tests one at 1:4 scale; T0 tests the glass.

Why it is designed like this: see the page “The hexagon panel” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

hex panel cabin side

What you are looking at

The panel from above the cabin side. The frame depth and the six ribs are easier to read in the line drawing; the render shows the glass.

Dimensions shown (mm)

- 4,500 corner to corner
- 3,897 across the flats
- 2,250 edge

Part of

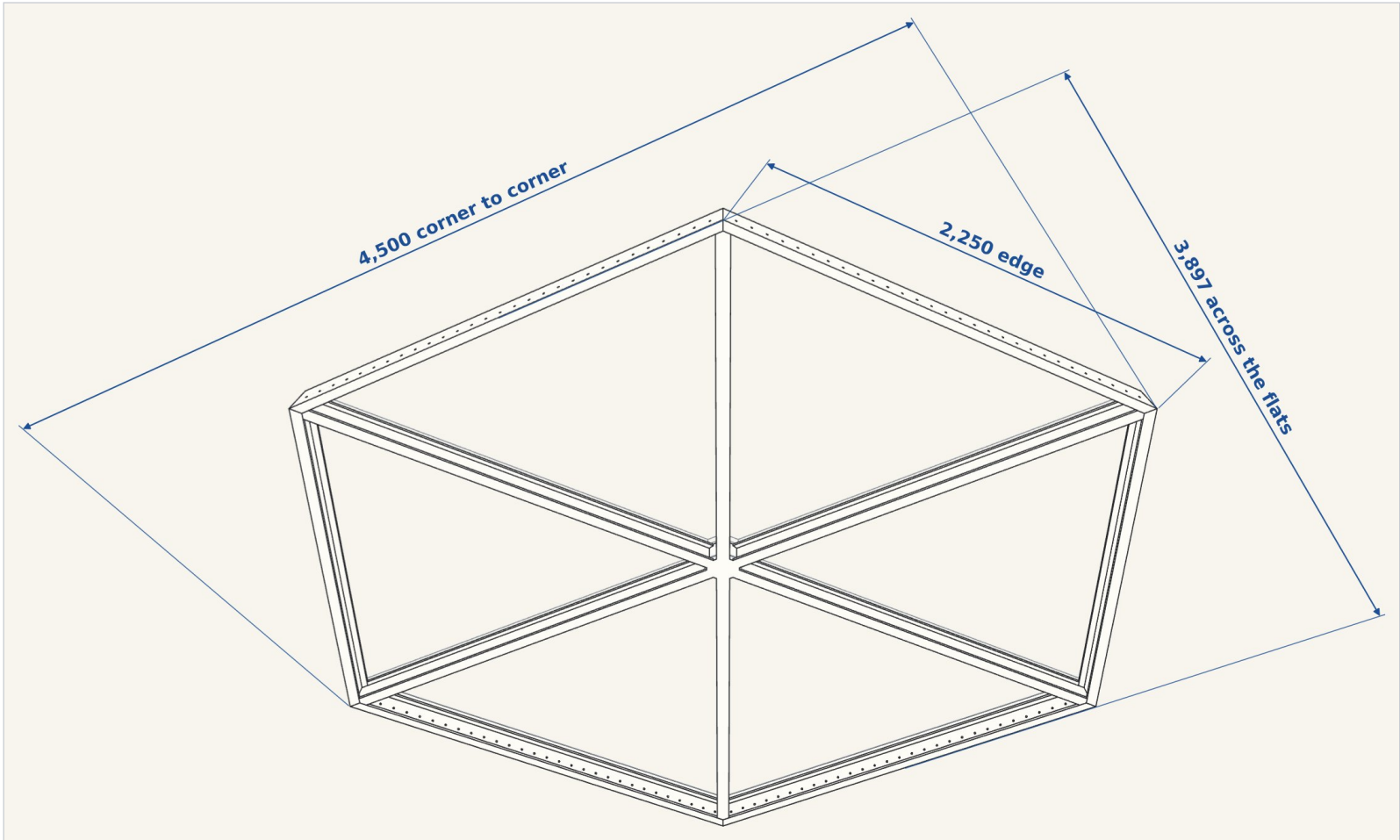
20 of these make up most of the shell.

Testing

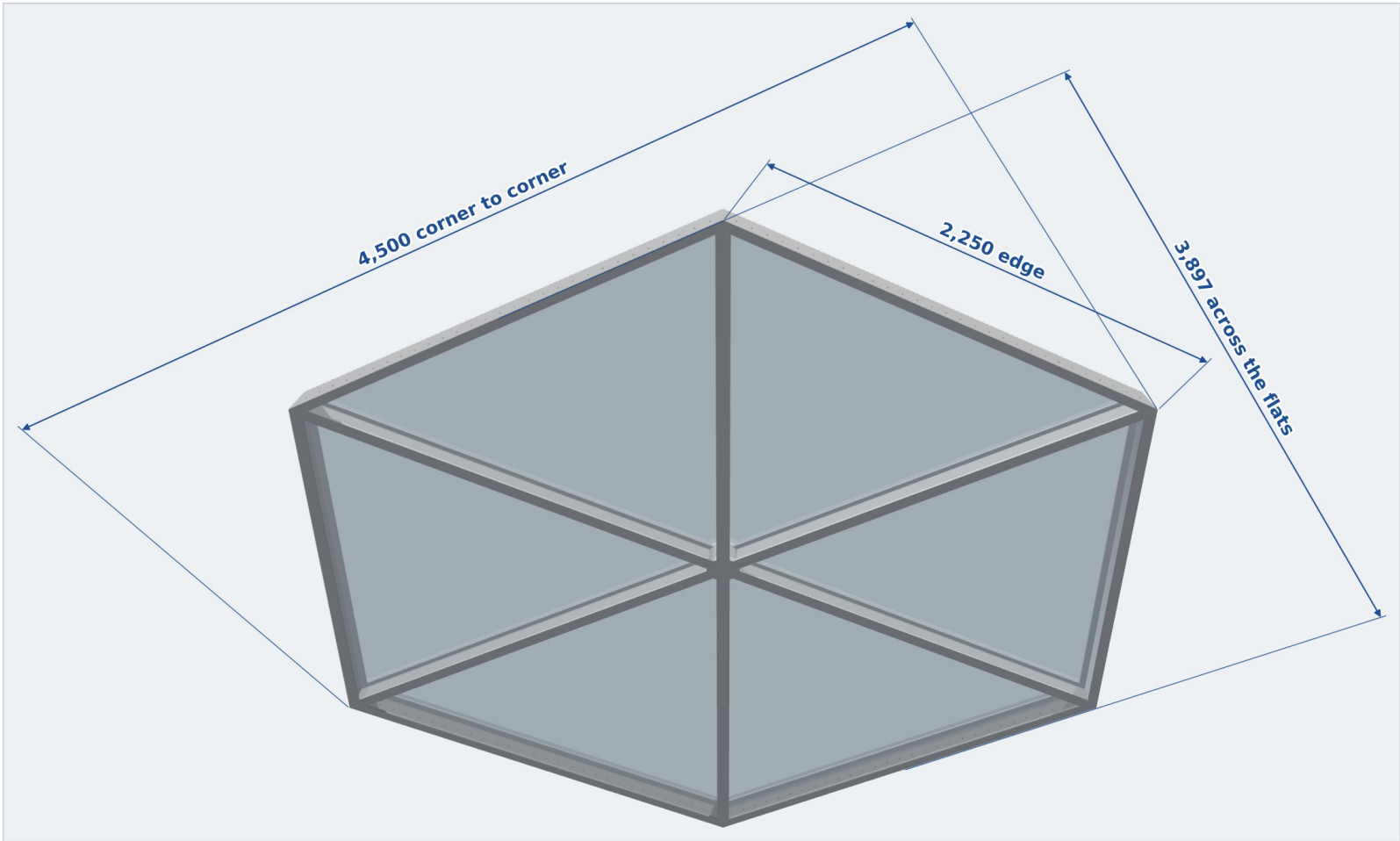
T2 tests one at 1:4 scale; T0 tests the glass.

Why it is designed like this: see the page “The hexagon panel” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

hex panel exterior side

What you are looking at

The panel from the outside, tilted. The outer flanges make a continuous rim; the panes sit under it.

Dimensions shown (mm)

- 4,500 corner to corner
- 3,897 across the flats
- 2,250 edge

Part of

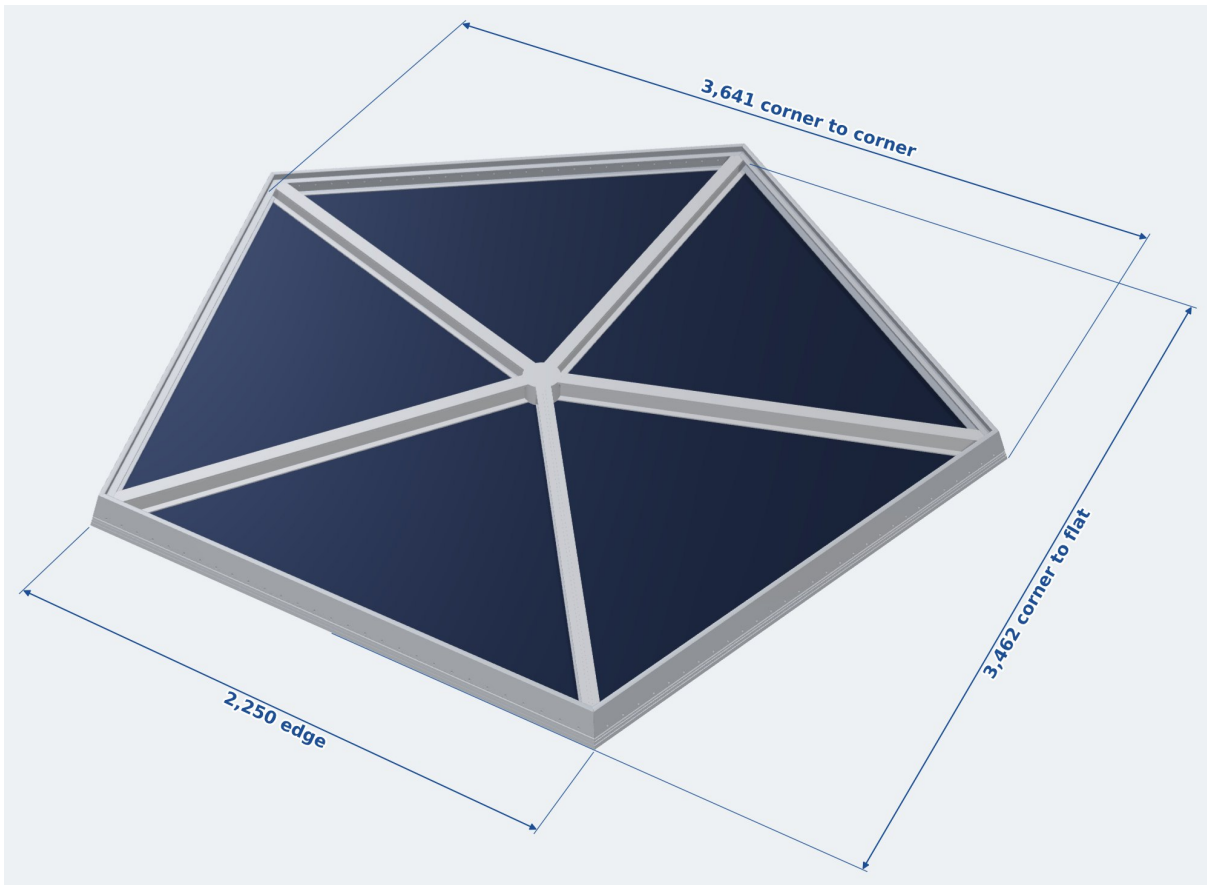
20 of these make up most of the shell.

Testing

T2 tests one at 1:4 scale; T0 tests the glass.

Why it is designed like this: see the page “The hexagon panel” at the start of this part.

The pentagon panel



What it is

One solid panel, not a window: 3,641 mm corner to corner, 3,462 mm from a corner to the opposite flat, 140 mm deep. In the design it houses solar panels and a deployable shield that folds out over the five hexagons around it. The model has only its structure so far: five edge frames, five ribs to a hub, and a 4 mm 6061-T6 face plate. The housing, the solar panels and the shields are not drawn yet.

Why it is designed like this

- Putting the solar panels and the shields on the pentagons keeps all 20 hexagons clear as windows.
- Each pentagon sits in the middle of five hexagons, so a shield stowed there can fold out over its neighbours. Every hexagon borders three pentagons.
- Same frames, same joint and same depth as the hexagons, so every seam in the shell is the same.
- Its frames are always the grooved B type, so all five frames of every pentagon are one part.
- Open: where the housing sits (inside the 140 mm frame depth or proud of the shell), and what a shield folding out puts into the pentagon frame and the corners.

Where it sits in the whole

12 of these: one at the top (PENT-00), one at the bottom, the rest spread evenly between the hexagons.

How we test it

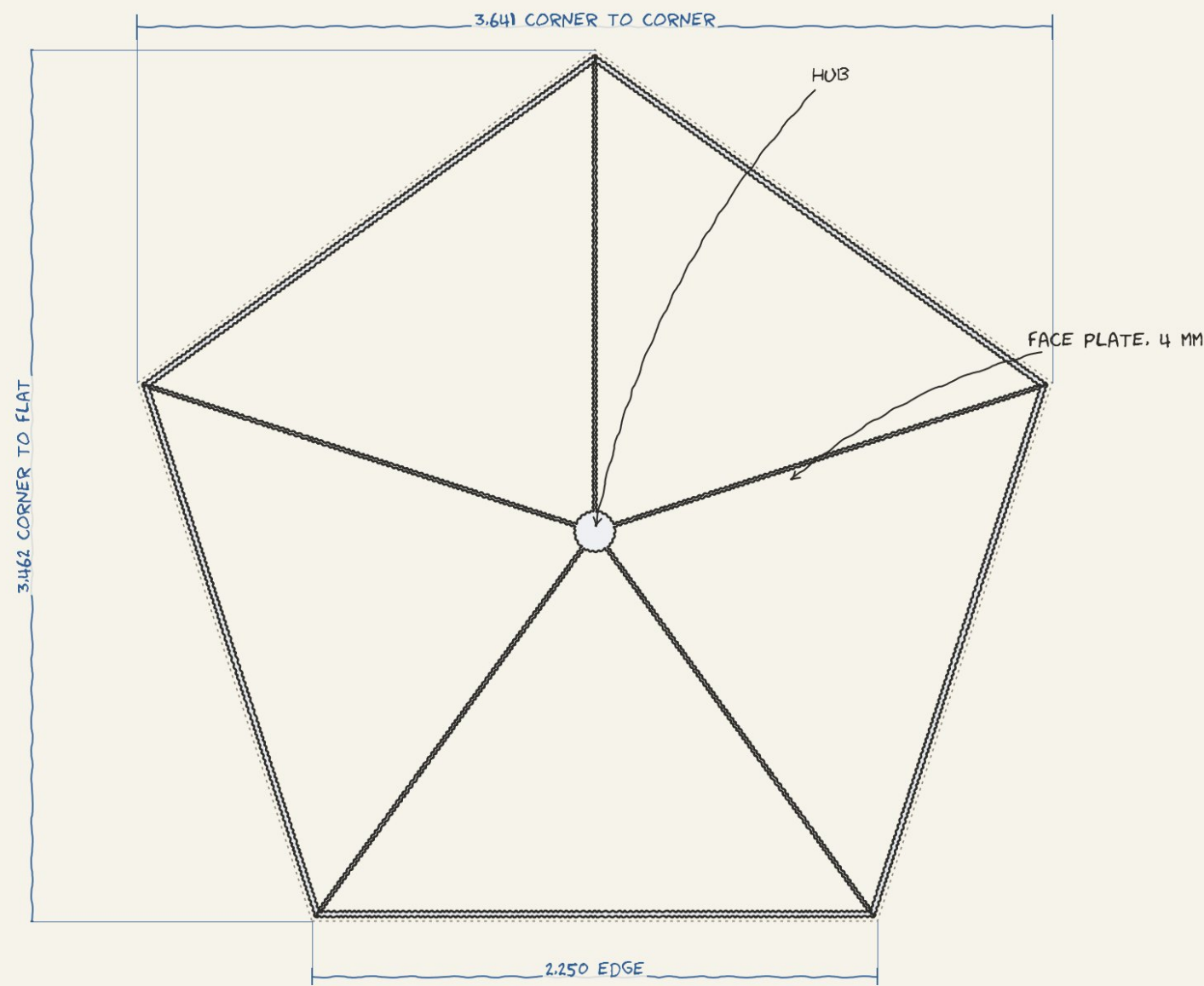
- Not tested on its own this year. Its seams are the same as the hexagon seams, so T1 covers them.
- The housing, the solar panels and the shield mechanism get their own design and tests later. A shield folding out over five hexagons loads the pentagon frame in ways T0 to T2 do not cover.

Numbers to remember

- About 0.88 MN pushing out on each pentagon at 1 atm

PENTAGON PANEL. PLAN FROM THE CABIN SIDE

CUT AT Z = 60. SOLID, NOT A WINDOW; 4 MM FACE PLATE; THE SOLAR PANEL AND SHIELD HOUSING IS NOT DRAWN YET.



HCSA REV 5 | SKETCH FROM THE MODEL, 26 SEP 2026

2D SKETCH

pent panel plan

What you are looking at

The pentagon panel from the cabin side, cut 60 mm in. Five frames, five ribs to the hub, and the face plate behind them. The solar panel and shield housing that goes on this panel is not drawn yet.

Dimensions shown (mm)

- 3,641 corner to corner
- 3,462 corner to flat
- 2,250 edge

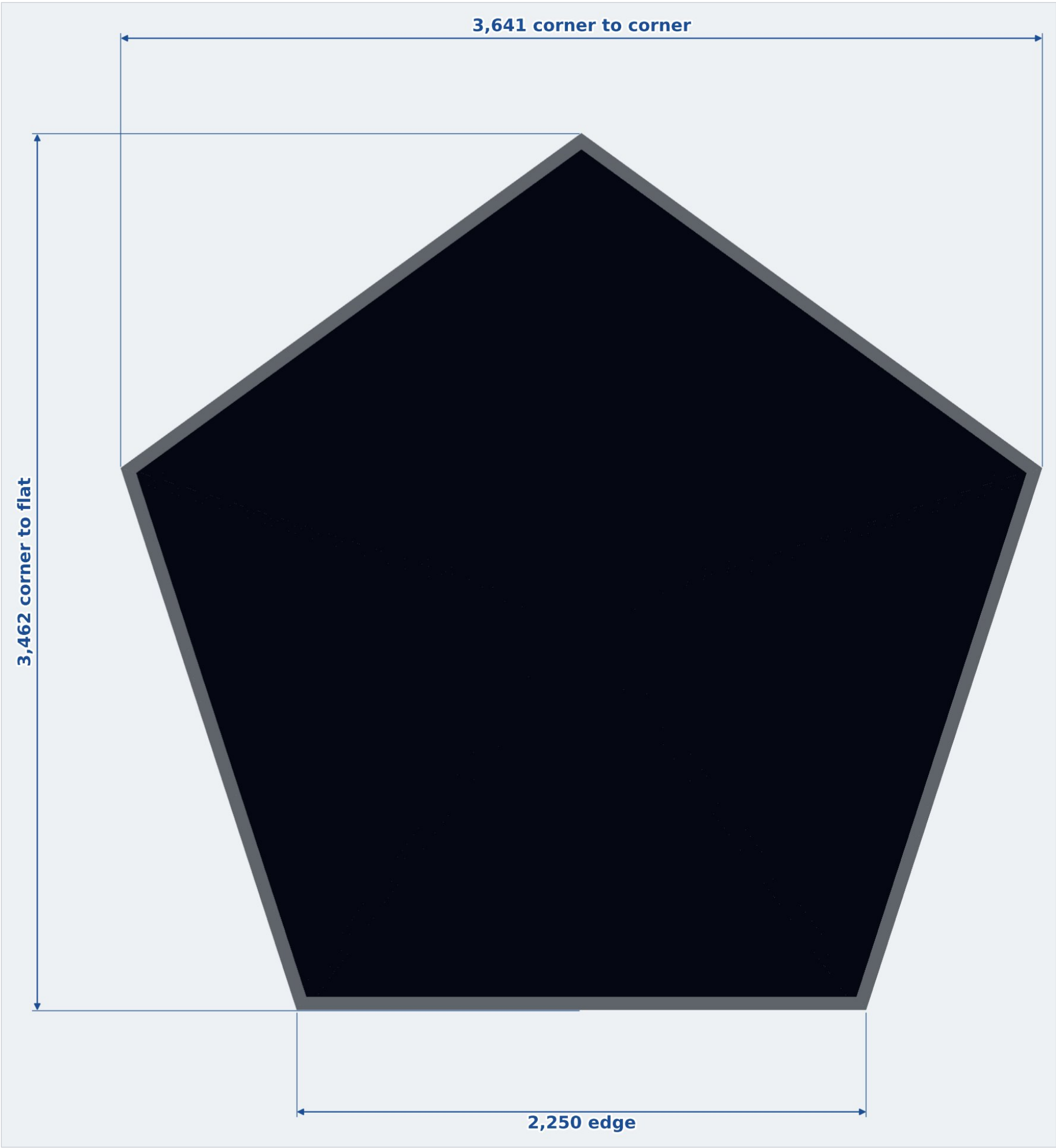
Part of

12 of these: one at the top (PENT-00), one at the bottom, the rest spread evenly between the hexagons.

Testing

Not tested on its own; its seams are covered by T1.

Why it is designed like this: see the page “The pentagon panel” at the start of this part.



2D RENDER

pent panel plan exterior

What you are looking at

The pentagon from outside: the face plate (drawn dark so it does not read as glass) inside the rim of the frames. In the design this is where the solar panels and the stowed shield sit.

Dimensions shown (mm)

- 3,641 corner to corner
- 3,462 corner to flat
- 2,250 edge

Part of

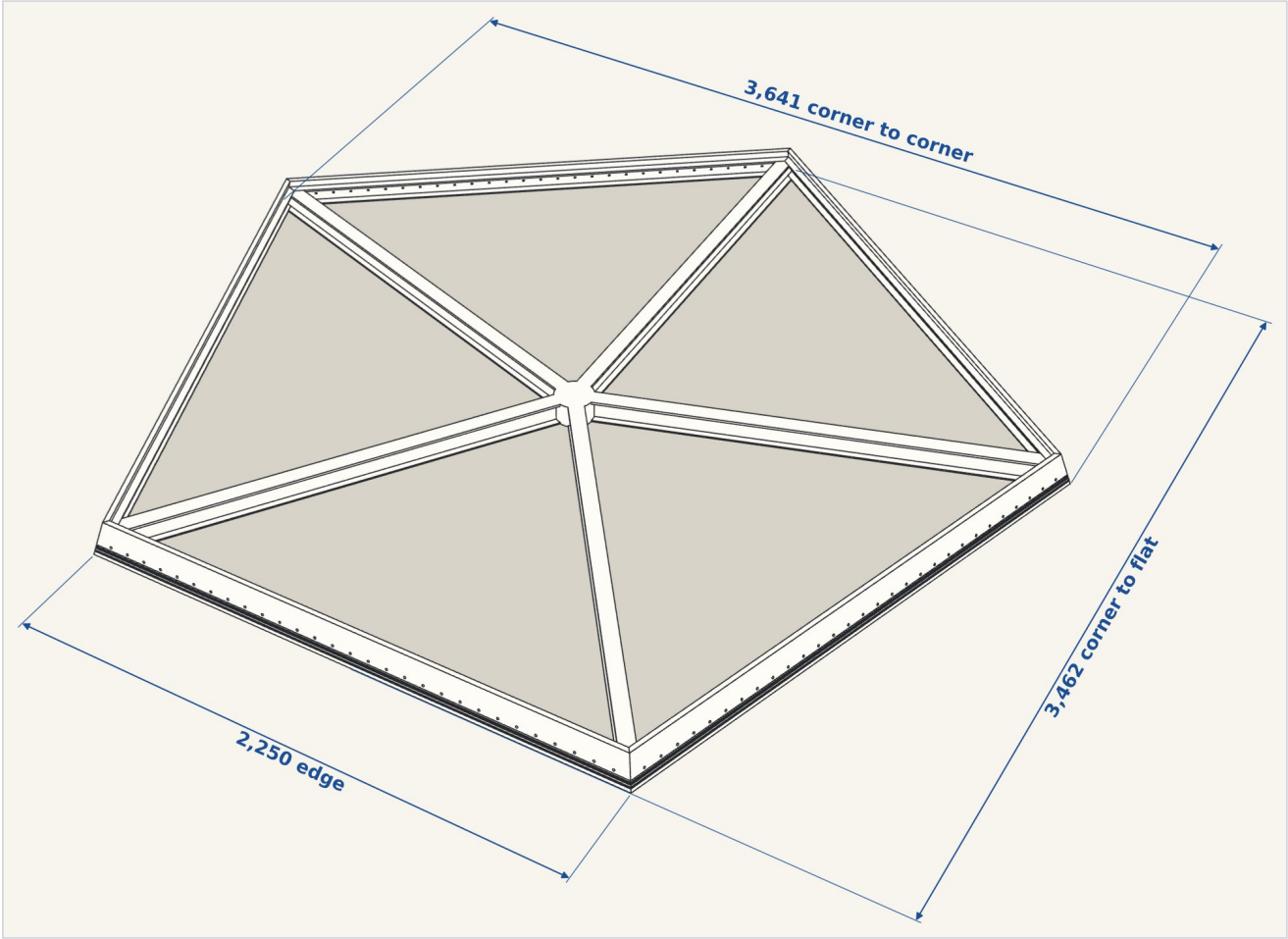
12 of these: one at the top (PENT-00), one at the bottom, the rest spread evenly between the hexagons.

Testing

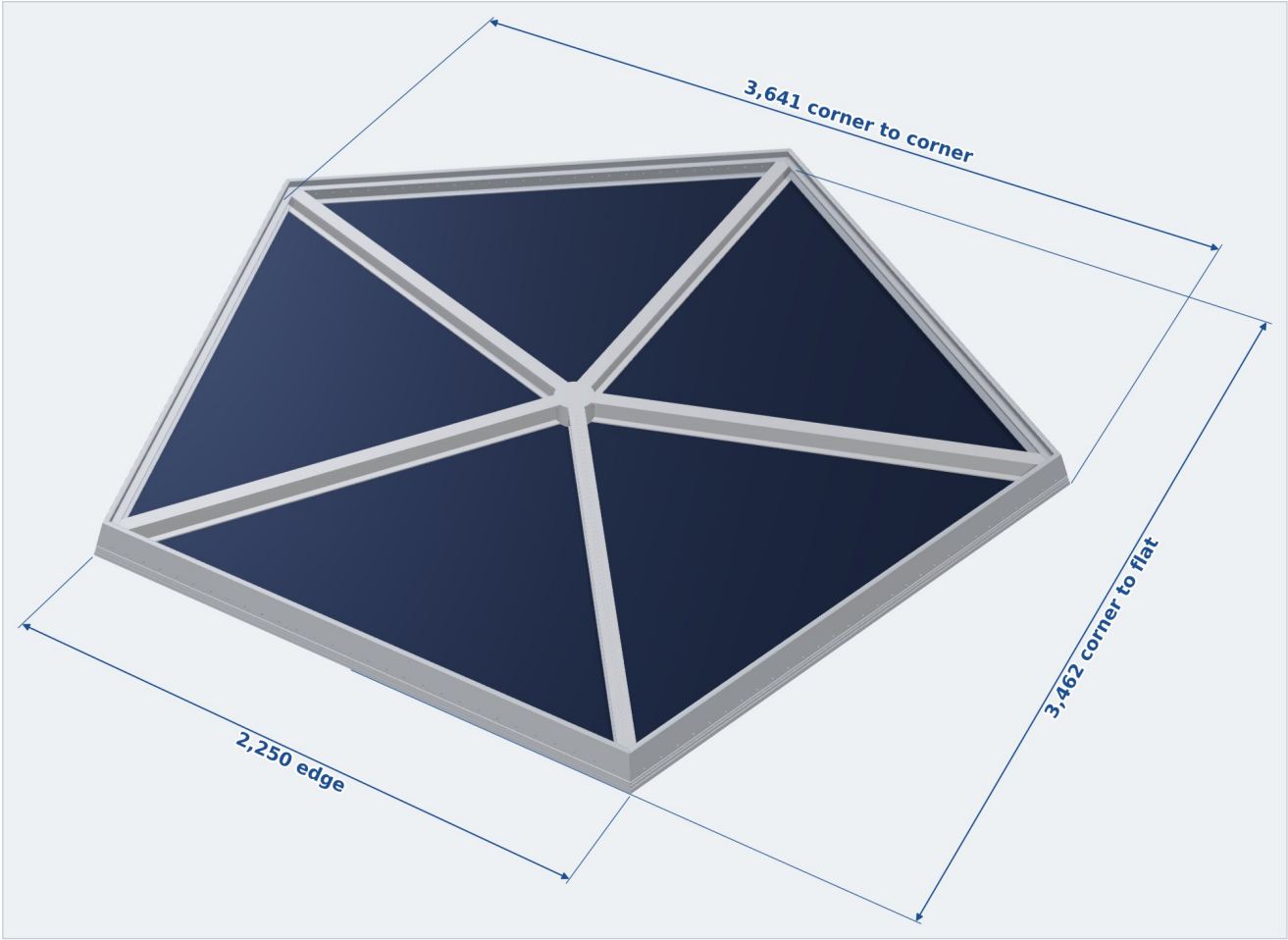
Not tested on its own; its seams are covered by T1.

Why it is designed like this: see the page “The pentagon panel” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

pent panel cabin side

What you are looking at

The pentagon from the cabin side, tilted, showing the ribs standing up from the plate. The plate is shaded (dark in the render) to show it is solid, not glass.

Dimensions shown (mm)

- 3,641 corner to corner
- 3,462 corner to flat
- 2,250 edge

Part of

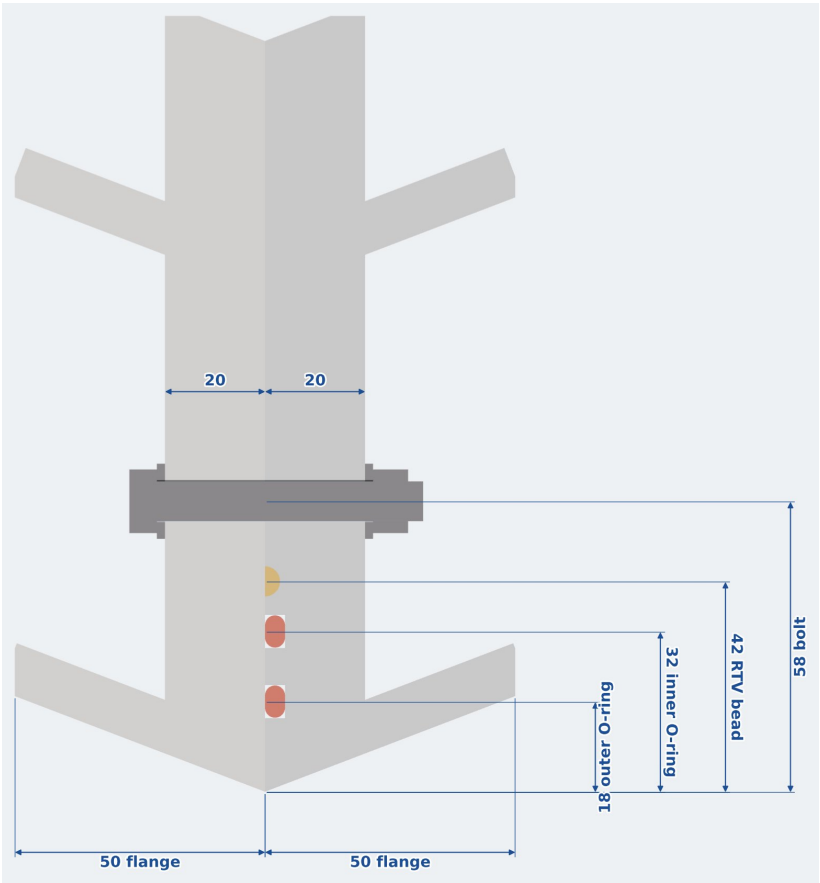
12 of these: one at the top (PENT-00), one at the bottom, the rest spread evenly between the hexagons.

Testing

Not tested on its own; its seams are covered by T1.

Why it is designed like this: see the page “The pentagon panel” at the start of this part.

The simple joint, in section



What it is

A cut through one seam at a bolt. Two frames sit face to face on the joint plane. Each frame has a 20 mm web, an outer flange 50 x 10 mm at the outside face and a seat lip 50 x 10 mm that holds the pressure pane. One 5/16 in bolt every 75 mm. Frame B has two O-ring grooves and the RTV bead; Frame A is flat.

Why it is designed like this

- Two flat faces bolted together is the simplest joint that still seals. Nothing slides or keys into anything, so it is easy to machine and to check.
- Three seals in a row: two O-rings and an RTV bead. One can fail without a leak. A small port between the O-rings lets each one be leak-checked on its own.
- All the grooves are in one frame (B), so the two frames never have to line up groove to groove. A pentagon is always B; between two hexagons the lower-numbered one is A.
- The joint face lies on the plane through the seam and the centre of the shell. Then the three joint faces at every corner meet on one line and the corner closes. It is why the flanges sit at an angle (20.9 degrees on a hexagon) and the section looks like a Y.
- Dropped from the older Rev C joint: tongue and groove, key boss, cable conduit, looped groove ends and guide pins.

Where it sits in the whole

This section runs the full length of all 90 seams.

How we test it

- T1 pulls short slices of it apart and checks the seal while it is loaded.
- The 550 mm joint article checks that it goes together and holds pressure.

Numbers to remember

- Up to 277 kN/m across the joint at 1 atm (upper bound)
- 20.7 kN per bolt, almost three times the 7.4 kN in TRUTH
- Seals at 18, 32 and 42 mm and the bolt at 58 mm from the outside tip

2D SKETCH

simple joint section

What you are looking at

The joint cut through a bolt, drawn by hand at about 2:1. Cabin at the top, vacuum at the bottom; the dashed line is the joint plane. Frame A (left) is flat; Frame B (right) has the grooves. The blue numbers on the right give where each seal and the bolt sit, measured along the joint plane from the outside tip.

Dimensions shown (mm)

- 20 and 20: the two webs
- 50 and 50: the outer flanges, from the joint plane
- 140 deep, square to the panel face
- 18 outer O-ring, 32 inner O-ring, 42 RTV bead, 58 bolt: along the joint plane from the outside tip

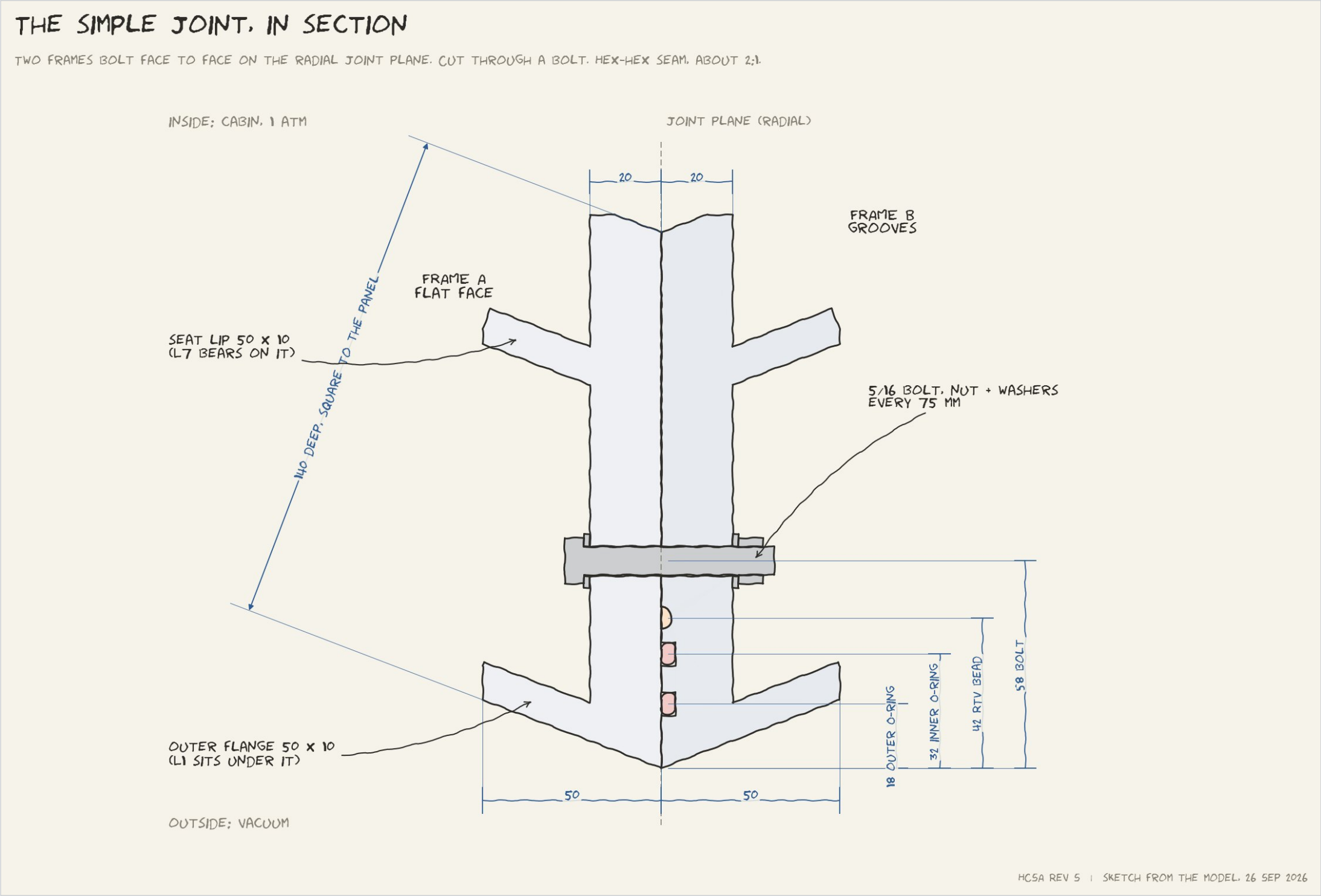
Part of

This section runs the full length of all 90 seams.

Testing

T1 seam slices and the 550 mm joint article.

Why it is designed like this: see the page “The simple joint, in section” at the start of this part.



2D RENDER

seam section

What you are looking at

The same cut, rendered. Red are the two O-rings, yellow is the RTV bead, dark grey is the bolt with its washers and nut. The Y shape comes from the flanges sitting at 20.9 degrees to the joint plane.

Dimensions shown (mm)

- 20
- 50 flange
- 18 outer O-ring
- 32 inner O-ring
- 42 RTV bead
- 58 bolt

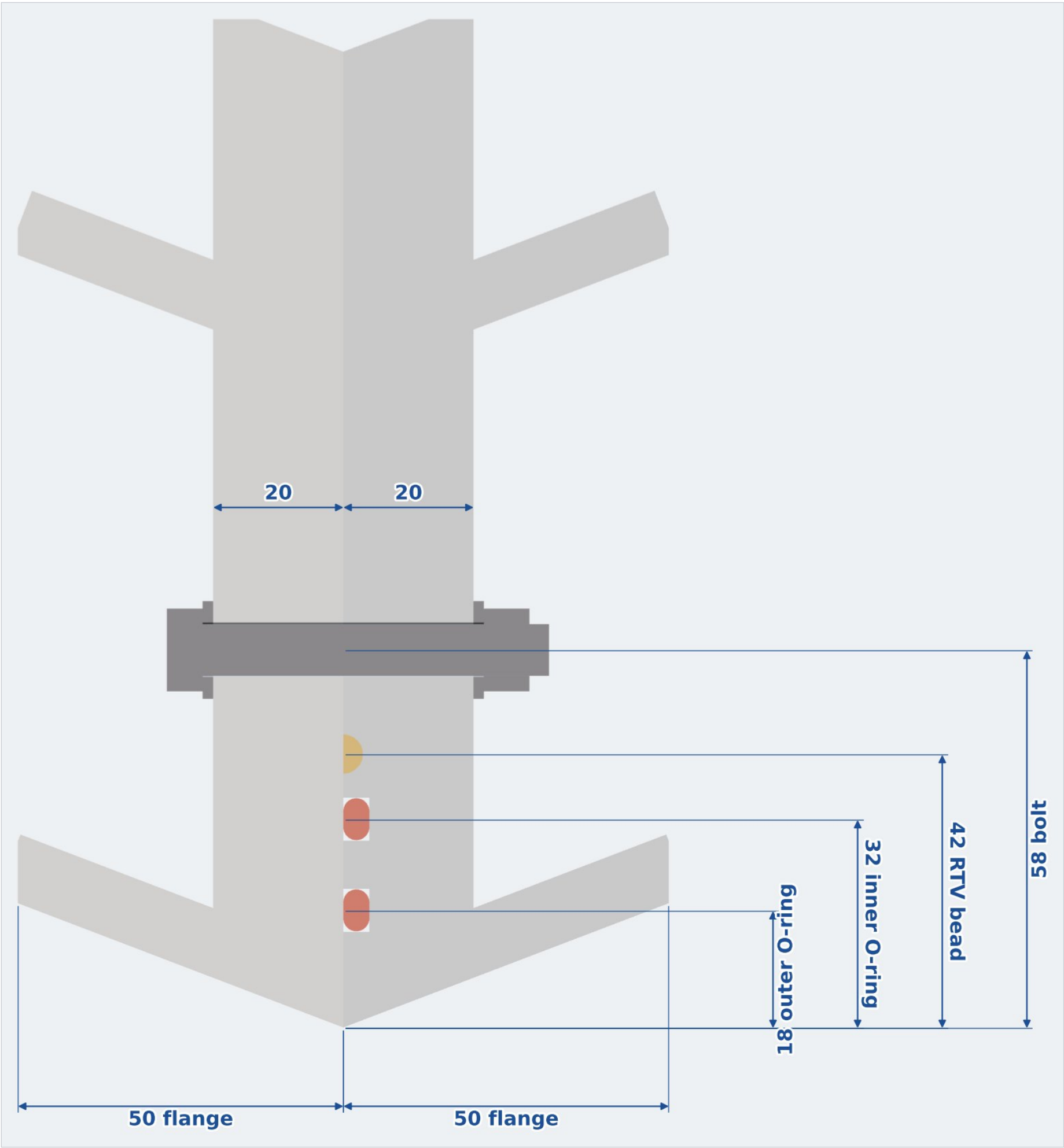
Part of

This section runs the full length of all 90 seams.

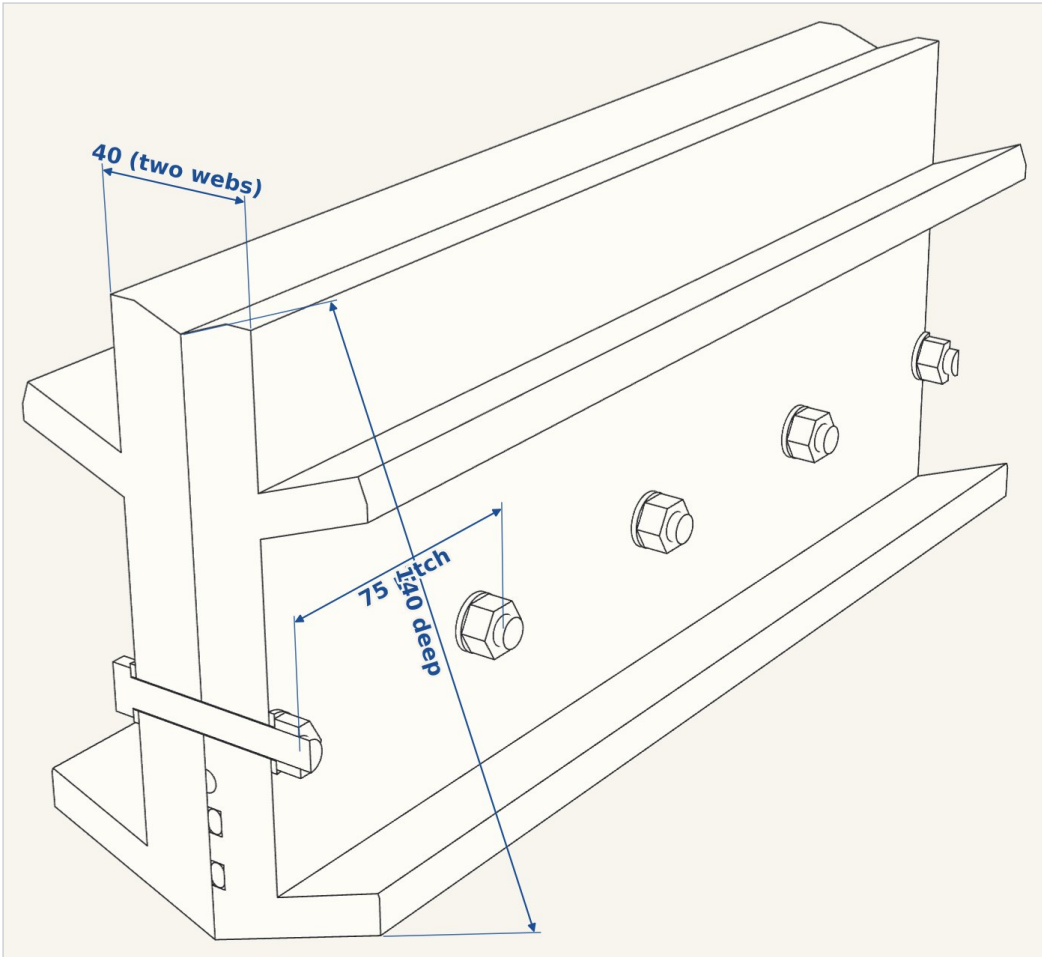
Testing

T1 seam slices and the 550 mm joint article.

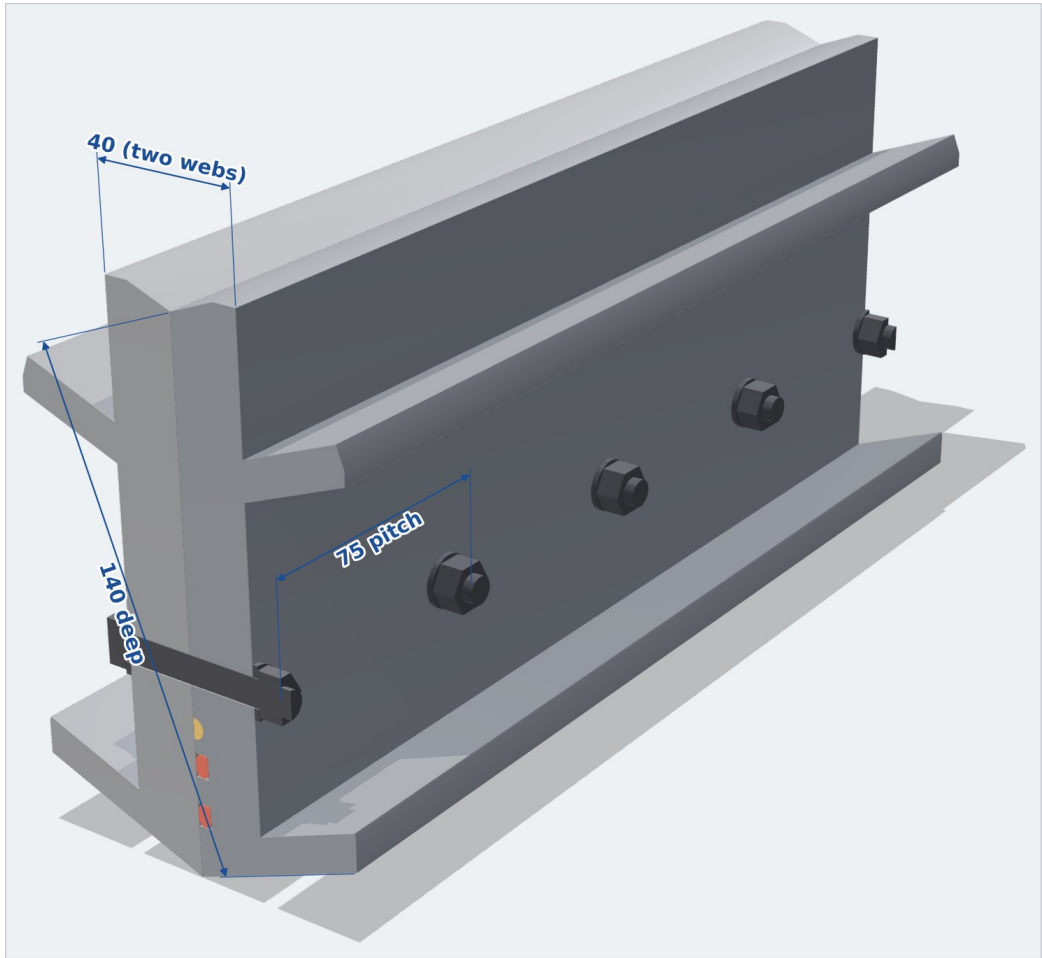
Why it is designed like this: see the page “The simple joint, in section” at the start of this part.



3D sketch



3D render



3D SKETCH AND 3D RENDER

seam closeup cut

What you are looking at

A short piece of seam cut through a bolt, so you see the section and the length together: webs face to face, the bolt row, and the O-rings and bead on the cut face.

Dimensions shown (mm)

- 75 pitch
- 140 deep
- 40 (two webs)

Part of

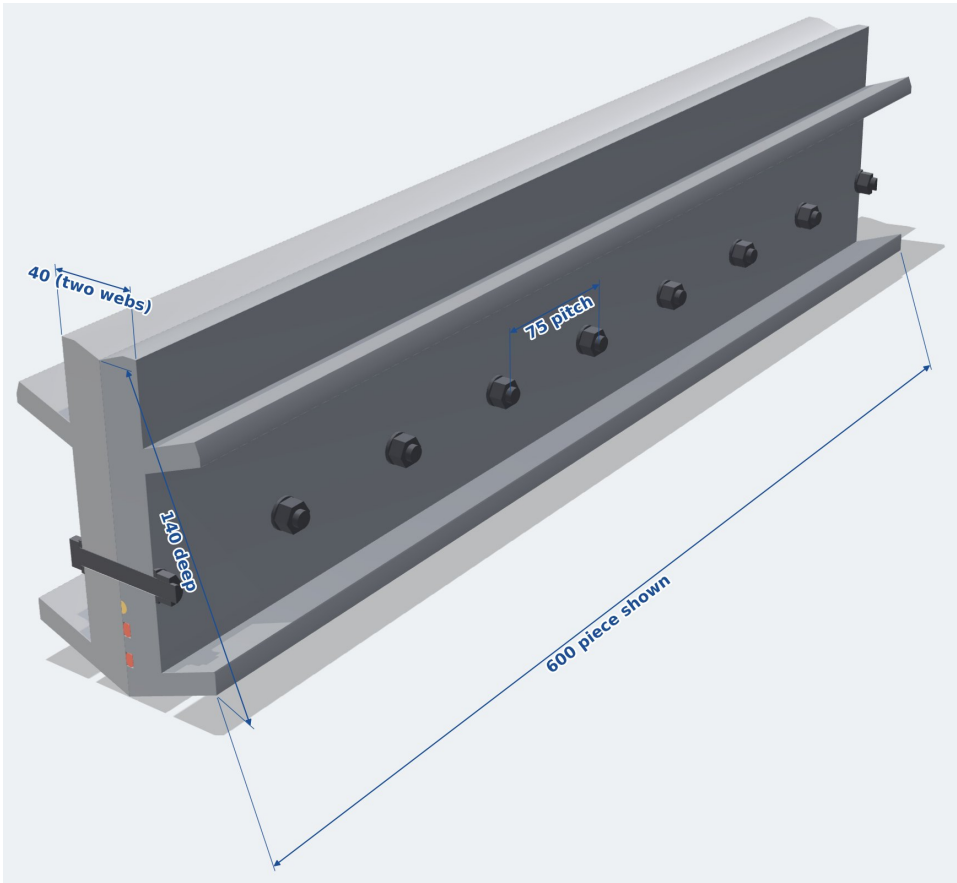
This section runs the full length of all 90 seams.

Testing

T1 seam slices and the 550 mm joint article.

Why it is designed like this: see the page “The simple joint, in section” at the start of this part.

One seam along its length



What it is

One full seam is 2,250 mm long with 29 bolts at 75 mm. The end bolts are 75 mm from each corner. The O-ring grooves run the whole length and stop 30 mm short of each corner.

Why it is designed like this

- The 75 mm bolt spacing is set in TRUTH.
- 29 bolts, not 30: with 30, the end nuts of two hexagon-pentagon seams run into each other at every pentagon corner. With 29 there is 75 mm to each corner and nothing touches. TRUTH says about 30, so this needs a decision.

Where it sits in the whole

90 of these, one along every edge of the shell.

How we test it

- The T1 slices have 1, 2 and 3 bolts. If the load and gap per bolt stay the same as the slice gets longer, the whole 29-bolt seam behaves like the slices.

Numbers to remember

- 620 kN across a whole 2,250 mm seam at 1 atm (upper bound)
- 29 bolts per seam, 2,610 in the shell

2D SKETCH

seam elevation

What you are looking at

One whole seam seen from the B side, cut just inside the B web so you look at the grooves. The dots are the 29 bolt holes; the end ones are 75 mm from each corner.

Dimensions shown (mm)

- 2,250 edge, corner to corner
- 75 pitch between bolts
- 75 from the end bolt to the corner

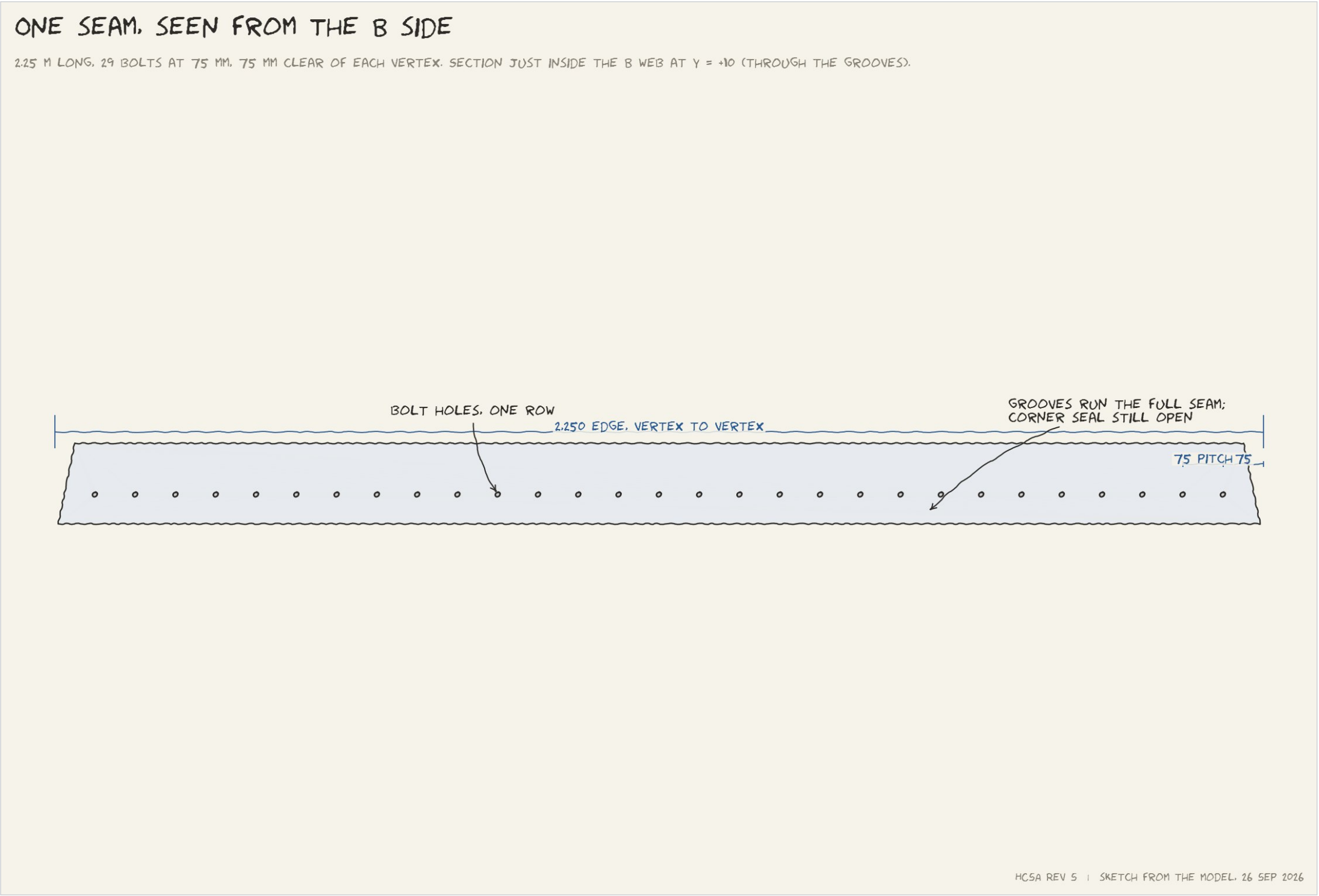
Part of

90 of these, one along every edge of the shell.

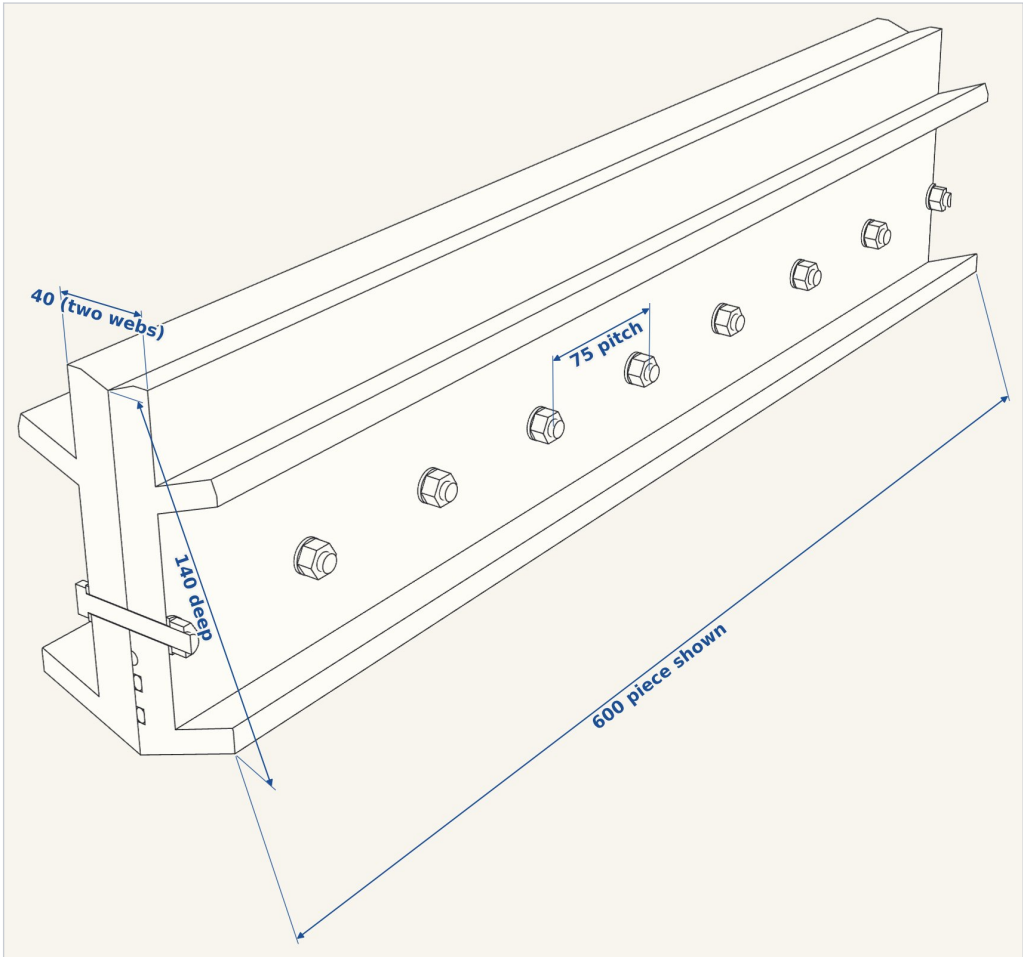
Testing

T1 size series with 1, 2 and 3 bolts.

Why it is designed like this: see the page “One seam along its length” at the start of this part.



3D sketch



3D SKETCH AND 3D RENDER

seam closeup

What you are looking at

A 600 mm piece of a seam. The nuts line up along one side at 75 mm; the outer flanges are at the bottom and the seat lips at the top.

Dimensions shown (mm)

- 600 piece shown
- 75 pitch
- 140 deep
- 40 (two webs)

Part of

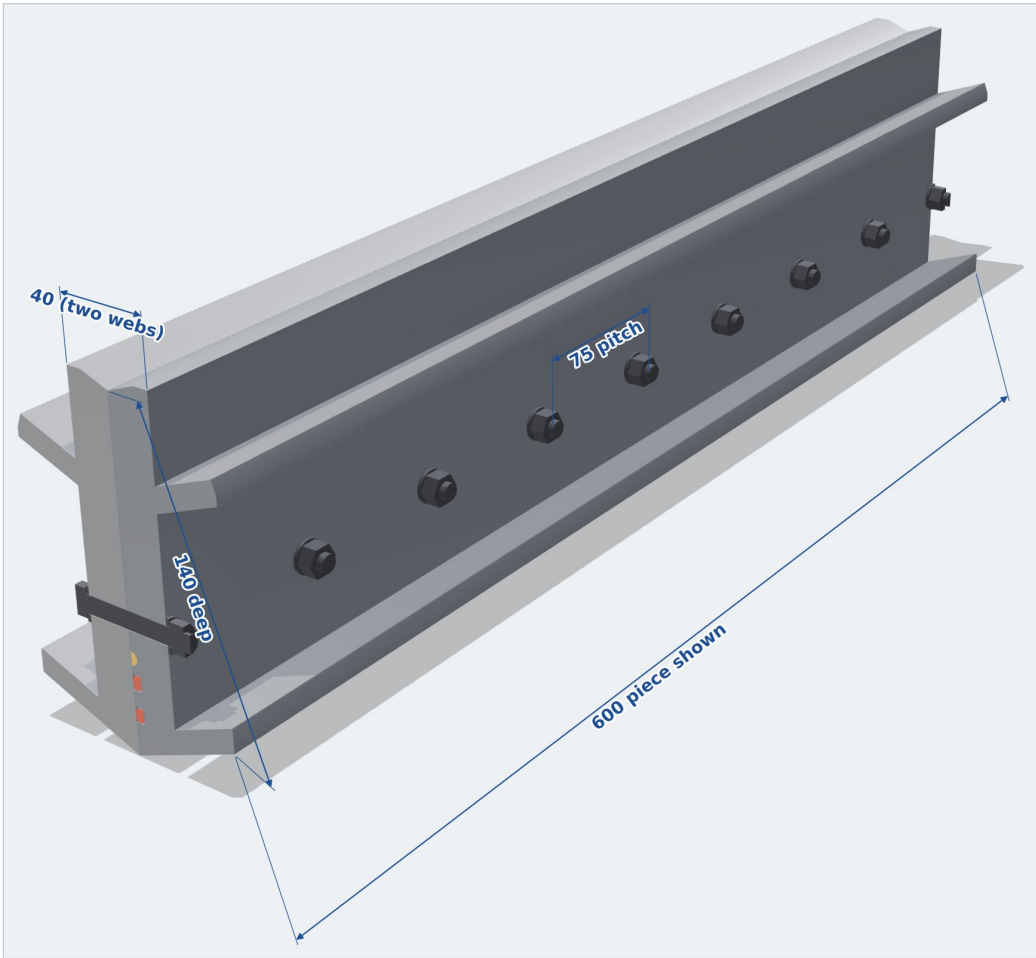
90 of these, one along every edge of the shell.

Testing

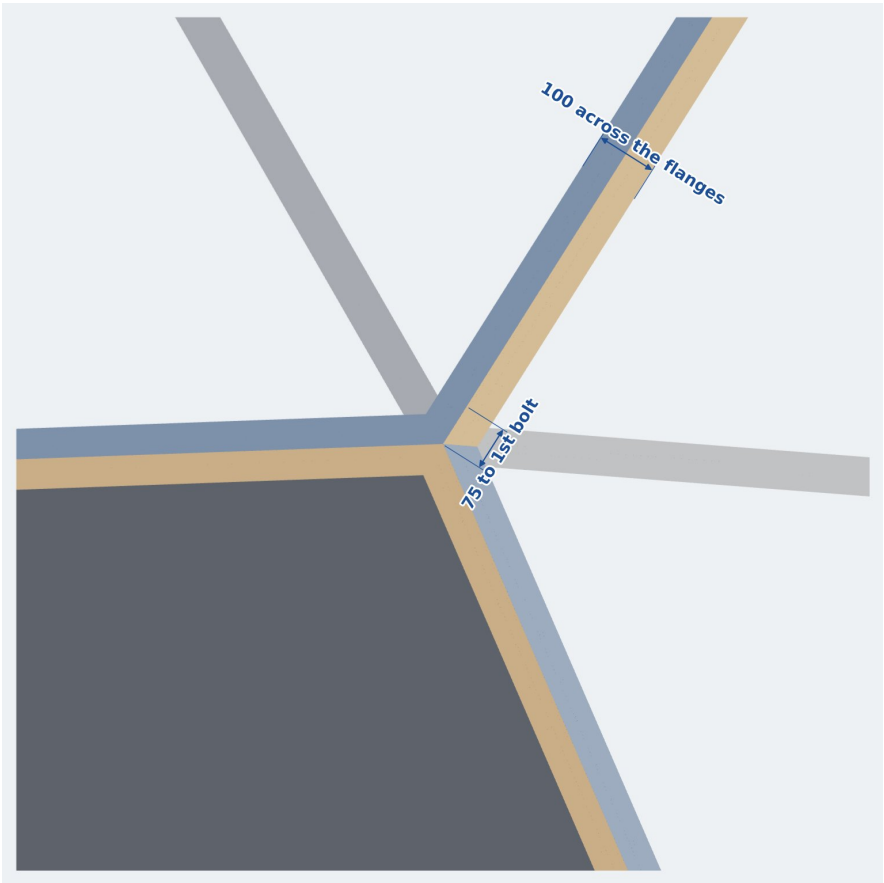
T1 size series with 1, 2 and 3 bolts.

Why it is designed like this: see the page “One seam along its length” at the start of this part.

3D render



The corner, where three panels meet



What it is

The corner where HEX-09, HEX-14 and PENT-10 meet, cut out as a 1.4 m square around the corner point. The panes are left out so the frames show.

Why it is designed like this

- Because every joint face lies on a plane through the centre of the shell, the three seams meet on one line through the corner and close with no gap.
- The ribs stop short of the corner and have pockets cut round the end bolts, so nothing clashes.
- The corner seal is still open (item O10): the O-rings of the three seams stop 30 mm from the corner, and nothing yet seals the small triangle where they meet.

Where it sits in the whole

There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

How we test it

- A corner test (T3) comes after T1 and T2 agree with the model, and after the corner seal is designed.
- If the in-plane load runs along the seams, each corner takes about 621 kN, which could make the corners the most loaded places in the shell.

Numbers to remember

- 100 mm across the two outer flanges of a seam
- 75 mm from the corner to the first bolt

2D SKETCH

corner node

What you are looking at

The corner seen from outside, looking straight down the line from the shell centre through the corner. The three seams and the hex ribs meet at the point in the middle.

Dimensions shown (mm)

- 100 across the two outer flanges of a seam
- 75 from the corner to the first bolt

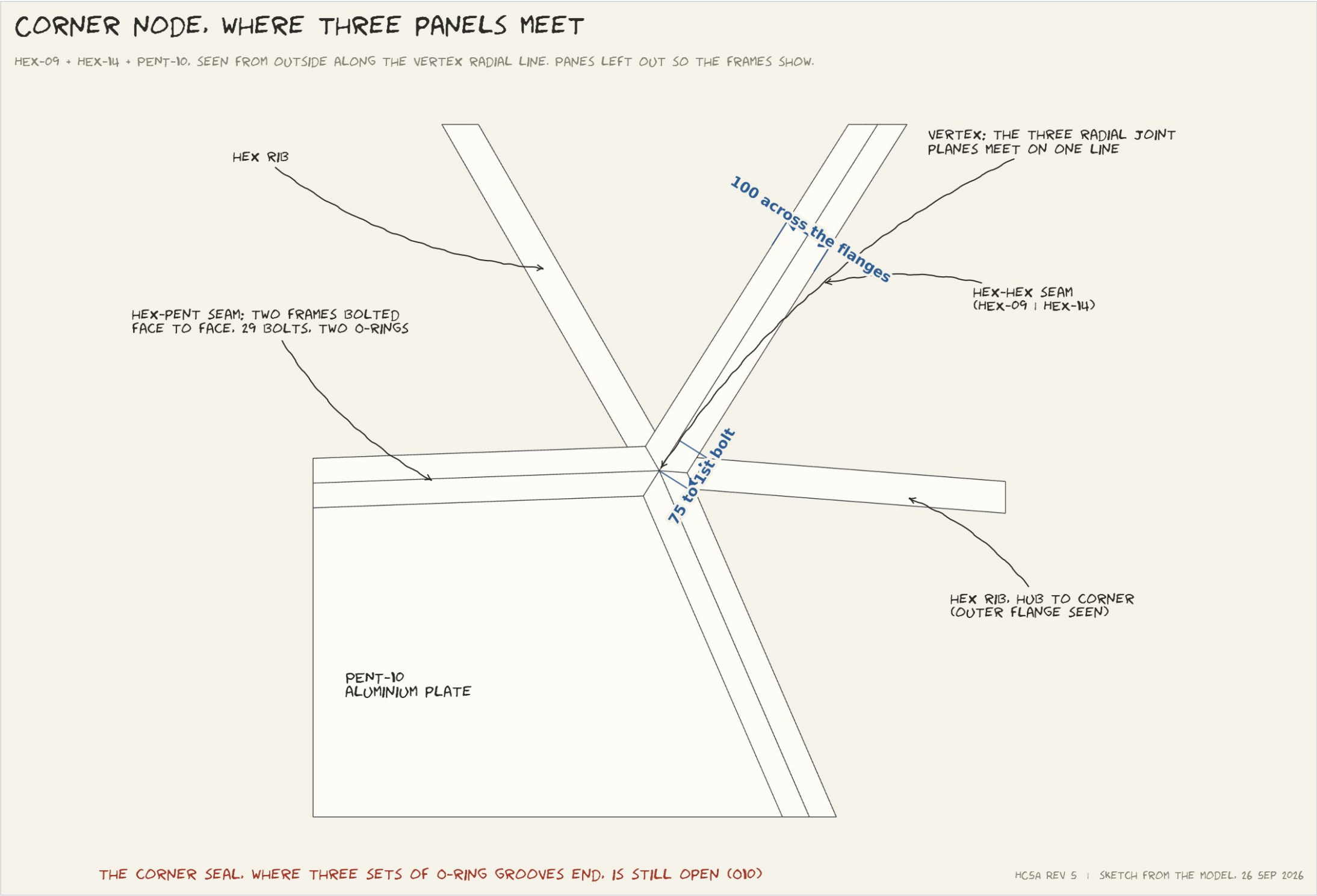
Part of

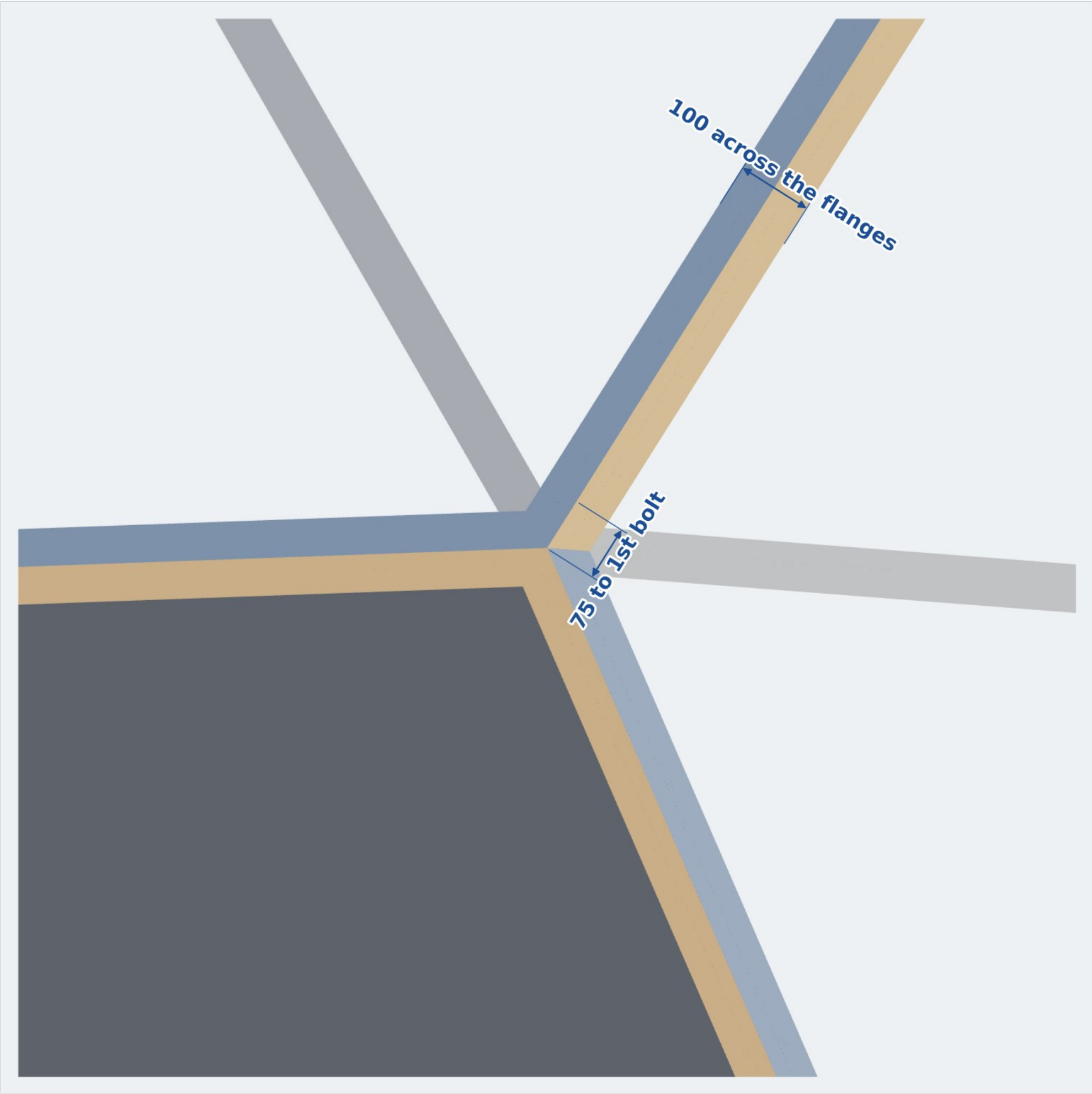
There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

Testing

A corner test (T3) later, once the corner seal is designed.

Why it is designed like this: see the page “The corner, where three panels meet” at the start of this part.





2D RENDER

corner node plan outside

What you are looking at

The same view, coloured: A frames blue, B frames tan, ribs grey, the pentagon plate dark. Each seam is one blue frame against one tan frame, and the pentagon's frames are tan on both of its seams.

Dimensions shown (mm)

- 100 across the flanges
- 75 to 1st bolt

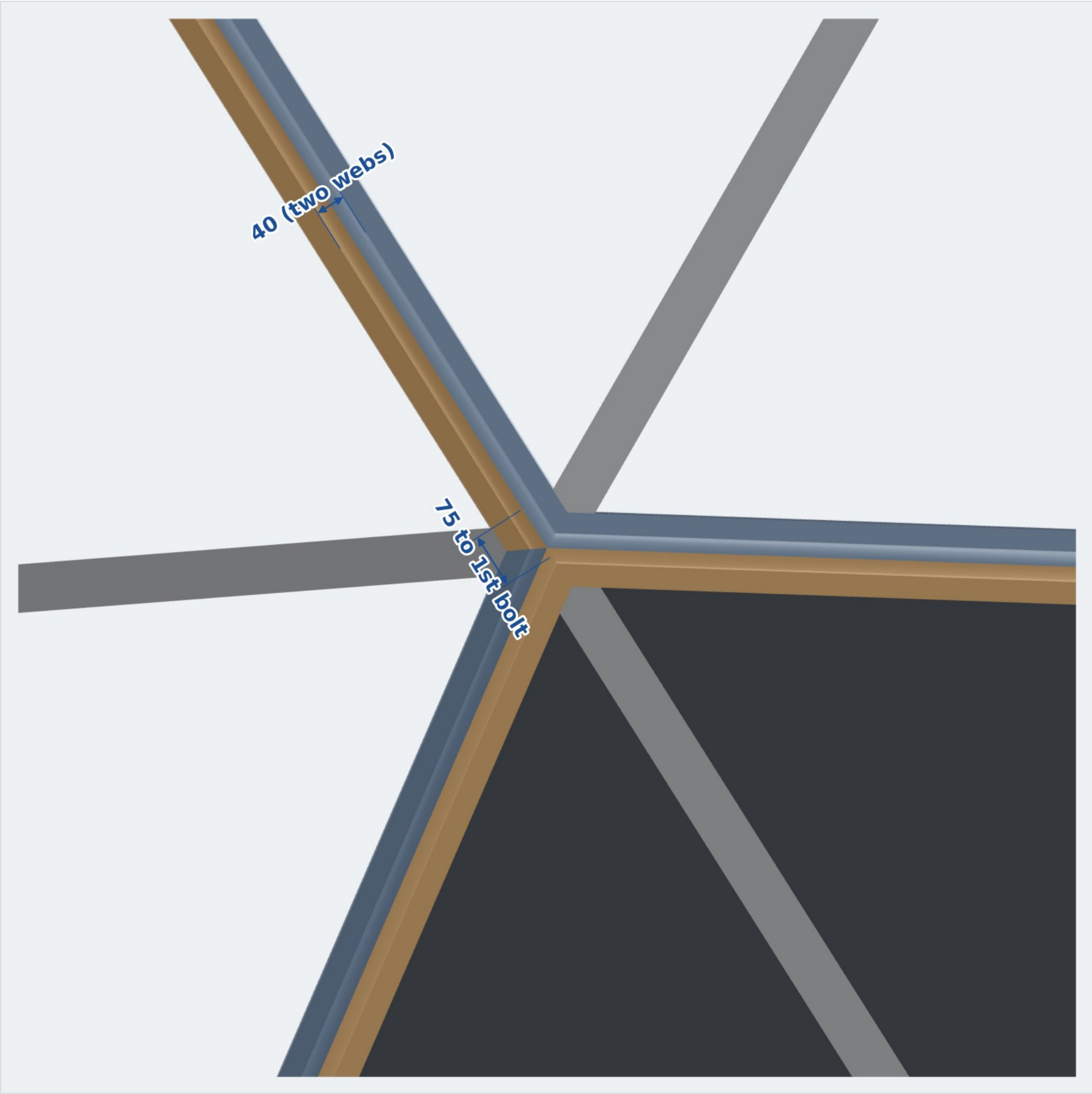
Part of

There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

Testing

A corner test (T3) later, once the corner seal is designed.

Why it is designed like this: see the page “The corner, where three panels meet” at the start of this part.



2D RENDER

corner node plan inside

What you are looking at

The corner from the cabin side. Here you see the inner ends of the webs, the seat lips and the three ribs running into the corner.

Dimensions shown (mm)

- 40 (two webs)
- 75 to 1st bolt

Part of

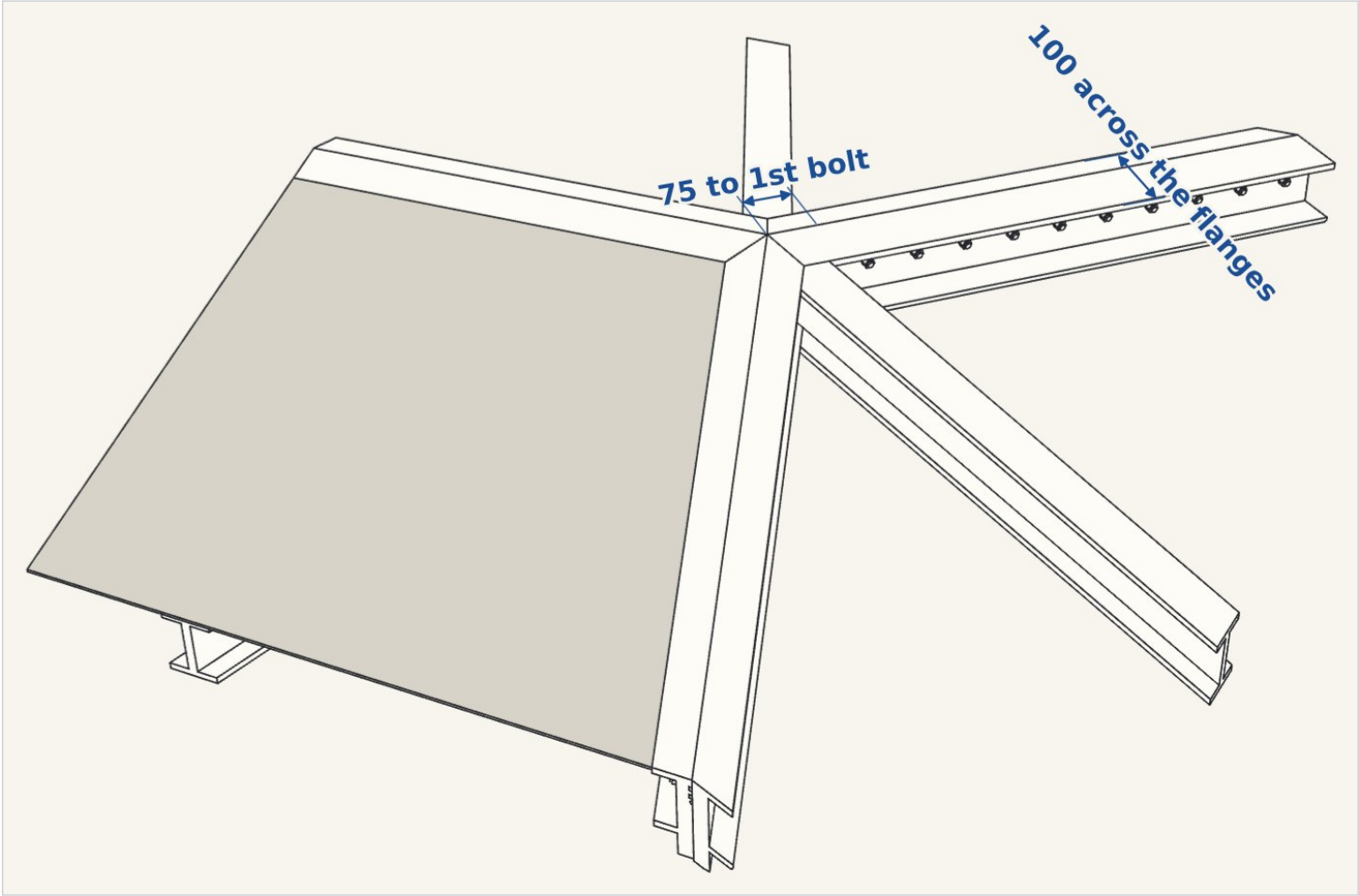
There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

Testing

A corner test (T3) later, once the corner seal is designed.

Why it is designed like this: see the page “The corner, where three panels meet” at the start of this part.

3D sketch



3D SKETCH AND 3D RENDER

corner node outside

What you are looking at

The corner from outside, tilted. The three seams fan out from the corner; the pentagon plate fills one side.

Dimensions shown (mm)

- 100 across the flanges
- 75 to 1st bolt

Part of

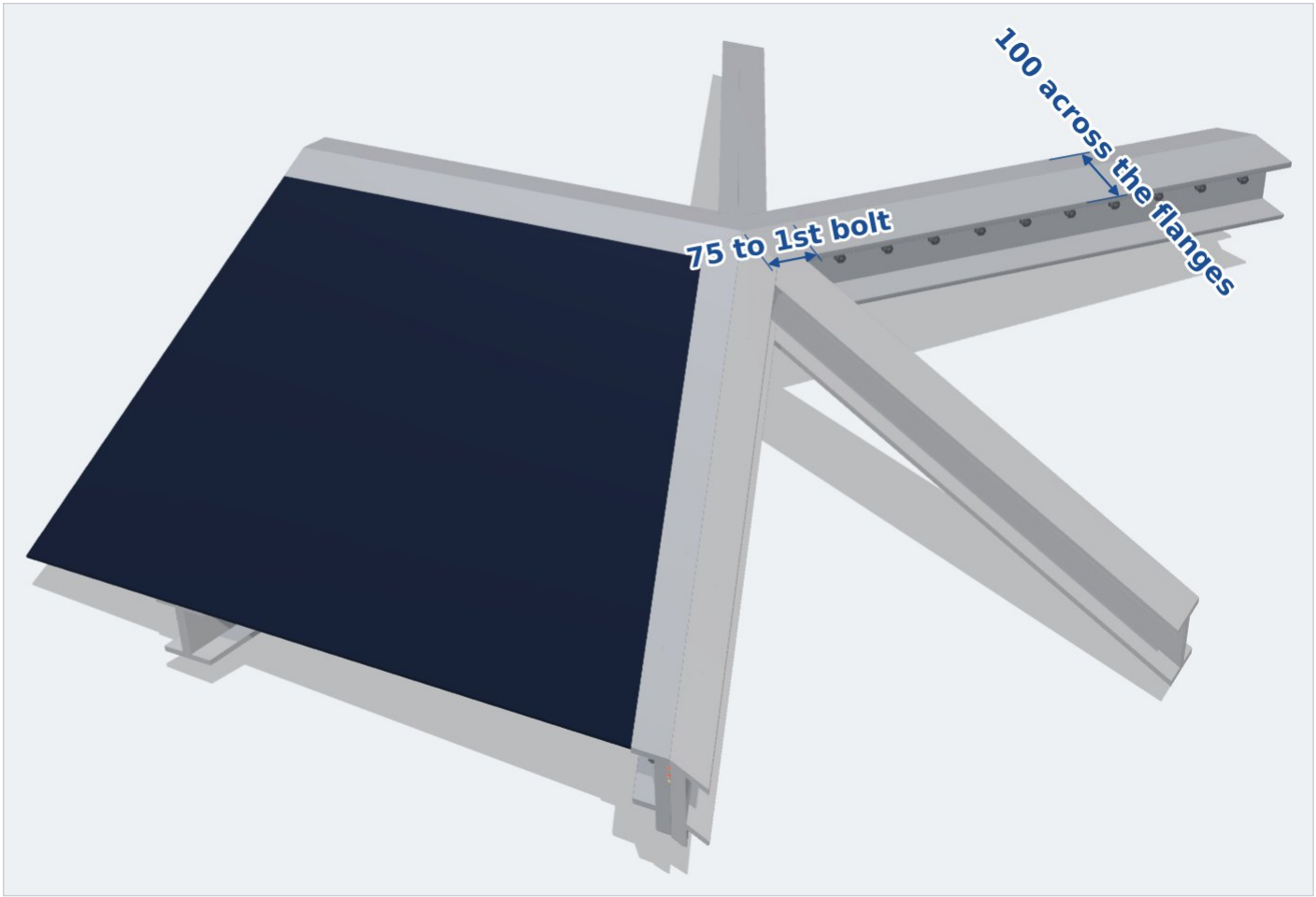
There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

Testing

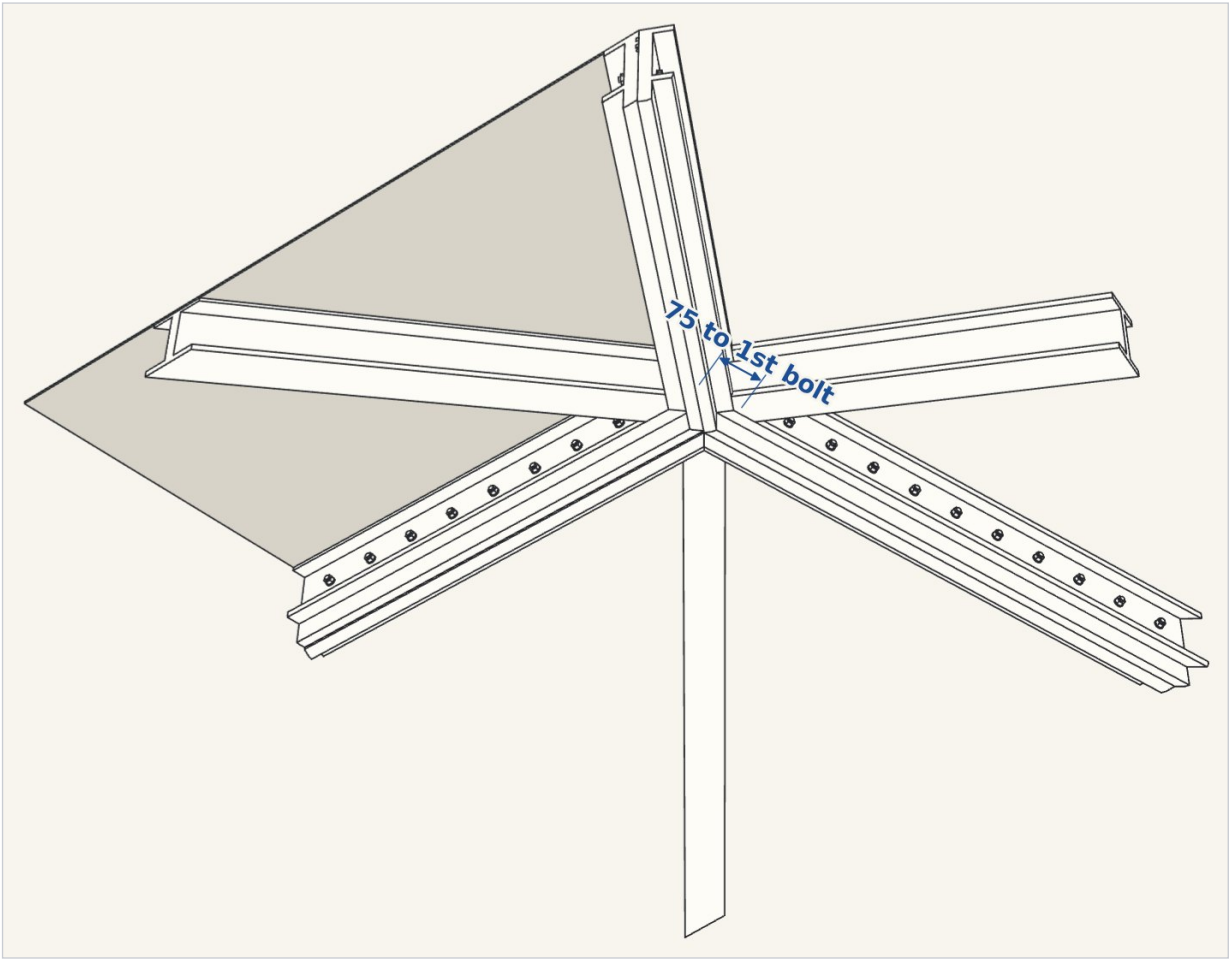
A corner test (T3) later, once the corner seal is designed.

Why it is designed like this: see the page “The corner, where three panels meet” at the start of this part.

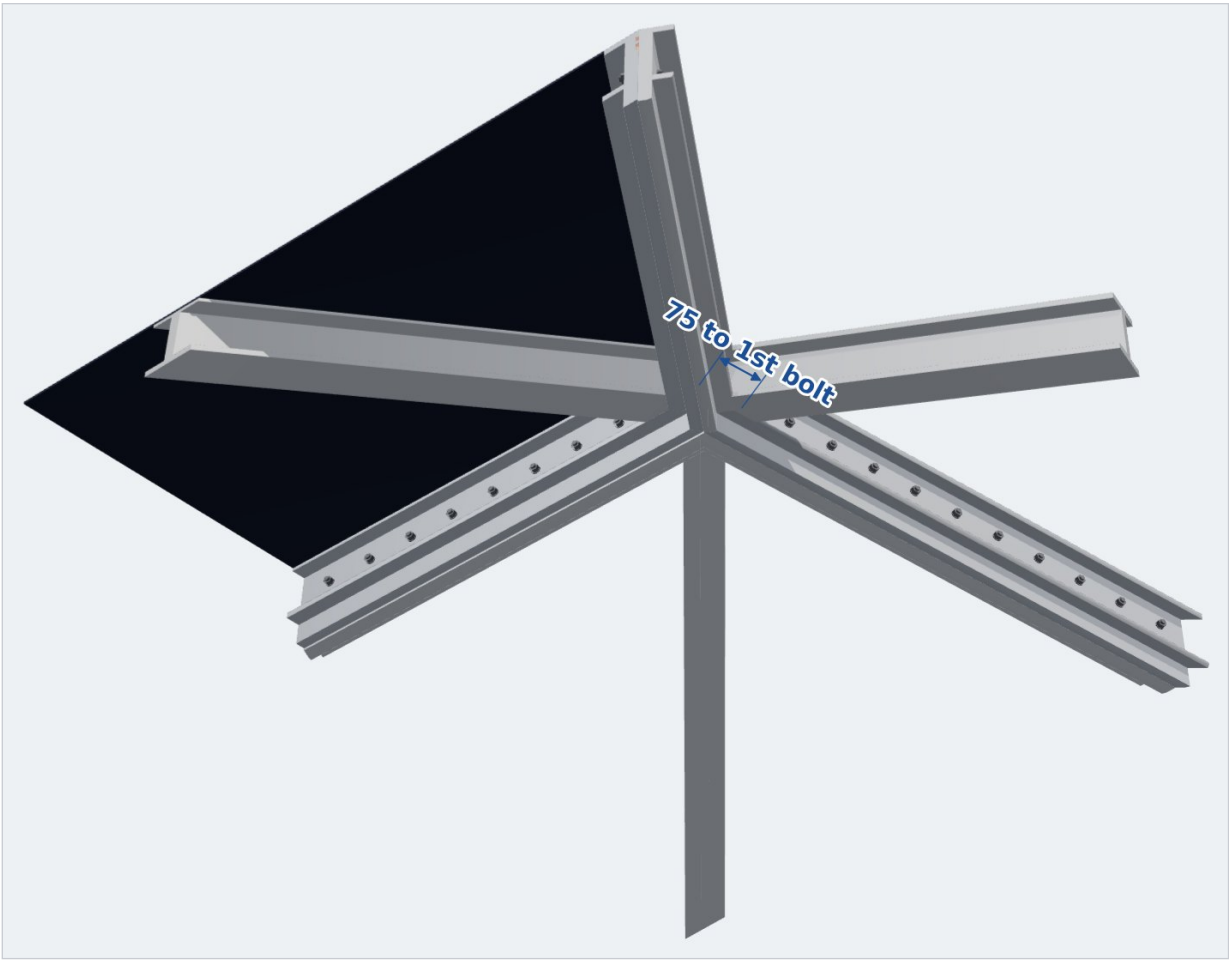
3D render



3D sketch



3D render



3D SKETCH AND 3D RENDER

corner node inside

What you are looking at

The corner from inside. The bolt rows run along the three seams toward the corner; the end bolts stop 75 mm short of it.

Dimensions shown (mm)

- 75 to 1st bolt

Part of

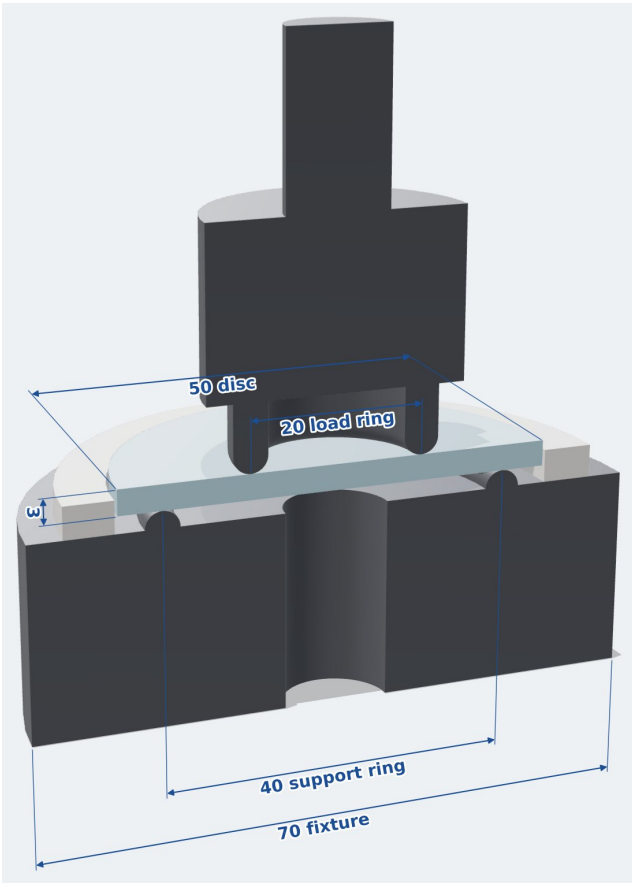
There are 60 corners in the shell, all like this one (two hexagons and a pentagon).

Testing

A corner test (T3) later, once the corner seal is designed.

Why it is designed like this: see the page “The corner, where three panels meet” at the start of this part.

T0: glass coupons, ring-on-ring



What it is

A 50 mm glass disc, 3 mm thick, sits on a 40 mm support ring and is pressed from above by a 20 mm load ring (ASTM C1499). The fixture is 70 mm across.

Why it is designed like this

- The pane stress (about 53 MPa at 1 atm) is the number we trust least. Glass breaks at its worst flaw, so its strength scatters and drops as the piece gets bigger.
- Ring-on-ring loads the middle of the disc evenly in every direction, like a pane bending under pressure, and away from the edges, so the edge finish does not decide the result.
- 30 or more discs per glass give a Weibull strength (ASTM C1239) that can be scaled up to a 2 square metre pane. For example, discs at 150 MPa with a Weibull modulus of 7 mean about 42 MPa for a full pane, before any safety factor.
- One set at four loading speeds measures slow crack growth (ASTM C1368): glass under steady load gets weaker over time.

Where it sits in the whole

It tests the material of the glass panes, not a part of the structure.

How we test it

- Run in a small load frame (5 kN) in October.
- 0.9 kN puts 50 MPa in the middle of a disc; loads run up to about 1 to 4 kN.

Numbers to remember

- 50 mm disc, 3 mm thick
- 40 mm support ring, 20 mm load ring
- 30 or more discs per candidate glass

2D SKETCH

T0 section

What you are looking at

The ring-on-ring fixture cut in half. The disc sits on the support ring; the load ring presses down in the middle with force F.

Dimensions shown (mm)

- 50 disc
- 40 support ring
- 20 load ring
- 3 disc thickness
- 70 fixture

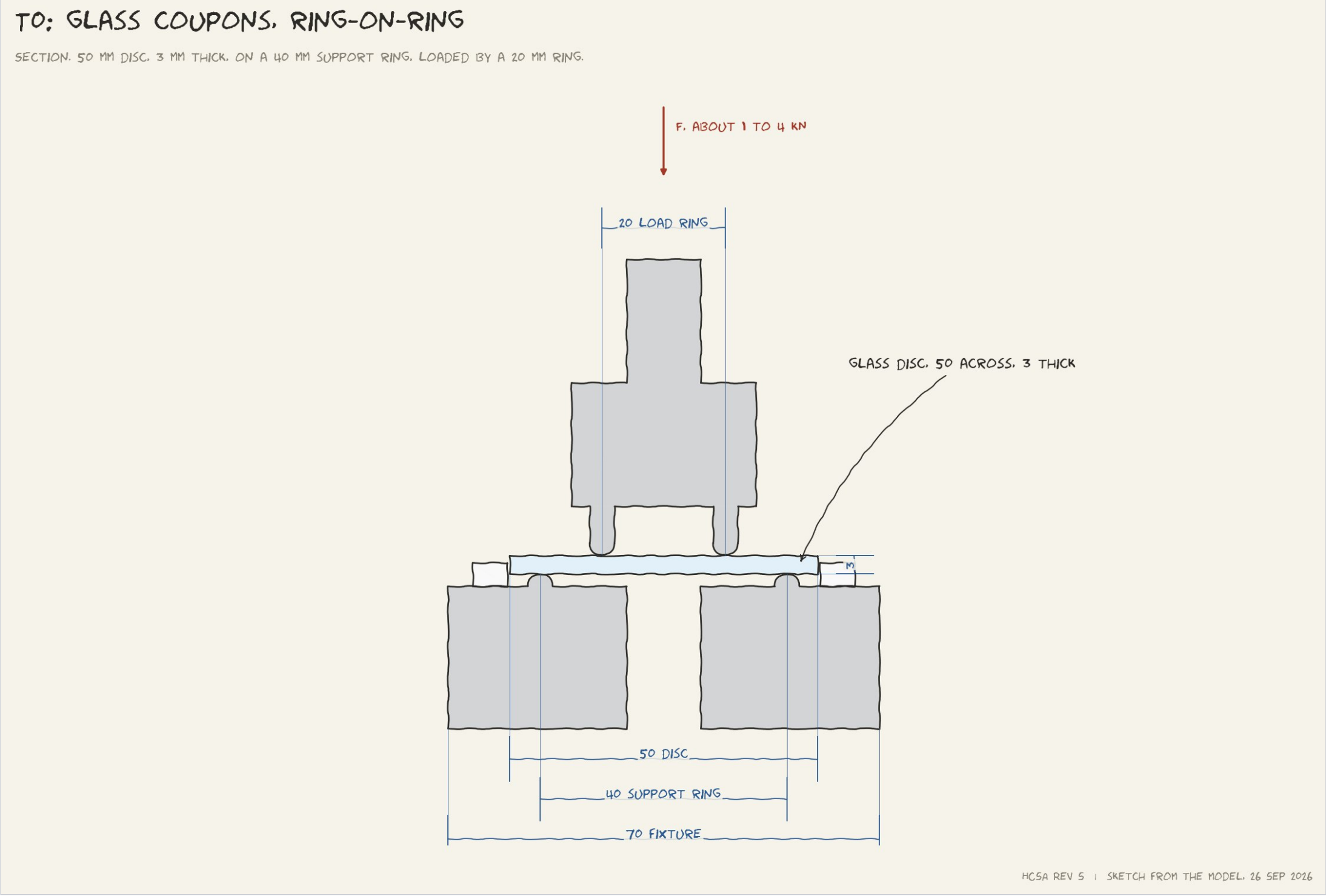
Part of

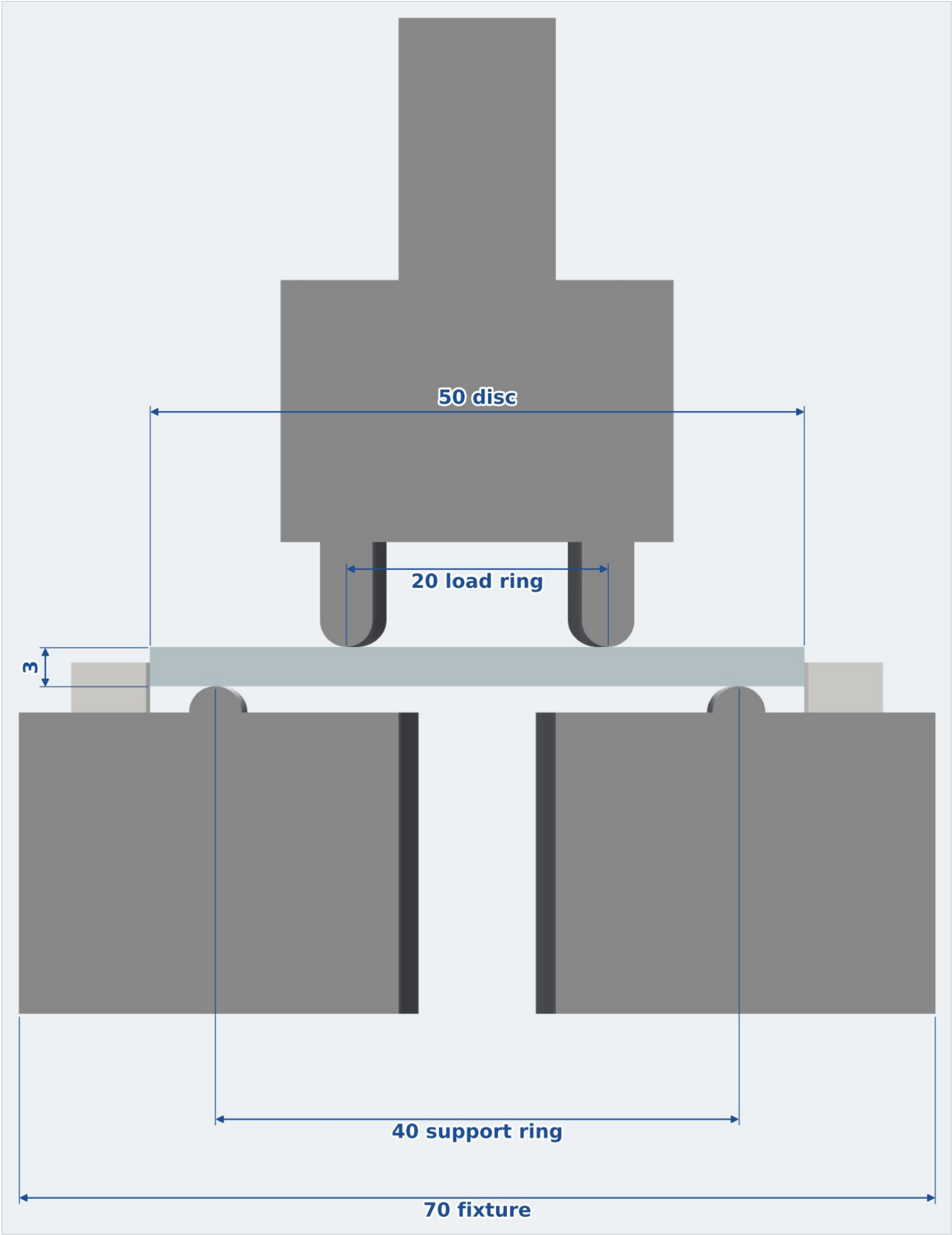
It tests the material of the glass panes, not a part of the structure.

Testing

This is the test: ring-on-ring to ASTM C1499, 30 or more discs per glass.

Why it is designed like this: see the page “T0: glass coupons, ring-on-ring” at the start of this part.





2D RENDER

T0 section

What you are looking at

The same cut, rendered. The light blue strip is the glass disc; you can see the rounded contact of each ring.

Dimensions shown (mm)

- 50 disc
- 40 support ring
- 20 load ring
- 3
- 70 fixture

Part of

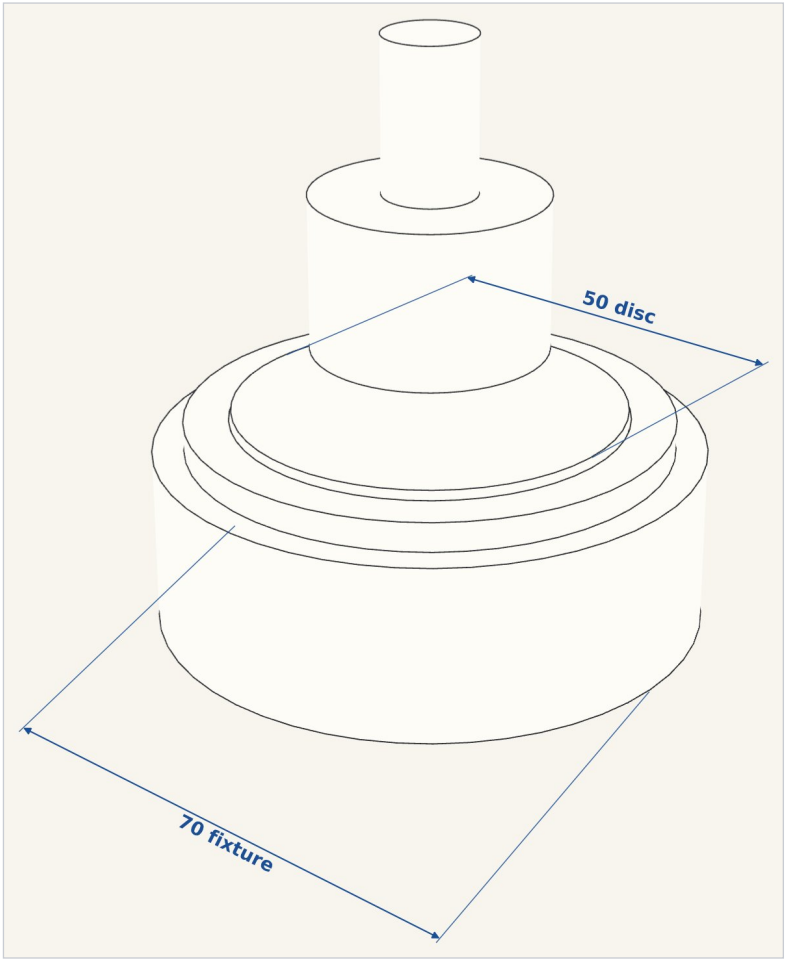
It tests the material of the glass panes, not a part of the structure.

Testing

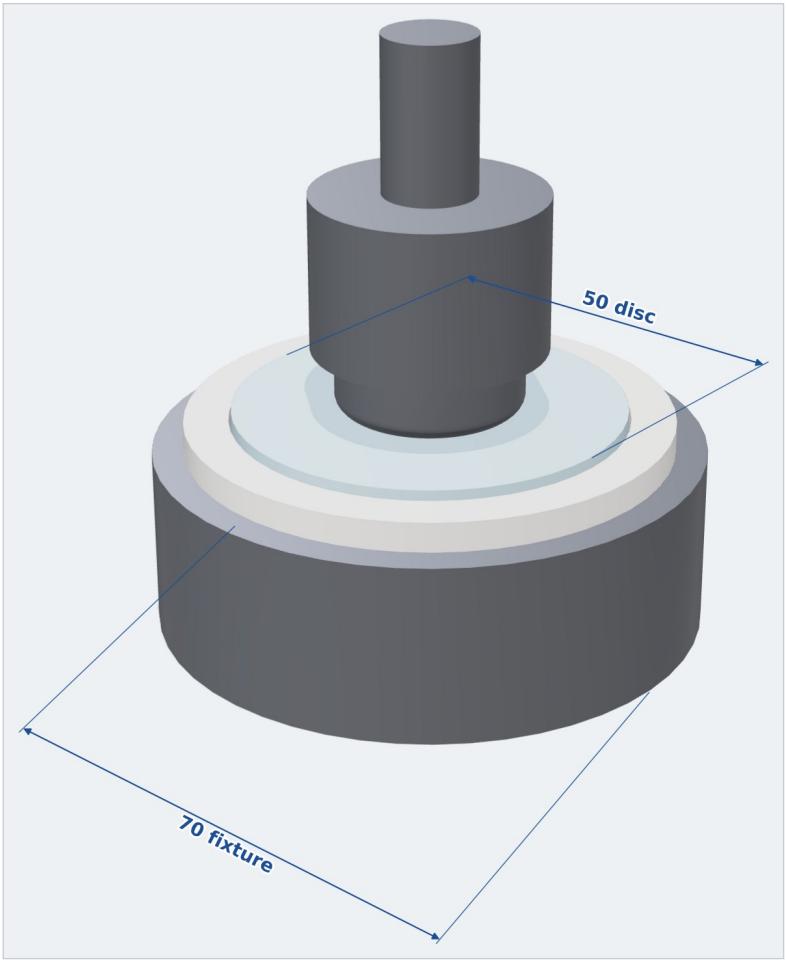
This is the test: ring-on-ring to ASTM C1499, 30 or more discs per glass.

Why it is designed like this: see the page “T0: glass coupons, ring-on-ring” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

T0 ring on ring

What you are looking at

The fixture assembled. The disc is the pale ring you can see between the support body and the load ring.

Dimensions shown (mm)

- 50 disc
- 70 fixture

Part of

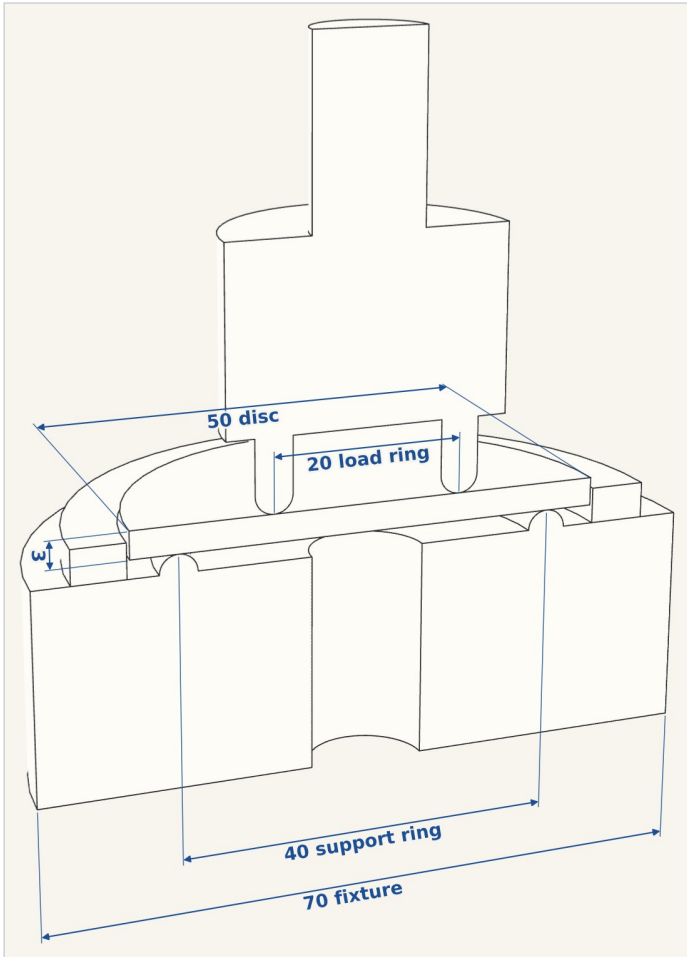
It tests the material of the glass panes, not a part of the structure.

Testing

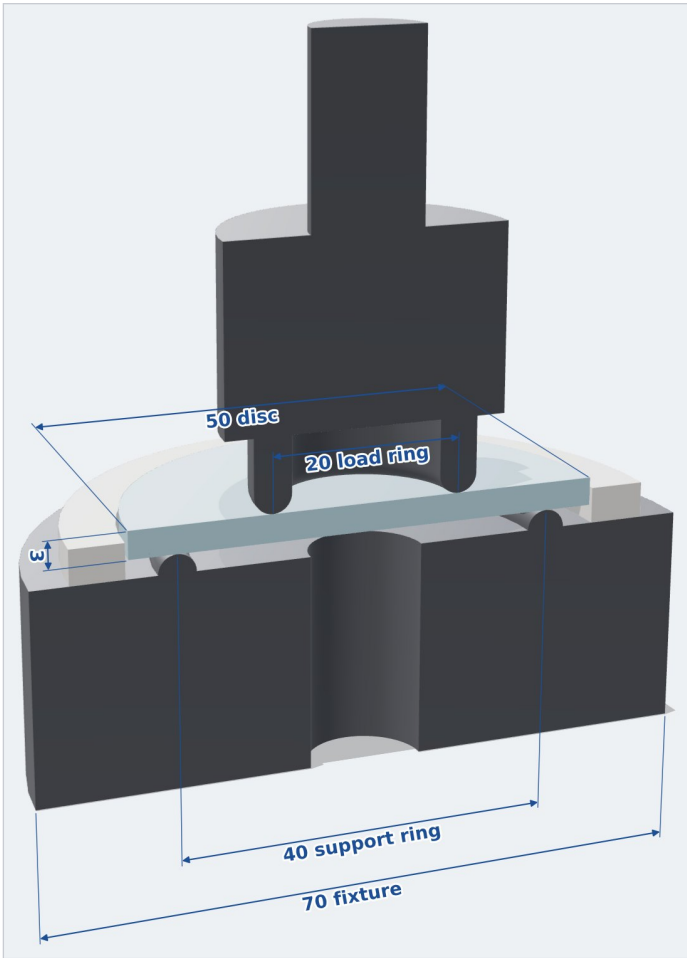
This is the test: ring-on-ring to ASTM C1499, 30 or more discs per glass.

Why it is designed like this: see the page “T0: glass coupons, ring-on-ring” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

T0 ring on ring cut

What you are looking at

The fixture cut in half, so the two rings and the disc show.
The load ring is half the diameter of the support ring.

Dimensions shown (mm)

- 50 disc
- 40 support ring
- 20 load ring
- 3
- 70 fixture

Part of

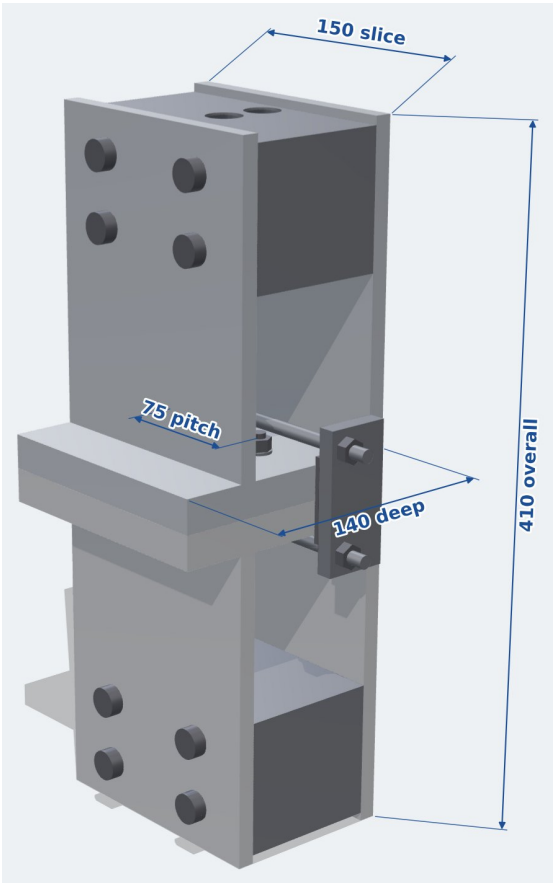
It tests the material of the glass panes, not a part of the structure.

Testing

This is the test: ring-on-ring to ASTM C1499, 30 or more discs per glass.

Why it is designed like this: see the page “T0: glass coupons, ring-on-ring” at the start of this part.

T1: seam slice pull test



What it is

A full-size slice of one seam: 75, 150 or 225 mm long (1, 2 or 3 bolts). The outer flange and the seat lip run out to 200 mm as grip legs, a steel pull block bolts between them, and end dams close the O-ring grooves at the cut ends. 410 mm overall.

Why it is designed like this

- Bolts, O-rings and grooves do not scale down, so the seam is tested at full size.
- A whole seam at 1 atm pulls about 620 kN, far beyond a lab frame. A 150 mm slice pulls 41.5 kN, which a 100 kN frame can do.
- Three lengths, after Tengxiao Li's advice: test a few sizes and see when the response per bolt stops changing.
- The pull goes in through the flange and lip, the same parts that carry the panel's load in the shell.
- The grooves run out of the cut ends, so rubber-faced end plates clamped by M8 rods close them and the space between the O-rings can hold pressure.

Where it sits in the whole

A short piece of the seam, cut from anywhere along its length.

How we test it

- Run 1: pull through both legs, midway between flange and lip.
- Run 2: remove the lip bolts and pull through the flange only, so the joint pries open.
- Measure load and stroke, the gap at the outside and cabin faces, the strain in one bolt, and the pressure between the O-rings.
- Pass: a straight-line response to the 1 atm load that returns to zero, no leak at 1 atm, inner O-ring gap under 0.33 mm (provisional), no yield below proof, the model within 10 percent.

Numbers to remember

- 150 mm slice: 41.5 kN at 1 atm, run to 2 atm (83 kN)
- 75 mm run to 2.4 atm, 225 mm to 1.5 atm

2D SKETCH

T1 slice section

What you are looking at

The 150 mm slice cut through a bolt. The joint is in the middle; the legs run out left and right to the steel pull blocks. The red arrows are the two pull positions: run 1 between flange and lip, run 2 on the flange only.

Dimensions shown (mm)

- 400 between the pull block ends
- 410 overall
- 140 deep

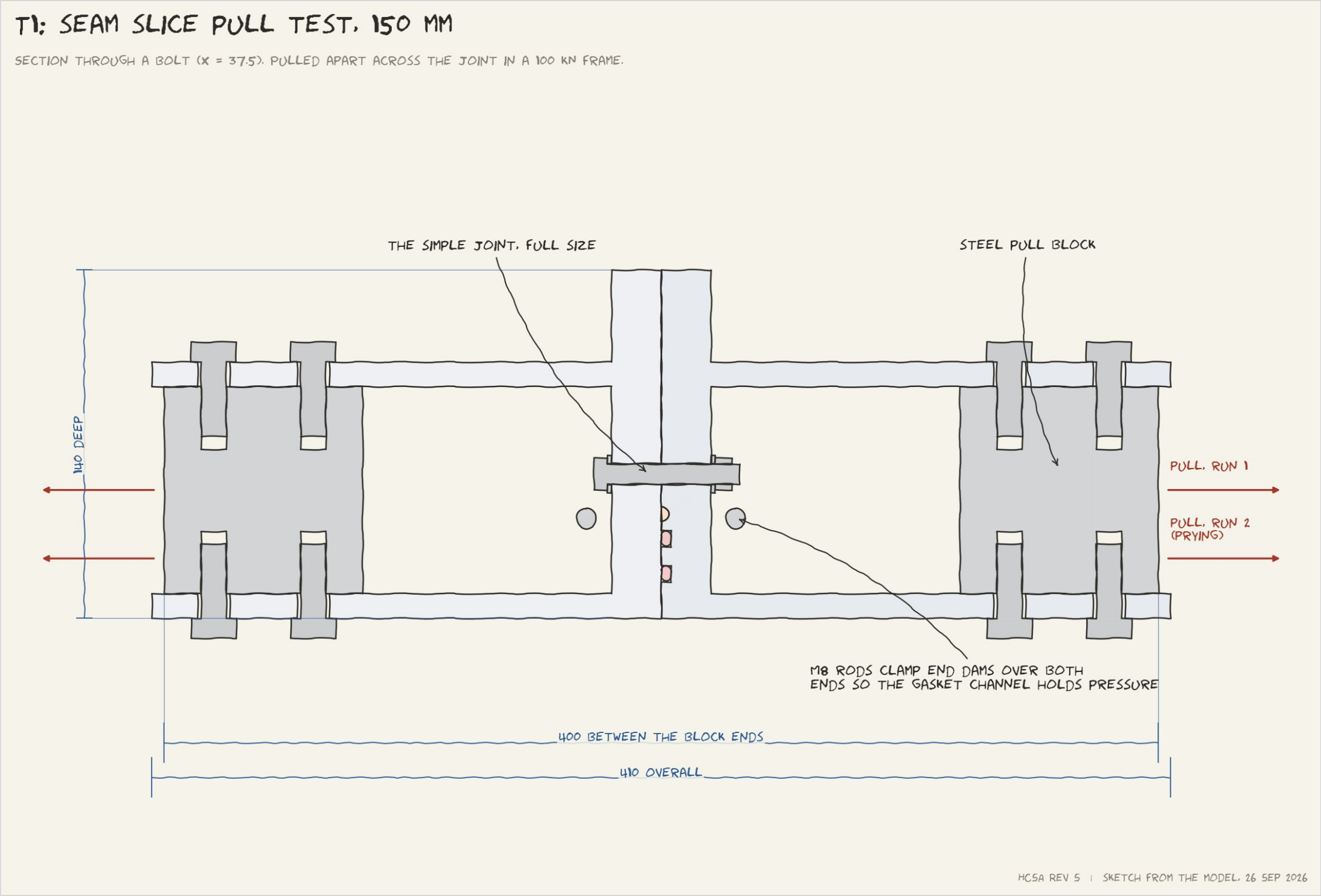
Part of

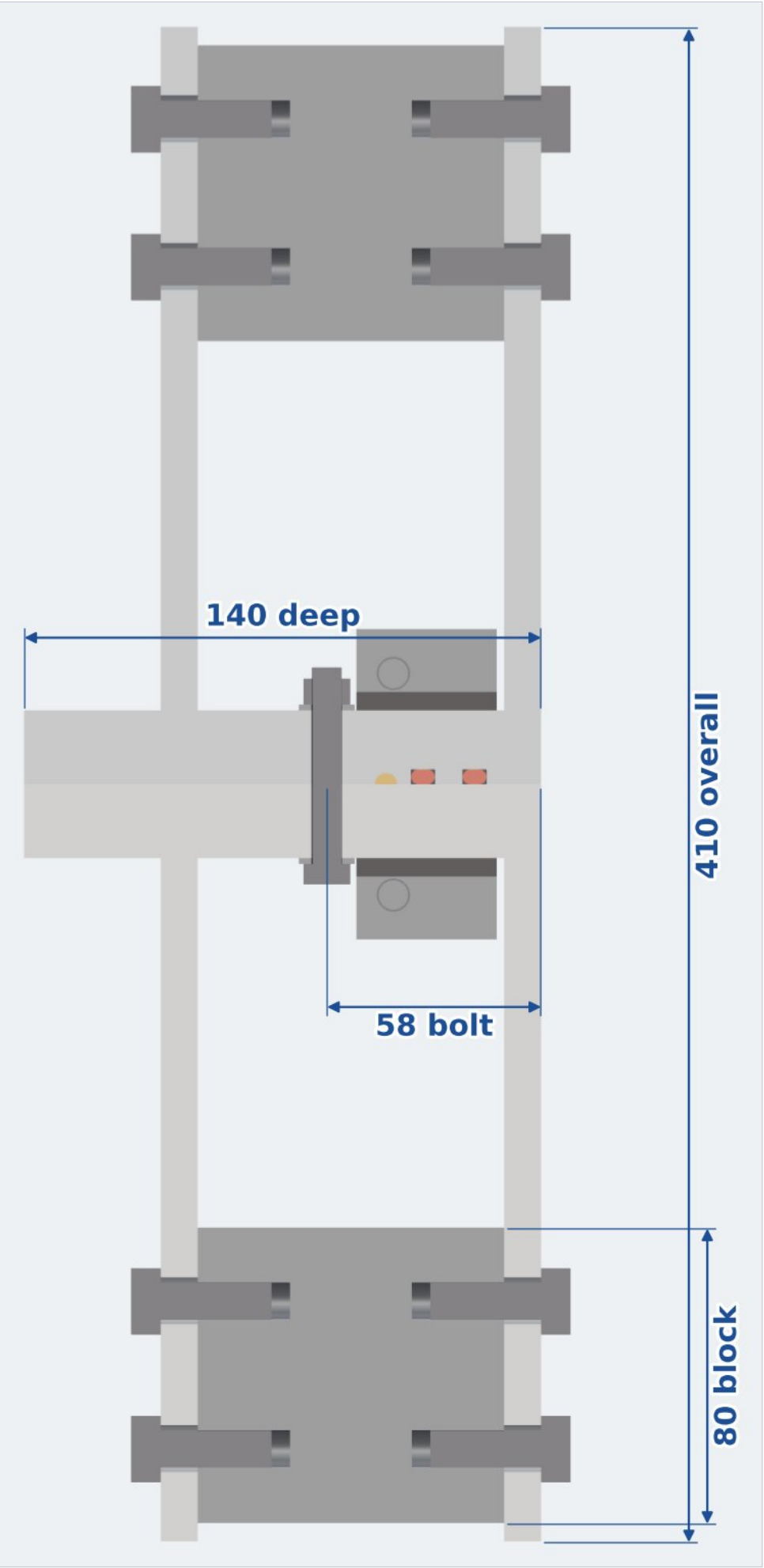
A short piece of the seam, cut from anywhere along its length.

Testing

This is the test: pulled apart in a 100 kN frame and leak-checked while loaded.

Why it is designed like this: see the page “T1: seam slice pull test” at the start of this part.





2D RENDER

T1 slice section

What you are looking at

The same cut, turned the way it hangs in the test frame, with the pull vertical. The pull blocks are at top and bottom; the joint is in the middle.

Dimensions shown (mm)

- 410 overall
- 140 deep
- 58 bolt
- 80 block

Part of

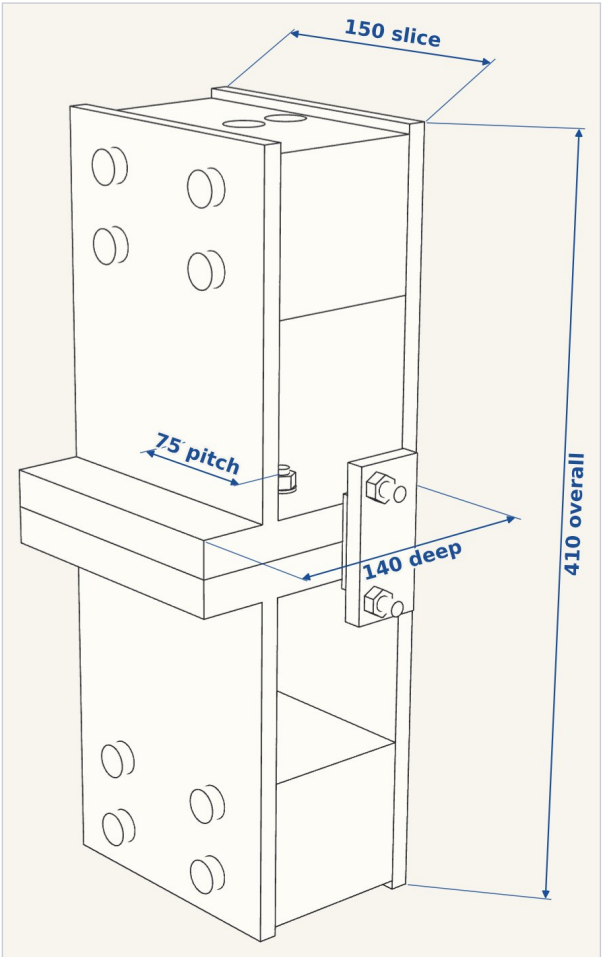
A short piece of the seam, cut from anywhere along its length.

Testing

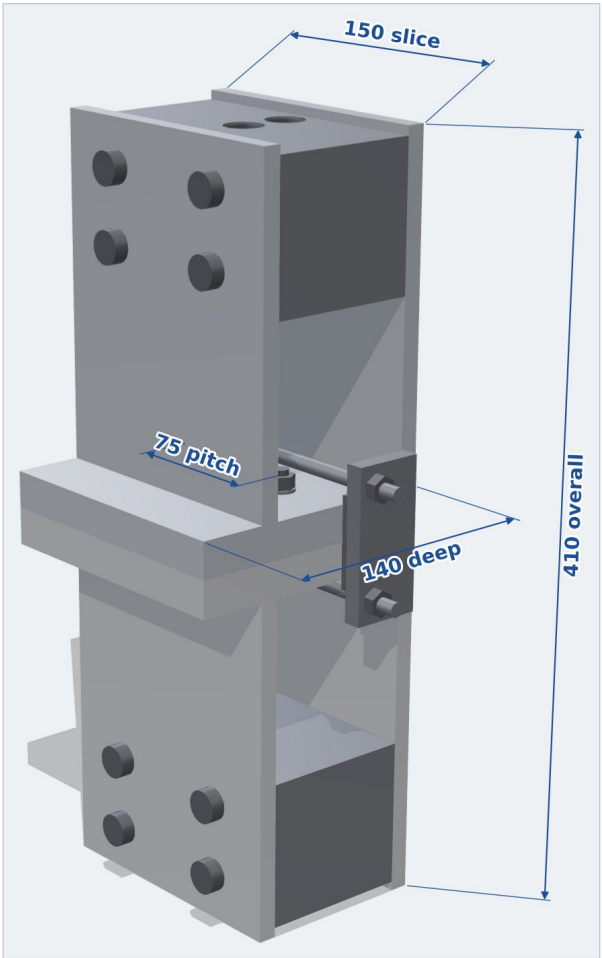
This is the test: pulled apart in a 100 kN frame and leak-checked while loaded.

Why it is designed like this: see the page “T1: seam slice pull test” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

T1 slice 150

What you are looking at

The 150 mm slice as it hangs in the frame. The two bolts are in the middle; the end dam and its rods are on the near face.

Dimensions shown (mm)

- 150 slice
- 410 overall
- 140 deep
- 75 pitch

Part of

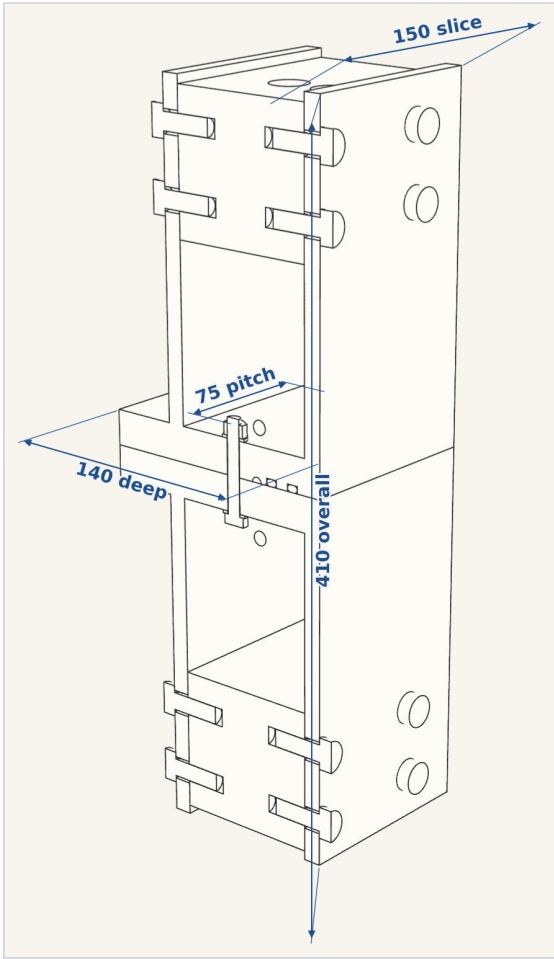
A short piece of the seam, cut from anywhere along its length.

Testing

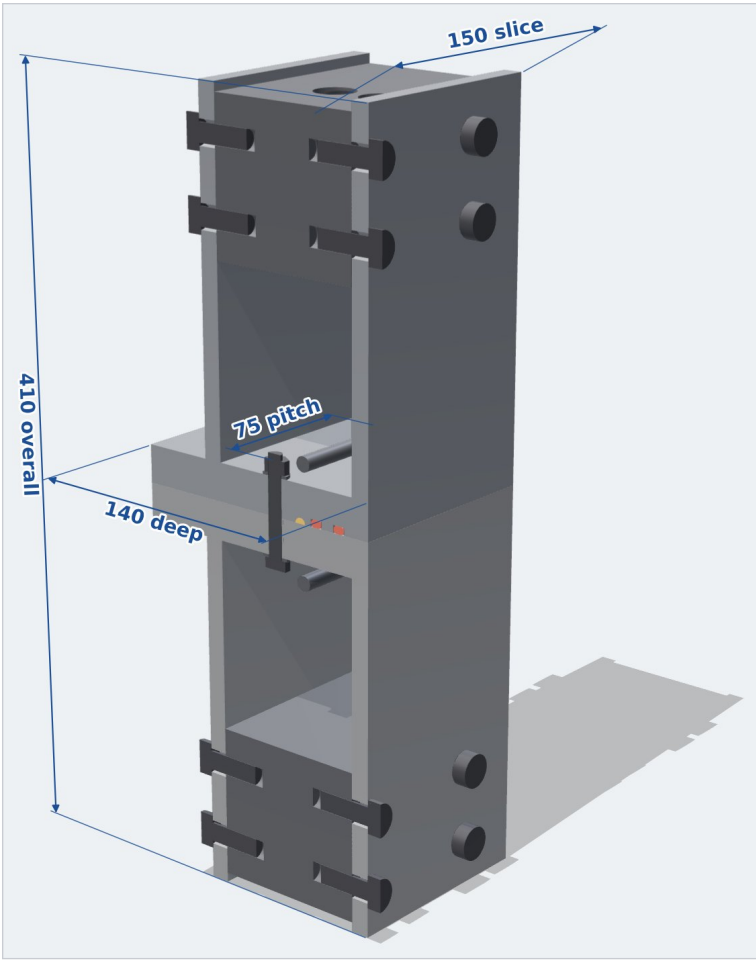
This is the test: pulled apart in a 100 kN frame and leak-checked while loaded.

Why it is designed like this: see the page “T1: seam slice pull test” at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

T1 slice 150 cut

What you are looking at

The same slice cut through a bolt, so the joint section, the bolt and the grip legs show together.

Dimensions shown (mm)

- 150 slice
- 410 overall
- 140 deep
- 75 pitch

Part of

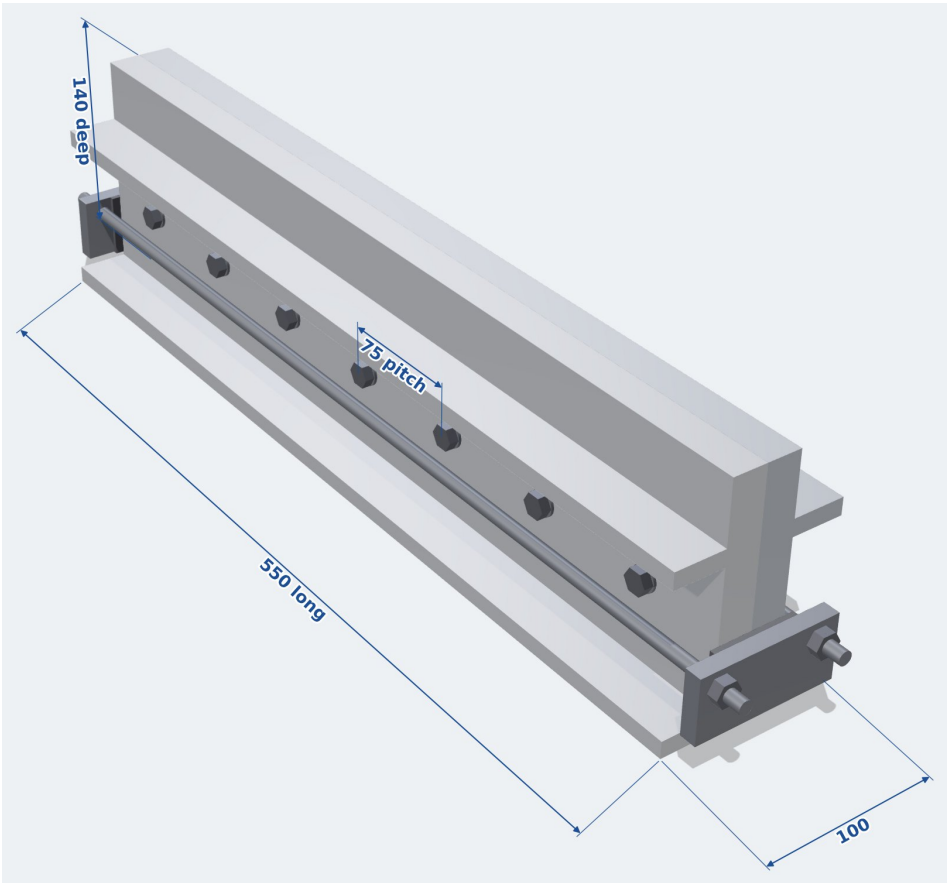
A short piece of the seam, cut from anywhere along its length.

Testing

This is the test: pulled apart in a 100 kN frame and leak-checked while loaded.

Why it is designed like this: see the page “T1: seam slice pull test” at the start of this part.

T1: the 550 mm joint article



What it is

550 mm of the joint with no grip legs: both frames, 7 bolts at 75 mm, both O-rings, the RTV bead and end dams. 100 mm across the flanges, 140 mm deep.

Why it is designed like this

- Pulled at the full seam load it would need 152 to 365 kN, too much for a 100 kN frame. So it is used for fit and seal, not strength.
- It is long enough to show whether a straight run of the joint goes together and seals the way the drawings say.

Where it sits in the whole

A longer piece of the same seam.

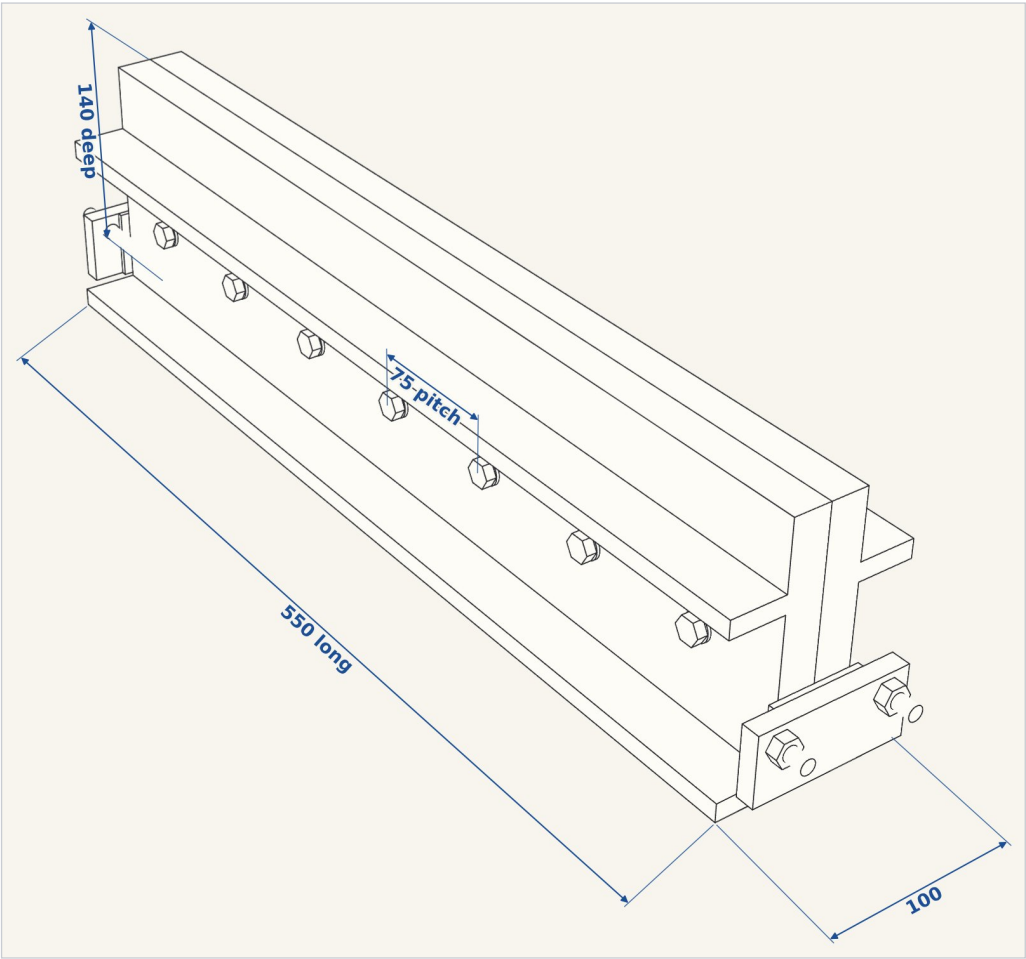
How we test it

- Dry fit: assemble, tighten the bolts, check how much the O-rings are squeezed.
- Leak test: 1 atm between the two O-rings, watch the pressure decay (load case LC5).

Numbers to remember

- 7 bolts
- 11 kg

3D sketch



3D SKETCH AND 3D RENDER

T1 joint article 550

What you are looking at

The 550 mm article: a straight run of the joint with 7 bolts, capped by end dams at both ends.

Dimensions shown (mm)

- 550 long
- 75 pitch
- 100
- 140 deep

Part of

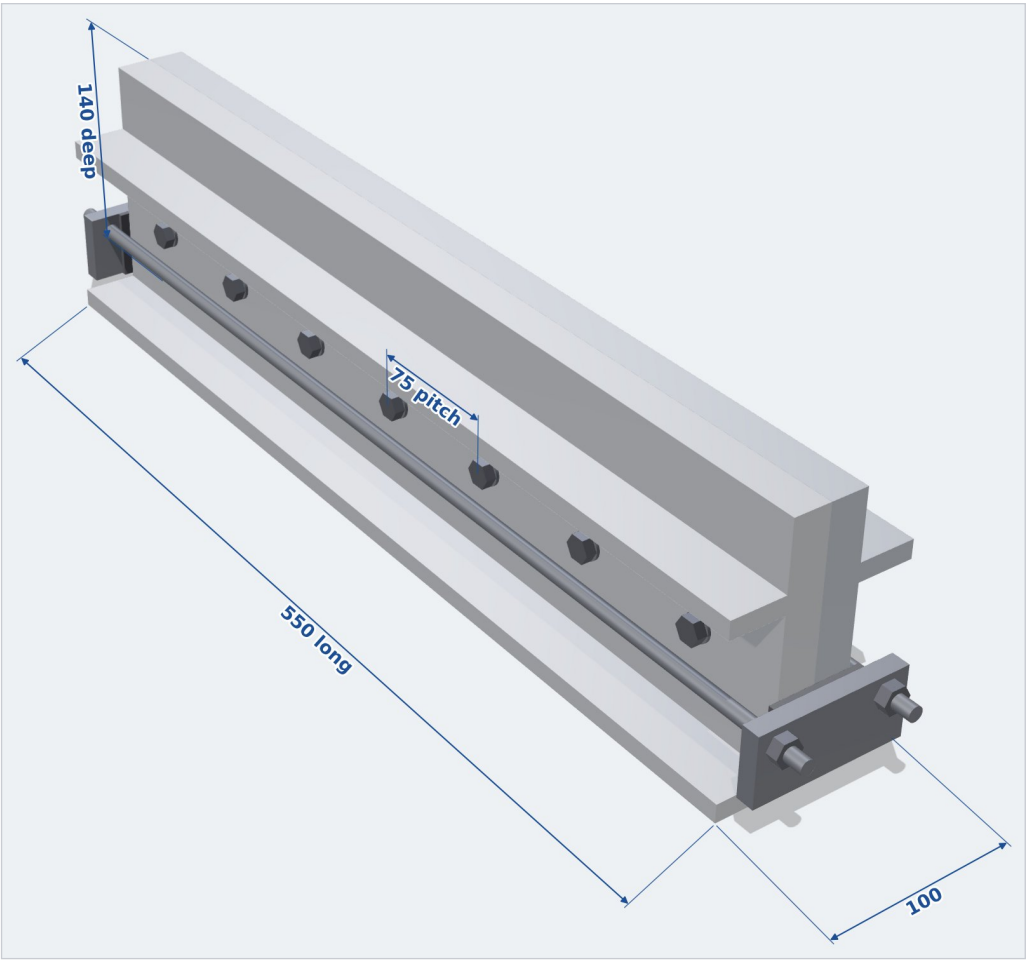
A longer piece of the same seam.

Testing

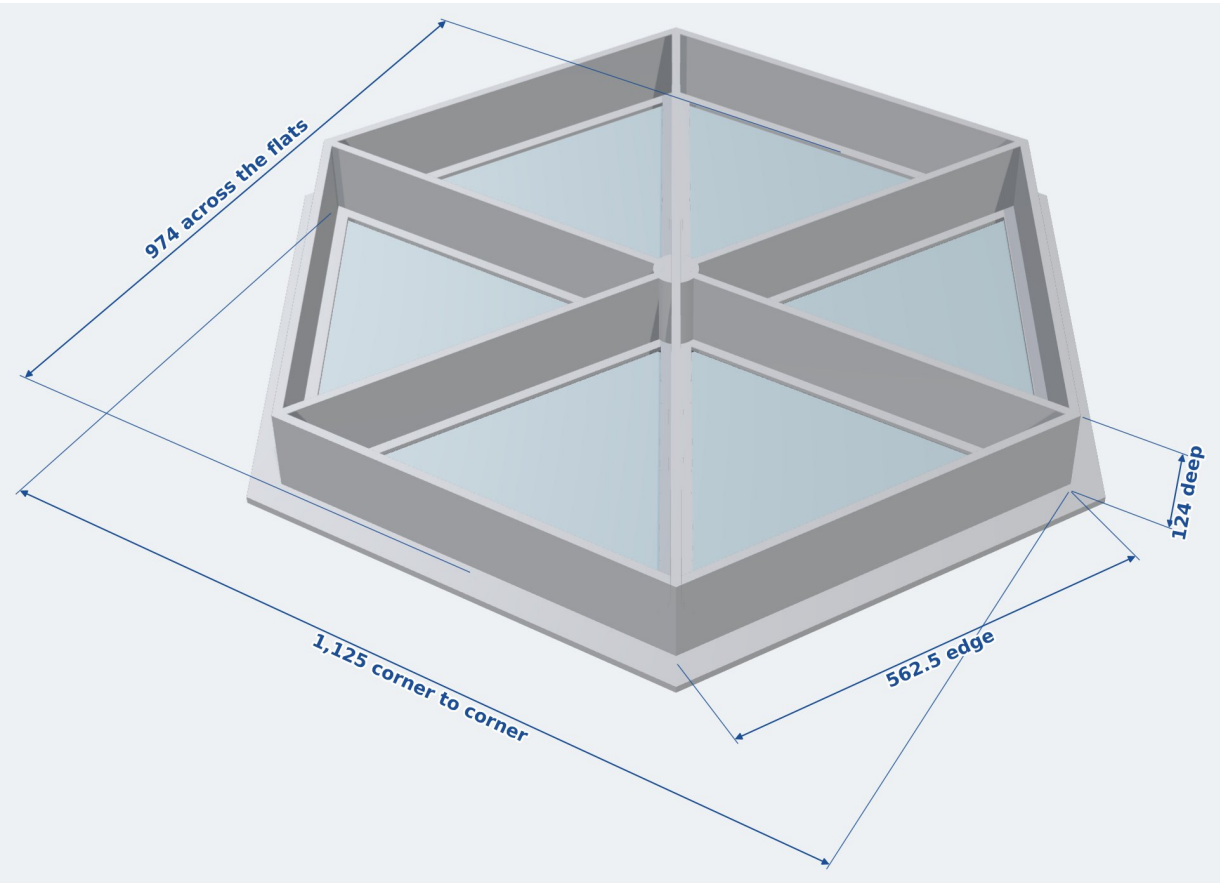
This is the test: dry fit, then a leak test between the O-rings.

Why it is designed like this: see the page “T1: the 550 mm joint article” at the start of this part.

3D render



T2: one panel at 1:4



What it is

A hexagon panel at quarter scale: 1,125 mm corner to corner, 974 mm across the flats, 124 mm deep. 12 mm frame webs with a 40 x 10 mm support flange for the rig, six T ribs 110 mm deep, and six 8 mm aluminium panes on neoprene setting strips, held by 3 x 10 mm beads. 47 kg.

Why it is designed like this

- Every length divided by 4 gives the same stresses at the same pressure, and a 1.1 m panel fits in a shop. The force at 1 atm is 83 kN instead of 1.33 MN.
- Aluminium panes bend almost like the 31 mm glass would at this scale (7.4 mm would match exactly), do not shatter, and can go past 1 atm.
- The ribs are new: sized by hand as spokes so they stay elastic to 2 atm (91 MPa at 1 atm, 183 MPa at 2 atm).
- Some things do not scale cleanly (setting strips, beads, web widths). That is one of the questions for Prof. Henann.

Where it sits in the whole

A model of one hexagon panel.

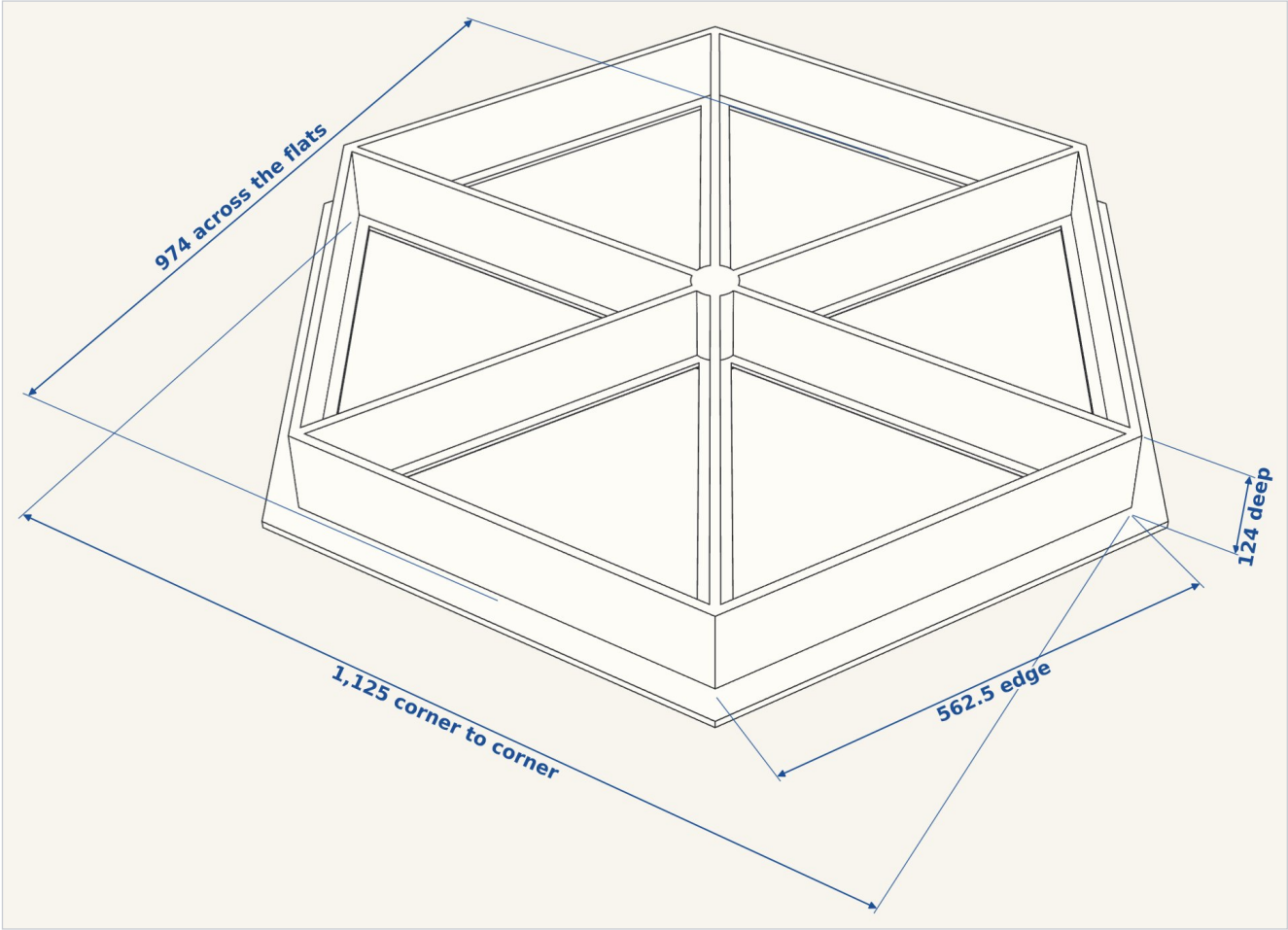
How we test it

- Loaded with water in the T2 rig (next pages).
- Gauges: strain rosettes on a pane centre (both faces) and the opposite pane, rib flanges at mid-length, the frame web at mid-edge and near a corner; deflection at the hub and a pane centre.

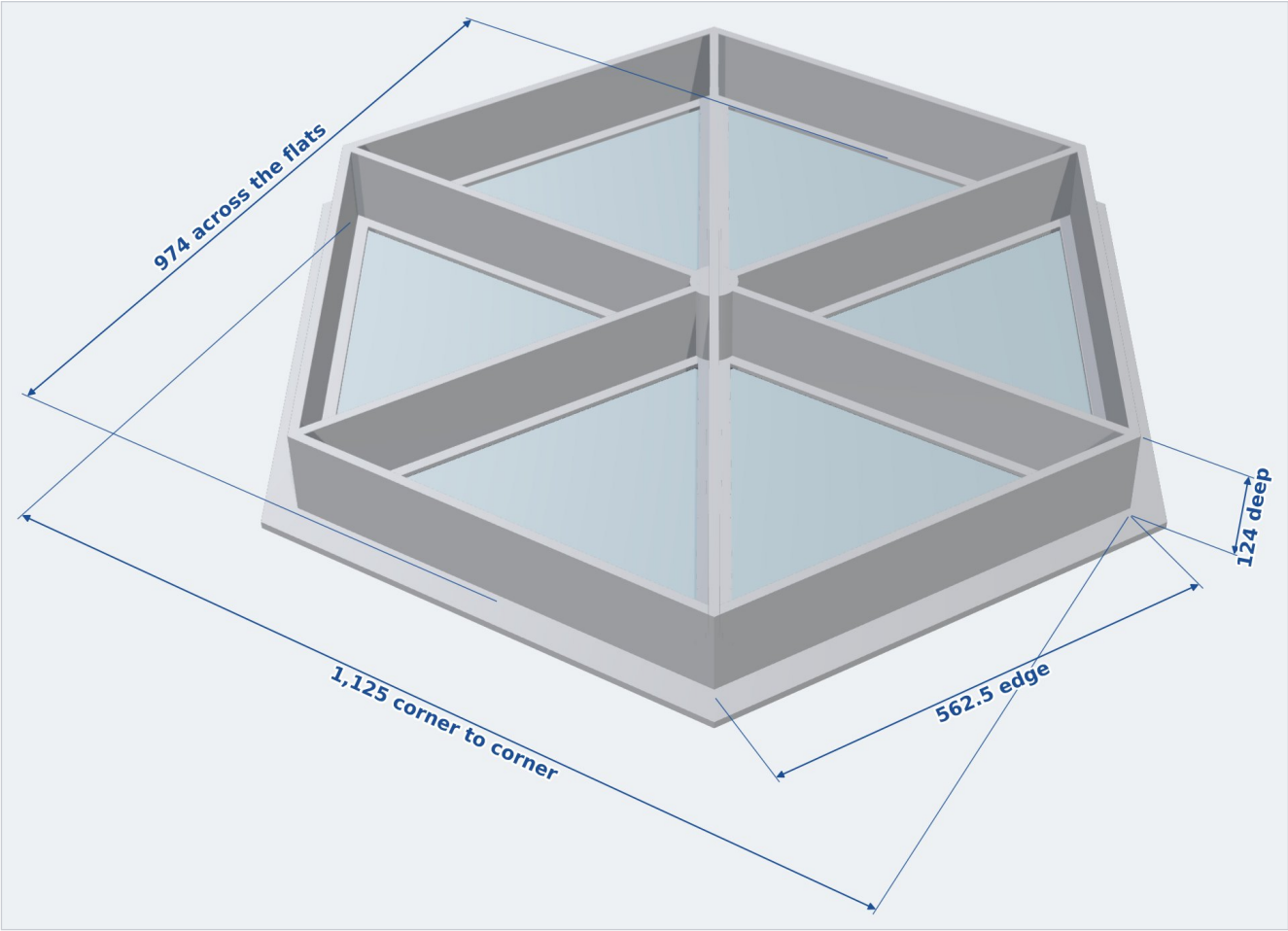
Numbers to remember

- 83 kN at 1 atm, 167 kN at 2 atm
- Pane stress 53 MPa at 1 atm, the same as full size

3D sketch



3D render



3D SKETCH AND 3D RENDER

T2 panel 1to4

What you are looking at

The 1:4 panel from above. Deep T ribs, a frame with a wide support flange all round, and the six aluminium panes.

Dimensions shown (mm)

- 1,125 corner to corner
- 974 across the flats
- 562.5 edge
- 124 deep

Part of

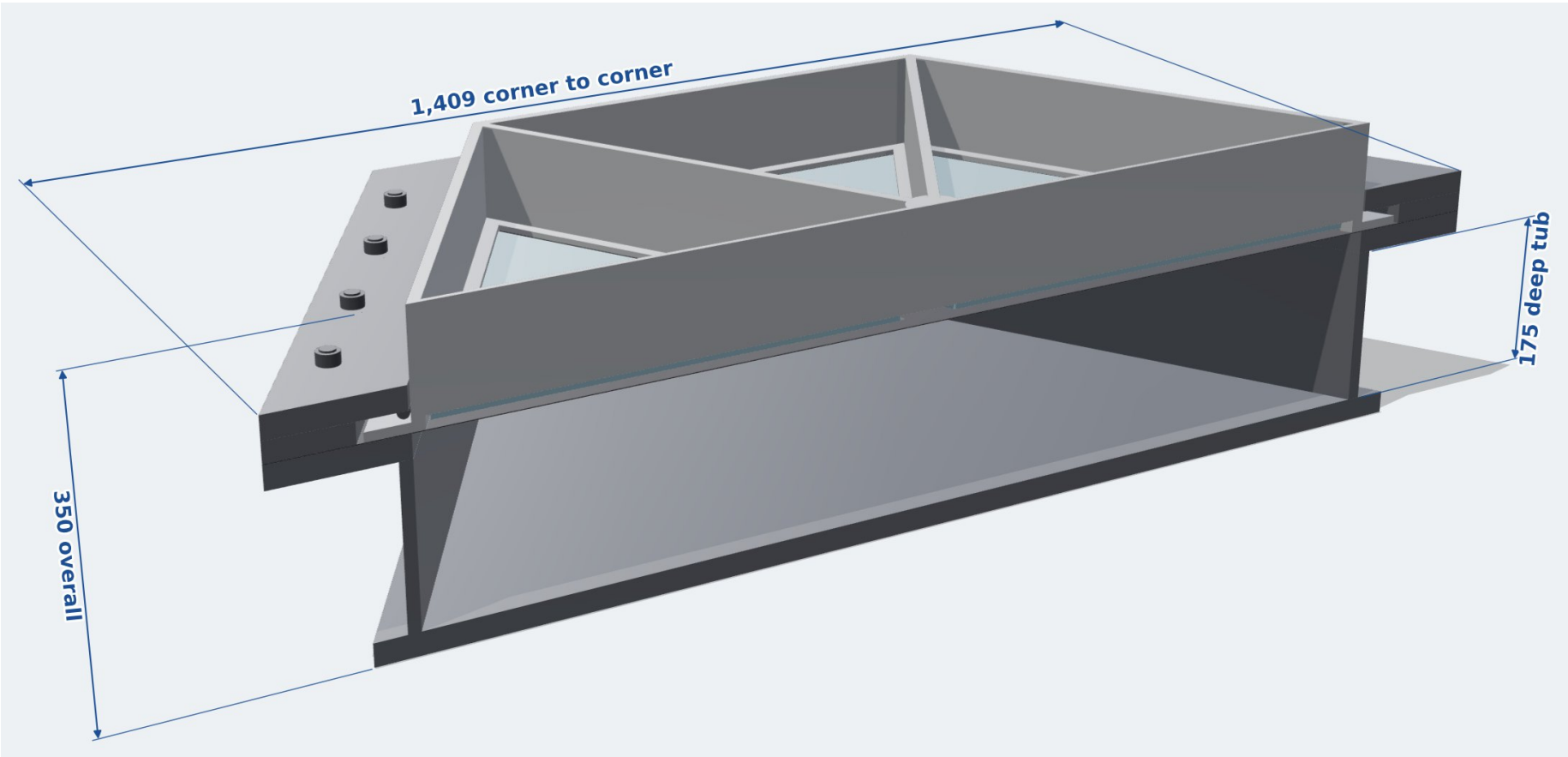
A model of one hexagon panel.

Testing

This is the test article: loaded with water in the T2 rig.

Why it is designed like this: see the page “T2: one panel at 1:4” at the start of this part.

T2: the hexagonal water rig



What it is

A hexagonal steel tub (12 mm walls, 25 mm floor, 175 mm deep) with a 25 mm box flange. The panel sits cabin side down on a 1 mm rubber membrane, the membrane has its own clamp ring, and a hexagonal top ring is bolted down with 24 M16 bolts. 1,409 x 1,220 mm, 350 mm tall; about 490 kg of steel and 165 L of water.

Why it is designed like this

- Water hardly compresses, so if something breaks almost no stored energy is released. Air at the same pressure would store far more.
- The rings are hexagonal so they follow the panel all the way round.
- The membrane carries the water pressure onto the panes and keeps the water in. It does no structural work.
- The membrane has its own clamp ring so its seal holds even when the panel lifts.
- Two edge conditions. Pinned: the panel's support flange bears on 12 mm round bars, so the edge can rotate like a hinge. Clamped: the top ring bears straight on the flange, so the edge cannot rotate. The real edge in the shell is somewhere between, and the pane only meets its bending limit if it acts close to clamped.

Where it sits in the whole

Test equipment, not part of the facility.

How we test it

- Write down the model prediction for every gauge before any water goes in.
- Fill and bleed all the air; check the relief valve lifts at 220 kPa.
- Pinned first: 0 to 1 atm in 0.25 atm steps, hold 2 minutes a step, unload the same way, three times. Then clamped, the same.
- Only then 1.5 and 2 atm, in 0.1 atm steps above 1 atm. Dial gauges read against a bridge clamped to the panel frame, so rig movement drops out.

Numbers to remember

- 700 kg when full
- 24 M16 bolts, 3.5 kN each at 1 atm
- Relief valve at 220 kPa

2D SKETCH

T2 rig section

What you are looking at

The rig cut through two opposite edges, pinned setup. Water below, the panel cabin side down on the membrane, the round bar under the top ring at each edge.

Dimensions shown (mm)

- 1,220 across the flats of the rings
- 998 across the tub
- 974 across the panel
- 175 deep tub
- 350 overall

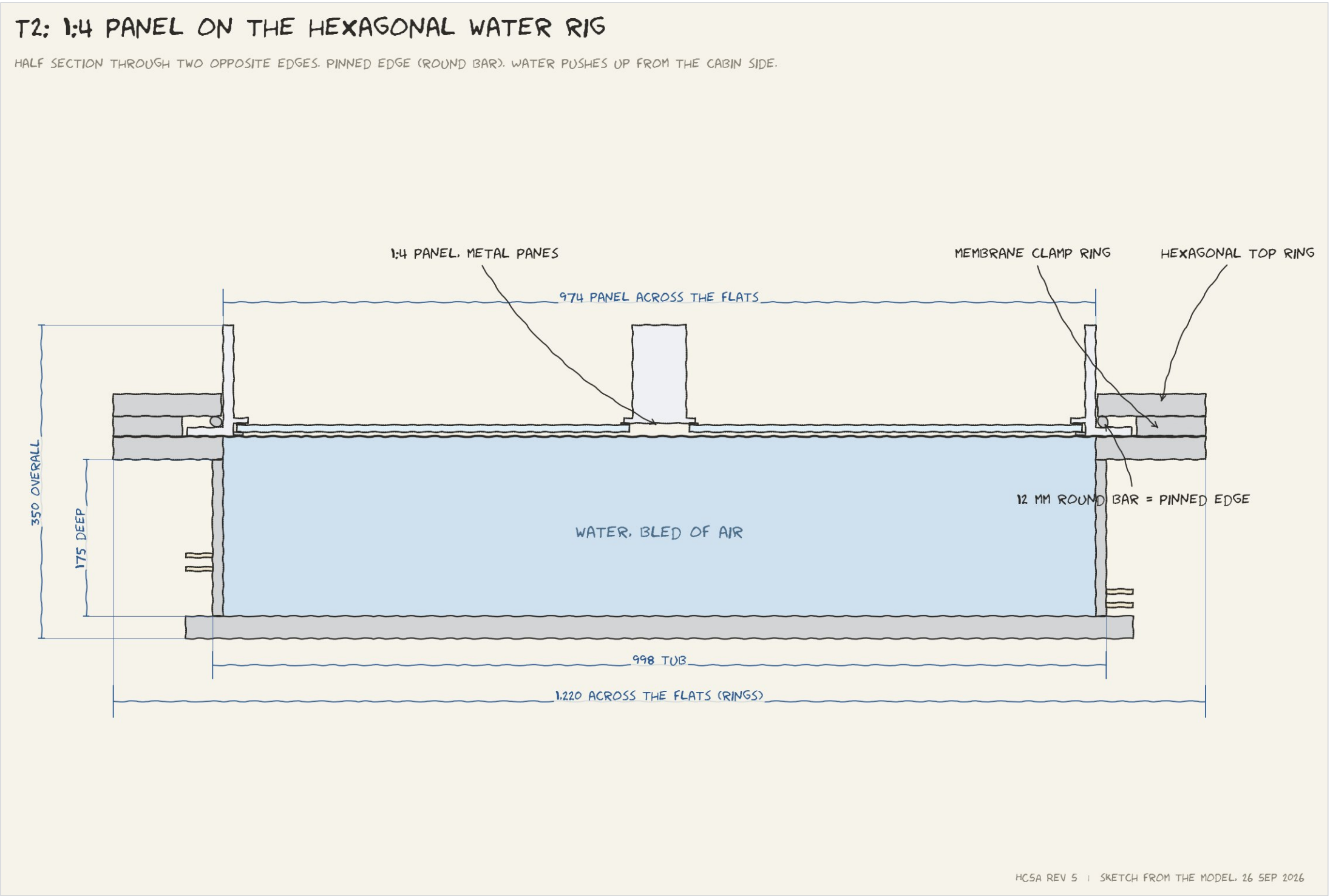
Part of

Test equipment, not part of the facility.

Testing

This is the test rig: pinned and clamped runs up to 2 atm.

Why it is designed like this: see the page “T2: the hexagonal water rig” at the start of this part.



2D RENDER

T2 rig pinned section

What you are looking at

The pinned setup cut through two opposite edges, rendered. Look for the small circle at each edge: that is the 12 mm round bar the panel pivots on.

Dimensions shown (mm)

- 1,220 across the flats (rings)
- 998 tub
- 974 panel
- 175 deep
- 350 overall

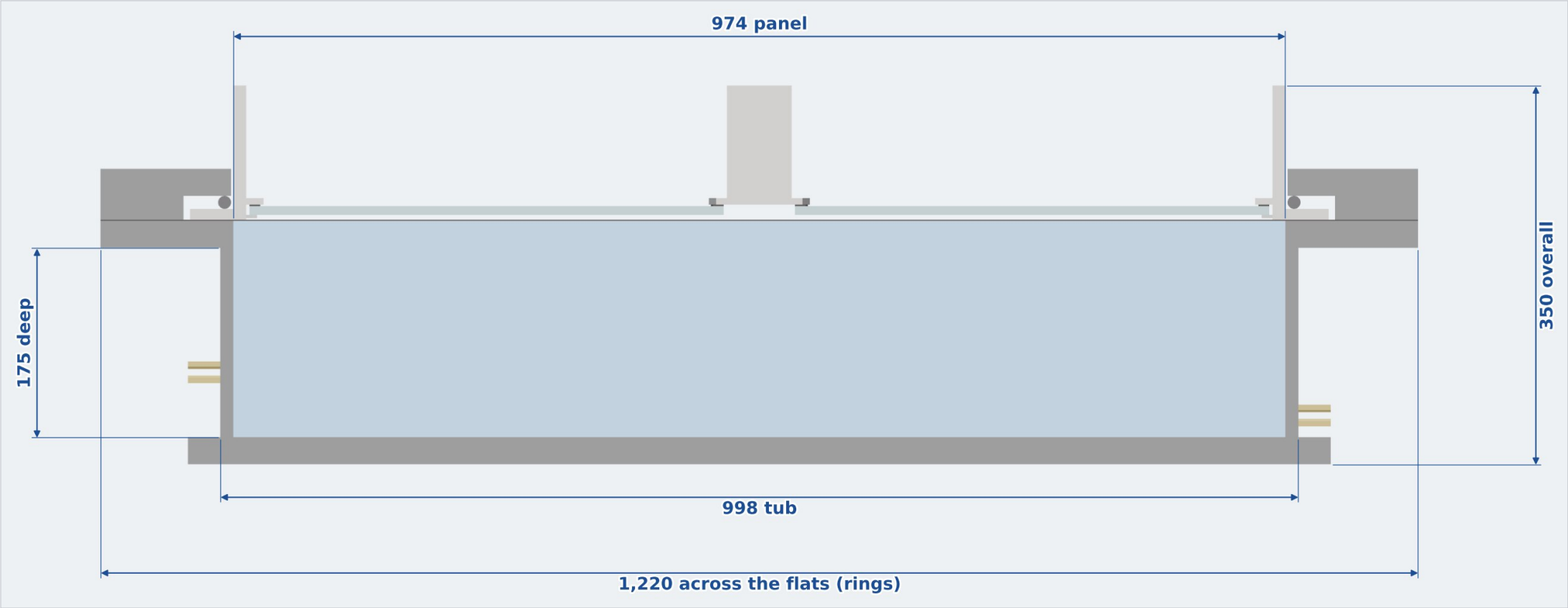
Part of

Test equipment, not part of the facility.

Testing

This is the test rig: pinned and clamped runs up to 2 atm.

Why it is designed like this: see the page “T2: the hexagonal water rig” at the start of this part.



2D RENDER

T2 rig clamped section

What you are looking at

The clamped setup, same cut. No round bar: the top ring sits straight on the panel's support flange, and the membrane clamp ring is thinner to suit.

Dimensions shown (mm)

- 1,220 across the flats (rings)
- 998 tub
- 974 panel
- 175 deep
- 350 overall

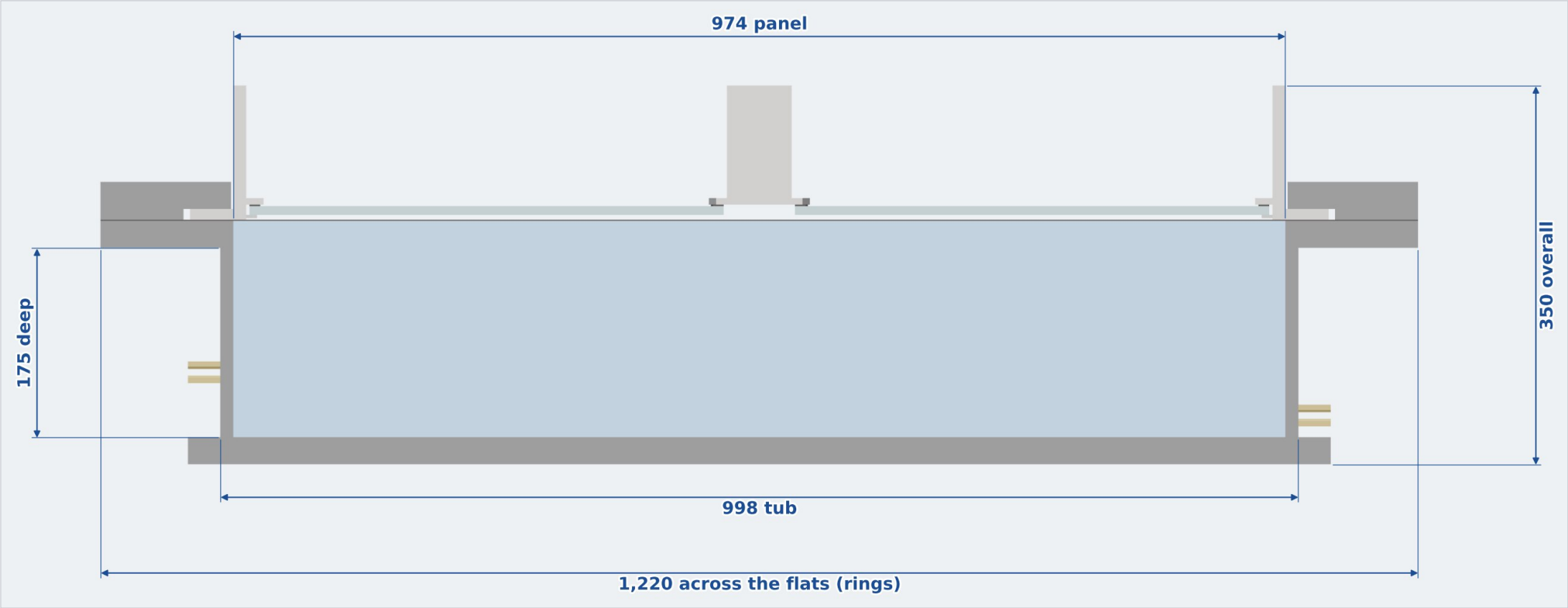
Part of

Test equipment, not part of the facility.

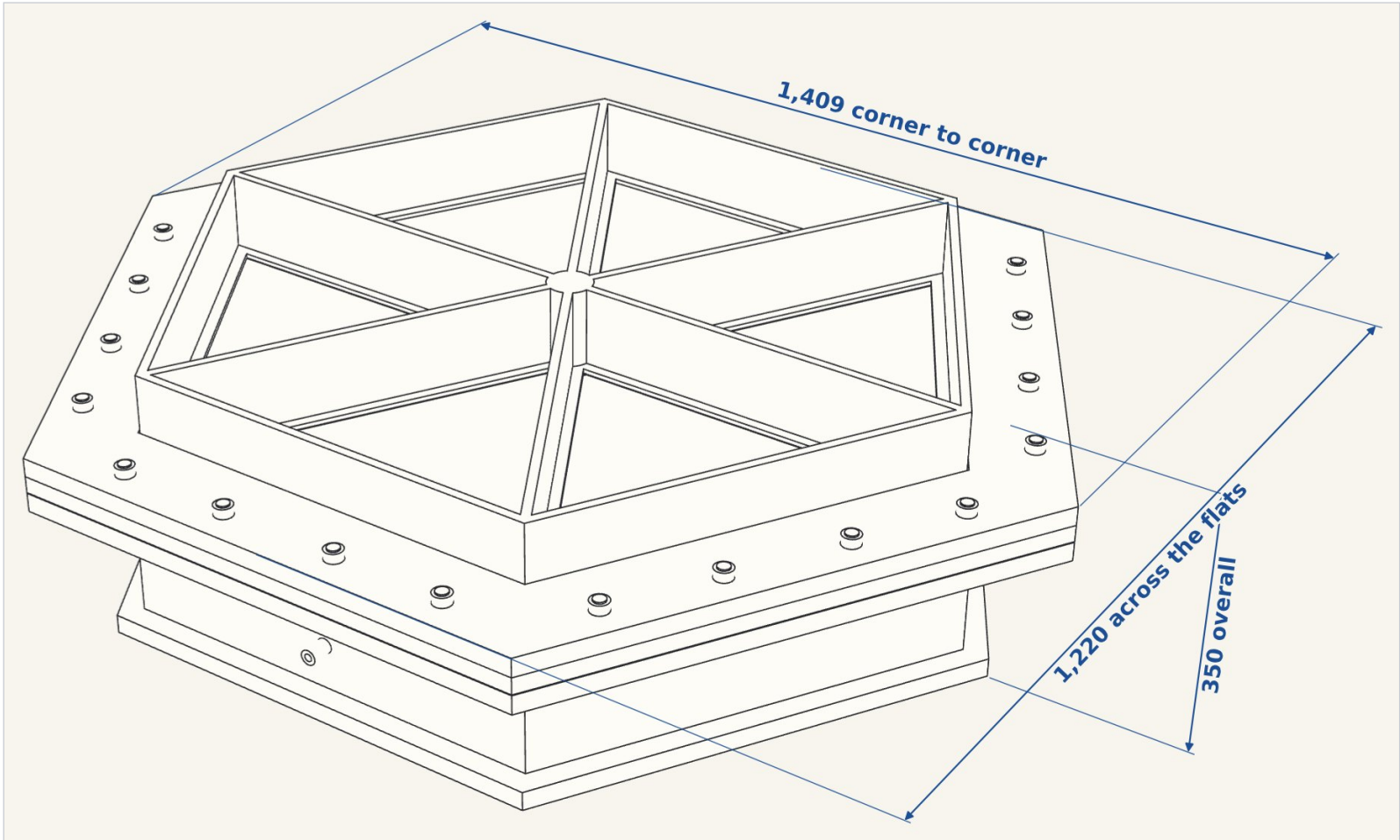
Testing

This is the test rig: pinned and clamped runs up to 2 atm.

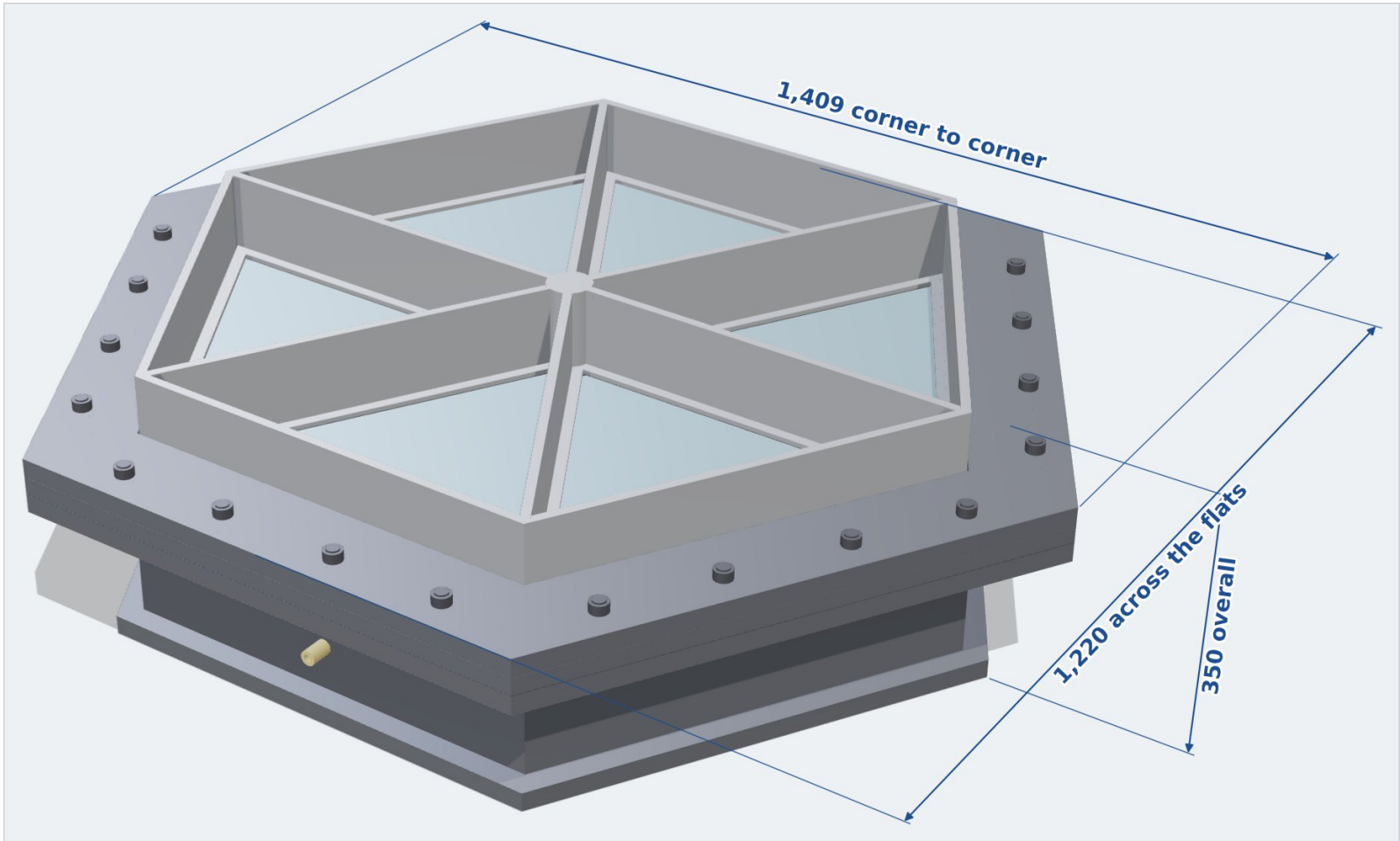
Why it is designed like this: see the page “T2: the hexagonal water rig” at the start of this part.



3D sketch



3D render



3D SKETCH AND 3D RENDER

T2 rig pinned

What you are looking at

The rig assembled with the panel in place. The 24 bolts run round the hexagonal top ring.

Dimensions shown (mm)

- 1,409 corner to corner
- 1,220 across the flats
- 350 overall

Part of

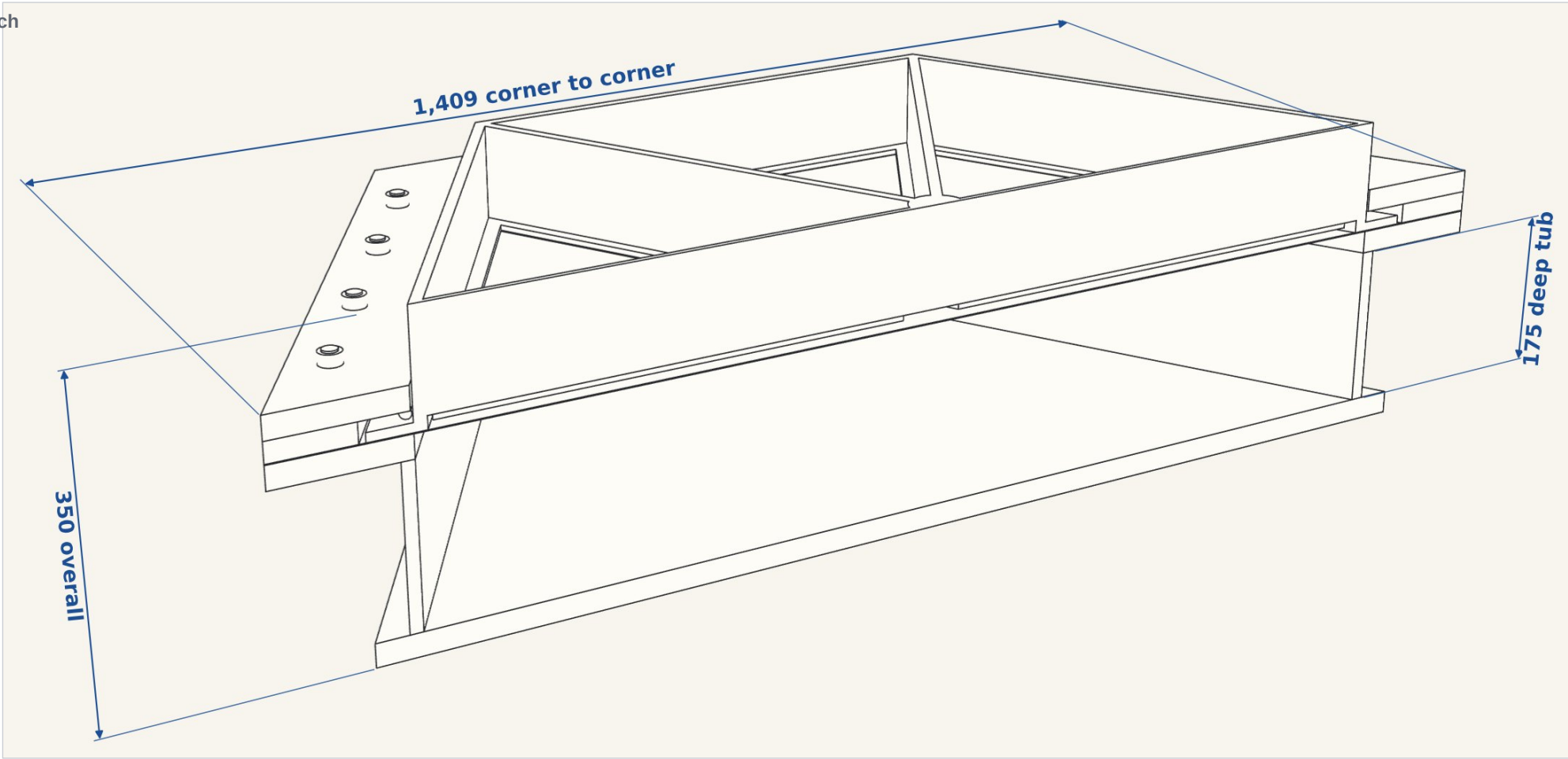
Test equipment, not part of the facility.

Testing

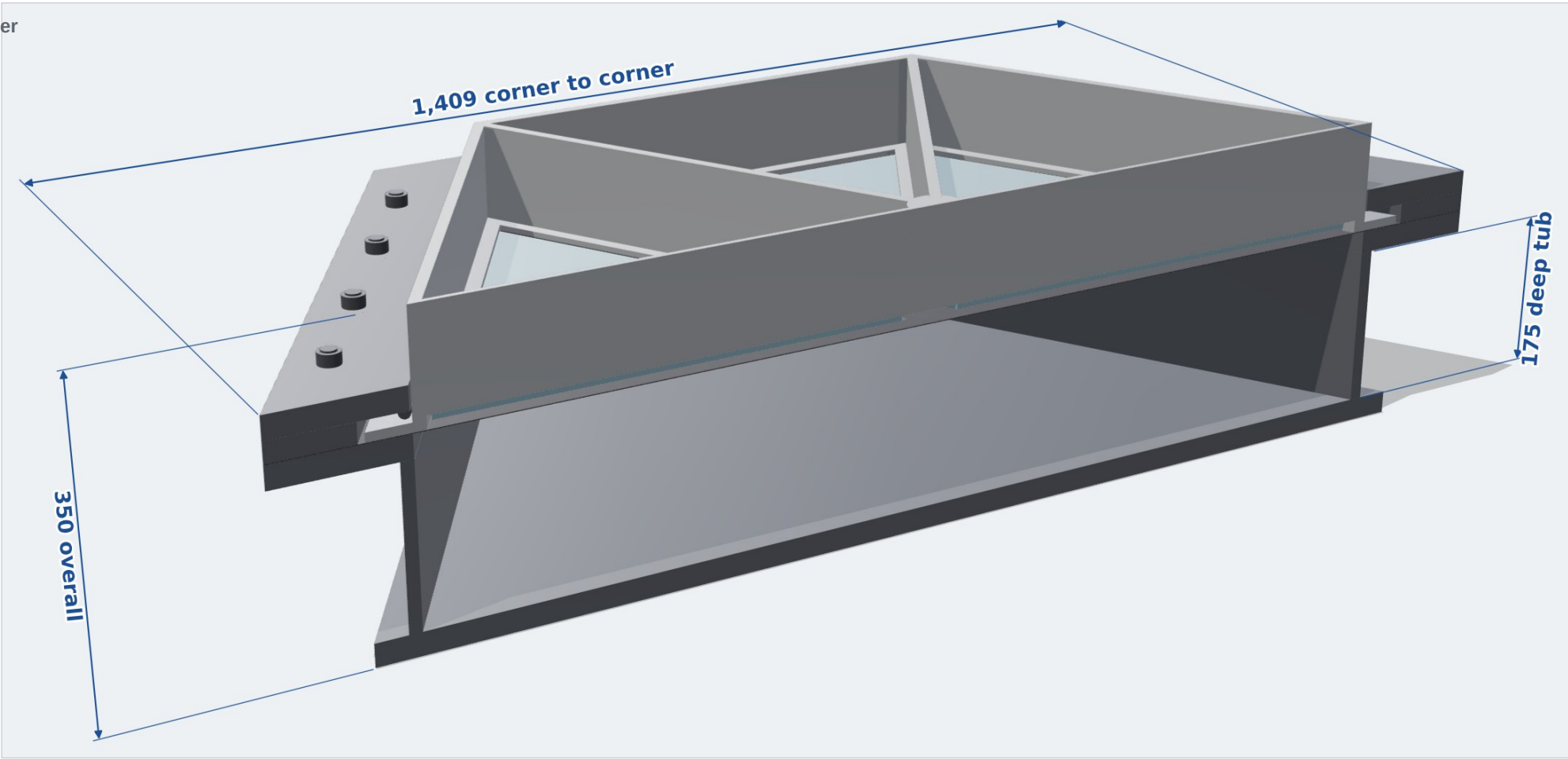
This is the test rig: pinned and clamped runs up to 2 atm.

Why it is designed like this: see the page "T2: the hexagonal water rig" at the start of this part.

3D sketch



3D render



3D SKETCH AND 3D RENDER

T2 rig pinned cut

What you are looking at

The pinned rig cut in half (water hidden). You can see the tub, the flange, the membrane clamp ring and the top ring stacked at the edge.

Dimensions shown (mm)

- 1,409 corner to corner
- 350 overall
- 175 deep tub

Part of

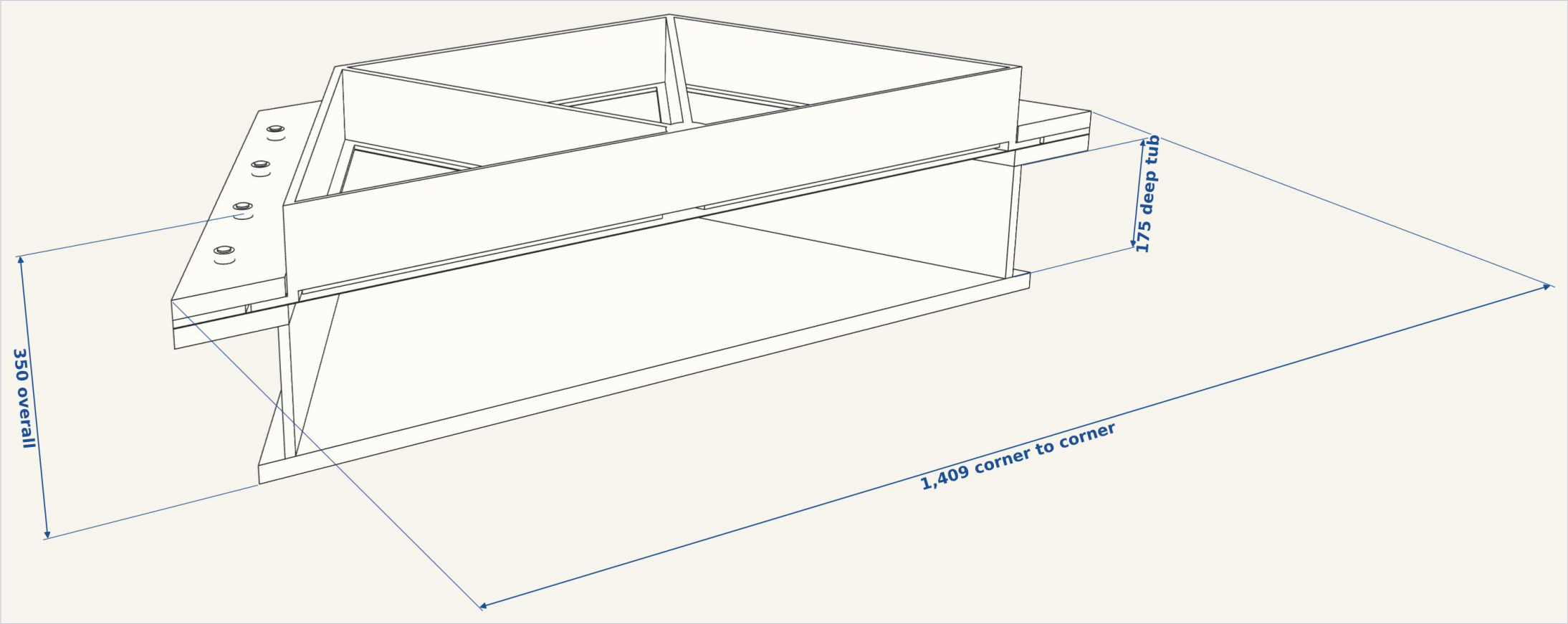
Test equipment, not part of the facility.

Testing

This is the test rig: pinned and clamped runs up to 2 atm.

Why it is designed like this: see the page “T2: the hexagonal water rig” at the start of this part.

3D sketch



3D SKETCH AND 3D RENDER

T2 rig clamped cut

What you are looking at

The clamped rig cut in half. The top ring sits lower, straight on the panel flange.

Dimensions shown (mm)

- 1,409 corner to corner
- 350 overall
- 175 deep tub

Part of

Test equipment, not part of the facility.

Testing

This is the test rig: pinned and clamped runs up to 2 atm.

Why it is designed like this: see the page "T2: the hexagonal water rig" at the start of this part.

3D render

